Though we'll need more than 60 Snowball devices, C is the only option that works. The internet uplink could transport less than 2 PB in 6 months (otherwise, say with a 10 Gb uplink, D would work).

upvoted 6 times

Cyberkayu 2 years, 5 months ago

transfer 5 PB data in 1Gbps link, assume 0 overhead and drop packet, need 485 days, 10 hours, 50 minutes, 40 seconds to complete.

Snowball it is. C

upvoted 1 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

C
https://docs.aws.amazon.com/storagegateway/latest/tgw/using-tape-gateway-snowball.html

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

Migrate petabyte-scale data stored on physical tapes to AWS using AWS Snowball
https://aws.amazon.com/snowball/#:~:text=Migrate-,petabyte%2Dscale,-data%20stored%20on

upvoted 2 times

hungta 2 years, 6 months ago

Selected Answer: C

5 PB data is too huge for using 1Gbps uplink. With this uplink, it takes more than 1 year to migrate this data.

upvoted 2 times

baba365 2 years, 8 months ago

Answer: D for most cost effective.

If you are looking for a cost-effective, durable, long-term, offsite alternative for data archiving, deploy a Tape Gateway. With its virtual tape library (VTL) interface, you can use your existing tape-based backup software infrastructure to store data on virtual tape cartridges that you create -

https://docs.aws.amazon.com/storagegateway/latest/tgw/WhatIsStorageGateway.html

upvoted 1 times

Devsin2000 2 years, 8 months ago

D
https://aws.amazon.com/storagegateway/vtl/
the bandwidth and available time is ample

upvoted 1 times

nnecode 2 years, 8 months ago

Selected Answer: A

The most cost-effective solution to meet the requirements is to read the data from the tapes on premises. Stage the data in a local NFS storage. Use AWS DataSync to migrate the data to Amazon S3 Glacier Flexible Retrieval.

This solution is the most cost-effective because it uses the least amount of bandwidth. AWS DataSync is a service that transfers data between on-premises storage and Amazon S3. It uses a variety of techniques to optimize the transfer speed and reduce c

upvoted 1 times

lemur88 2 years, 9 months ago

Selected Answer: C

Only thing that makes sense given the 1Gbps limitation

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Option C is likely the most cost-effective solution given the large data size and limited internet bandwidth. The physical data transfer and integration with the existing tape infrastructure provides efficiency benefits that can optimize the cost.

upvoted 3 times

barracouto 2 years, 9 months ago

Selected Answer: C

Went through this dump twice now. Exam is in about an hour. Will update with results.

upvoted 2 times

Vaishali12 2 years, 9 months ago

how was ur exam?
was these dump que helpful?

upvoted 1 times

riccardoto 2 years, 10 months ago

Finished the dump today - taking my exam tomorrow :-) Wish me luck!

upvoted 4 times

Ale1973 2 years, 10 months ago

My rational: question is about which solution will meet these requirements MOST cost-effectively, not MOST time or effectively, then, my response is D (using Tape Gateways)

upvoted 4 times

Question #584 Topic 1

A company is deploying an application that processes large quantities of data in parallel. The company plans to use Amazon EC2 instances for the workload. The network architecture must be configurable to prevent groups of nodes from sharing the same underlying hardware.

Which networking solution meets these requirements?

A. Run the EC2 instances in a spread placement group. **Most Voted**

B. Group the EC2 instances in separate accounts.

C. Configure the EC2 instances with dedicated tenancy.

D. Configure the EC2 instances with shared tenancy.

Correct Answer: A

Community vote distribution

A (87%) 13%

Comments

czyboi Highly Voted 2 years, 9 months ago

Selected Answer: A

A spread placement group is a group of instances that are each placed on distinct hardware.
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html

upvoted 12 times

Guru4Cloud Highly Voted 2 years, 9 months ago

Selected Answer: C

C is the correct answer.

Configuring the EC2 instances with dedicated tenancy ensures that each instance will run on isolated, single-tenant hardware. This meets the requirement to prevent groups of nodes from sharing underlying hardware.

A spread placement group only provides isolation at the Availability Zone level. Instances could still share hardware within an AZ.

upvoted 5 times

pentium75 2 years, 5 months ago

No. C ensures that your EC2 instances run on hardware that is not shared with other customers (!). It is still shared among YOUR instances.

upvoted 6 times

emakid Most Recent 1 year, 11 months ago

Selected Answer: A

Option A: Run the EC2 instances in a spread placement group.

Spread Placement Group: This placement group strategy ensures that EC2 instances are distributed across distinct hardware to reduce the risk of correlated failures. Instances in a spread placement group are placed on different underlying hardware, which aligns with the requirement to prevent groups of nodes from sharing the same underlying hardware. This is a good fit for the scenario where you need to ensure high availability and fault tolerance.

upvoted 2 times

Gape4 1 year, 11 months ago

Selected Answer: A

Spread – Strictly places a small group of instances across distinct underlying hardware to reduce correlated failures.

upvoted 2 times

Marco_St 2 years, 5 months ago

Selected Answer: A

dedicated tenancy cannot ensure the instances share the same hardware. So A
upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: A

A, spread placement group does exactly what is required here.
Not C and D, tenancy determines whether the hardware is shared with other customers or not, it has nothing to with your own instances sharing hardware. (On the contrary, dedicated tenancy would spread your EC2 instances across as little nodes as possible.)

Not B, accounts have nothing to do with the issue.
upvoted 5 times

maged123 2 years, 5 months ago

Selected Answer: A

Let's assume that you have two groups of instances, group A and group B and you have two physical hardware X and Y. With spread placement group, you can have group A of instances on hardware X and group B on hardware Y but this will not prevent hardware X to host other instances of other customers because your only requirement is to separate group A from group B. On the other hand, the dedicated tenancy means that AWS wil dedicate the physical hardware only for you. So, the correct answer is A.
upvoted 3 times

Murtadhaceit 2 years, 6 months ago

Question is ambiguous and confusing. Is it asking about the EC2 instance of the same application not sharing hardware? or EC2 instance not sharing hardware with other EC2 from other applications?
upvoted 2 times

Mikado211 2 years, 6 months ago

Selected Answer: A

Spread placement group allows you to isolate your instances on hardware level.
Dedicated tenancy allows you to be sure that you are the only customer on the hardware.

The correct answer is A.
upvoted 2 times

Mikado211 2 years, 6 months ago

A : Spread placement group
upvoted 2 times

lucasbg 2 years, 6 months ago

Selected Answer: A

Def is A: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html
upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

Keywords 'prevent groups of nodes from sharing the same underlying hardware'.
Spread Placement Group strictly places a small group of instances across distinct underlying hardware to reduce correlated failures.
upvoted 2 times

cciesam 2 years, 7 months ago

Selected Answer: A

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html
Each instances is placed on seven different racks, each rack has its own network and power source.
upvoted 2 times

wsdasdasdqwdaw 2 years, 7 months ago

Another tricky question, but I would go for A because:

Dedicated instances:
Dedicated Instances are EC2 instances that run on hardware that's dedicated to a single customer. Dedicated Instances that belong to different AWS accounts are physically isolated at a hardware level, even if those accounts are linked to a single payer account. However, Dedicated Instances might share hardware with other instances from the same AWS account that are not Dedicated Instances.
Which is not the desired option.

Spread – strictly places a small group of instances across distinct underlying hardware to reduce correlated failures.

That's why A.
upvoted 3 times

garuta 2 years, 8 months ago

Selected Answer: C

C is clear.
upvoted 1 times

pentium75 2 years, 5 months ago
"Dedicated tenancy" means that all your nodes run on hardware that is not shared with other customers. This is counter-productive to the objective here.
upvoted 2 times

Devsin2000 2 years, 8 months ago
A
When you launch a new EC2 instance, the EC2 service attempts to place the instance in such a way that all of your instances are spread out across underlying hardware to minimize correlated failures.
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html
upvoted 3 times

taustin2 2 years, 8 months ago
Selected Answer: A
Spread Placement Group strictly places a small group of instances across distinct underlying hardware to reduce correlated failures.
upvoted 1 times

Eminenza22 2 years, 9 months ago
Selected Answer: A
Option A is the correct answer. It suggests running the EC2 instances in a spread placement group. This solution is cost-effective and requires minimal development effort .
upvoted 3 times

Eminenza22 2 years, 9 months ago
The placement group reduces the risk of simultaneous failures by spreading the instances across distinct underlying hardware
upvoted 2 times

Question #585 *Topic 1*

A solutions architect is designing a disaster recovery (DR) strategy to provide Amazon EC2 capacity in a failover AWS Region. Business requirements state that the DR strategy must meet capacity in the failover Region.

Which solution will meet these requirements?

A. Purchase On-Demand Instances in the failover Region.

B. Purchase an EC2 Savings Plan in the failover Region.

C. Purchase regional Reserved Instances in the failover Region.

D. Purchase a Capacity Reservation in the failover Region. **Most Voted**

Correct Answer: D

Community vote distribution

D (93%) 7%

Comments

EllenLiu 1 year, 5 months ago

Selected Answer: D

it is called on-demand capacity reservation actually . no discount. must specify the following attributes:
Availability Zone & Instance type & os type & tenancy
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-capacity-reservations.html

upvoted 2 times

EllenLiu 1 year, 5 months ago

DR is a classic use case for capacity reservation

upvoted 2 times

emakid 1 year, 11 months ago

Selected Answer: D

Option D: Purchase a Capacity Reservation in the failover Region.

Capacity Reservation: Capacity Reservations ensure that you have reserved capacity in a specific region for your instances, regardless of whether you are using On-Demand or Reserved Instances. This is ideal for DR scenarios because it guarantees that the required EC2 capacity will be available when needed.

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

"Business requirements state that the DR strategy must meet capacity in the failover Region"
so only D meets these requirements
A. No reservation of capacity
B. Saving plans don't guarantee capacity
C. Can be possible but it's like an active instance so doesn't really make sense

upvoted 3 times

awsgeek75 2 years, 5 months ago

Correction on C (I mixed it up with tenancy!). Reserved instance are not really for capacity, its for type of instance which gives good discount but that is not required here.

upvoted 2 times

Derek_G 2 years, 5 months ago

Selected Answer: D

Purchase a Capacity Reservation in the failover Region:

A Capacity Reservation allows you to reserve a specific amount of EC2 instance capacity in a given region without purchasing specific instances. This reserved capacity is dedicated to your account and can be utilized for launching instances when needed. Capacity Reservations offer flexibility, allowing you to launch different instance types and sizes within the reserved capacity.

Purchase regional Reserved Instances in the failover Region:

Regional Reserved Instances involve paying an upfront fee to reserve a certain number of specific EC2 instances in a particular region. These reserved instances are of a predefined type and size, providing a more traditional reservation model. Regional Reserved Instances are specific to a designated region and ensure that the reserved instances of a particular specification are available when needed.

upvoted 4 times

TheLaPlanta 2 years, 2 months ago

What I don't get is... can't you accomplish that by using on-demand? I understood that you can scale infinitely

upvoted 2 times

Tanidanindo 2 years, 2 months ago

It won't guarantee that you have the capacity when you need it. If available, like in most cases, it'll work. But the scenario requires a guarantee that the capacity will be available.

upvoted 2 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

D

Ask is to reserve capacity with RI capacity is not reserved also you can reserve capacity along with RI but only in AZ .

https://repost.aws/knowledge-center/ri-reserved-capacity

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: D

Capacity Reservations mitigate against the risk of being unable to get On-Demand capacity in case there are capacity constraints. If you have strict capacity requirements, and are running business-critical workloads that require a certain level of long or short-term capacity assurance, create a Capacity Reservation to ensure that you always have access to Amazon EC2 capacity when you need it, for as long as you need it.

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: D

Capacity Reservations enable you to reserve capacity for your Amazon EC2 instances in a specific Availability Zone for any duration. This gives you the flexibility to selectively add capacity reservations and still get the Regional RI discounts for that usage. By creating Capacity Reservations, you ensure that you always have access to Amazon EC2 capacity when you need it, for as long as you need it.

upvoted 3 times

potomac 2 years, 7 months ago

Savings Plans does not provide a capacity reservation.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Capacity Reservations allocate EC2 capacity in a specific AWS Region for you to launch instances.

The capacity is reserved and available to be utilized when needed, meeting the requirement to provide EC2 capacity in the failover region.

Other options do not reserve capacity. On-Demand provides flexible capacity but does not reserve capacity upfront. Savings Plans and Reserved Instances provide discounts but do not reserve capacity.

Capacity Reservations allow defining instance attributes like instance type, platform, Availability Zone so the reserved capacity matches the production environment.

upvoted 4 times

Eminenza22 2 years, 9 months ago

Selected Answer: D

A regional Reserved Instance does not reserve capacity

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/reserved-instances-scope.html

upvoted 3 times

judyda 2 years, 9 months ago

Selected Answer: D

reserved instances for price discount. need capacity reservation.

upvoted 3 times

gispankaj 2 years, 9 months ago

Selected Answer: C

The Reserved Instance discount applies to instance usage within the instance family, regardless of size.

upvoted 2 times

pentium75 2 years, 5 months ago
"Reserved Instances are not physical instances, but rather a billing discount "

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-reserved-instances.html
upvoted 2 times

ErnShm 2 years, 9 months ago
D

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-capacity-reservations.html
upvoted 2 times

Question #586 | Topic 1

A company has five organizational units (OUs) as part of its organization in AWS Organizations. Each OU correlates to the five businesses that the company owns. The company's research and development (R&D) business is separating from the company and will need its own organization. A solutions architect creates a separate new management account for this purpose.

What should the solutions architect do next in the new management account?

A. Have the R&D AWS account be part of both organizations during the transition.

B. Invite the R&D AWS account to be part of the new organization after the R&D AWS account has left the prior organization. **Most Voted**

C. Create a new R&D AWS account in the new organization. Migrate resources from the prior R&D AWS account to the new R&D AWS account.

D. Have the R&D AWS account join the new organization. Make the new management account a member of the prior organization.

Correct Answer: B

Community vote distribution

B (90%) | 10%

Comments

awsgeek75 **Highly Voted** 2 years, 5 months ago

Selected Answer: B

An account can only join another org when it leaves the first org.
A is wrong as it's not possible
C that's a new account so not really a migration
D The R&D department is separating from the company so you don't want the OU to join via nesting

upvoted 9 times

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: B

B as exactly described here: https://repost.aws/knowledge-center/organizations-move-accounts

upvoted 7 times

emakid **Most Recent** 1 year, 11 months ago

Selected Answer: B

Option B: Invite the R&D AWS account to be part of the new organization after the R&D AWS account has left the prior organization is the appropriate approach. This option ensures that the R&D AWS account transitions smoothly from the old organization to the new one. The steps involved are:

Remove the R&D AWS account from the existing organization: This is done from the existing organization’s management account.

Invite the R&D AWS account to join the new organization: Once the R&D account is no longer part of the previous organization, it can be invited to and accepted into the new organization.

upvoted 2 times

Marco_St 2 years, 5 months ago

Selected Answer: B

https://aws.amazon.com/blogs/mt/migrating-accounts-between-aws-organizations-with-consolidated-billing-to-all-features/

upvoted 4 times

ale_brd_111 2 years, 5 months ago

Selected Answer: B

https://repost.aws/knowledge-center/organizations-move-accounts
Remove the member account from the old organization.
Send an invite to the member account from the new organization.
Accept the invite to the new organization from the member account.
upvoted 3 times

Derek_G 2 years, 5 months ago

Selected Answer: C

C is better. first migrate , then delete. avoid the data lose.
upvoted 1 times

pentium75 2 years, 5 months ago

What kind of "data lose" would happen when you change the account to a new organization? And why should you migrate ALL RESOURCES of the account to a new account?
upvoted 2 times

Derek_G 2 years, 5 months ago

C is better. first migrate , then delete. avoid the data lose.
upvoted 1 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

As per this document, B is clearly the answer.
https://repost.aws/knowledge-center/organizations-move-accounts#:~:text=In%20either%20case%2C-,perform%20these%20actions,-for%20each%20member
upvoted 2 times

Joben 2 years, 8 months ago

Selected Answer: B

In either case, perform these actions for each member account:
- Remove the member account from the old organization.
- Send an invite to the member account from the new organization.
- Accept the invite to the new organization from the member account.

https://repost.aws/knowledge-center/organizations-move-accounts
upvoted 7 times

Guru4Cloud 2 years, 8 months ago

Selected Answer: C

Creating a brand new AWS account in the new organization (Option C) allows for a clean separation and migration of only the necessary resources from the old account to the new.
upvoted 2 times

pentium75 2 years, 5 months ago

"A clean separation" is already existing, they have their own account. "Migration of only the necessary resources from the old account to the new" i: not asked for. They have an account in an existing organization, they need their own organization, thus move the existing account to a new organisation (B), done.
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

When separating a business unit from an AWS Organizations structure, best practice is to:

Create a new AWS account dedicated for the business unit in the new organization
Migrate resources from the old account to the new account
Remove the old account from the original organization
This allows a clean break between the organizations and avoids any linking between them after separation.
upvoted 1 times

pentium75 2 years, 5 months ago

Says who?
upvoted 2 times

ErnShm 2 years, 9 months ago

B
https://aws.amazon.com/blogs/mt/migrating-accounts-between-aws-organizations-with-consolidated-billing-to-all-features/
upvoted 3 times

gispankaj 2 years, 9 months ago

Selected Answer: B

account can leave current organization and then join new organization.
upvoted 4 times

Question #587 Topic 1

A company is designing a solution to capture customer activity in different web applications to process analytics and make predictions. Customer activity in the web applications is unpredictable and can increase suddenly. The company requires a solution that integrates with other web applications. The solution must include an authorization step for security purposes.

Which solution will meet these requirements?

A. Configure a Gateway Load Balancer (GWLB) in front of an Amazon Elastic Container Service (Amazon ECS) container instance that stores the information that the company receives in an Amazon Elastic File System (Amazon EFS) file system. Authorization is resolved at the GWLB.

B. Configure an Amazon API Gateway endpoint in front of an Amazon Kinesis data stream that stores the information that the company receives in an Amazon S3 bucket. Use an AWS Lambda function to resolve authorization.

C. Configure an Amazon API Gateway endpoint in front of an Amazon Kinesis Data Firehose that stores the information that the company receives in an Amazon S3 bucket. Use an API Gateway Lambda authorizer to resolve authorization. Most Voted

D. Configure a Gateway Load Balancer (GWLB) in front of an Amazon Elastic Container Service (Amazon ECS) container instance that stores the information that the company receives on an Amazon Elastic File System (Amazon EFS) file system. Use an AWS Lambda function to resolve authorization.

Correct Answer: C

Community vote distribution

C (86%) 14%

Comments

ralfj Highly Voted 2 years, 9 months ago

Selected Answer: C

https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-use-lambda-authorizer.html

upvoted 9 times

FlyingHawk Most Recent 1 year, 4 months ago

Selected Answer: C

https://docs.aws.amazon.com/firehose/latest/dev/basic-deliver.html

upvoted 1 times

emakid 1 year, 11 months ago

Selected Answer: C

Option C: Configure an Amazon API Gateway endpoint in front of an Amazon Kinesis Data Firehose that stores the information that the company receives in an Amazon S3 bucket. Use an API Gateway Lambda authorizer to resolve authorization.

This solution meets the requirements in the following ways:

Handles Unpredictable Traffic: Amazon Kinesis Data Firehose can handle variable amounts of streaming data and automatically scales to accommodate sudden increases in traffic.

Integration with Web Applications: Amazon API Gateway provides a RESTful API endpoint for integrating with web applications.

Authorization: An API Gateway Lambda authorizer provides the necessary authorization step to secure API access.

Data Storage: Amazon Kinesis Data Firehose can deliver data directly to an Amazon S3 bucket for storage, making it suitable for long-term analytics and predictions.

upvoted 4 times

Matte_ 2 years ago

Selected Answer: B

https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-use-lambda-authorizer.html
upvoted 2 times

MatAlves 1 year, 8 months ago
You cannot use Kinesis Data stream to store data in S3. You need Firehose for that.
upvoted 4 times

Mr_Marcus 2 years ago
Ummm. This link (Use API Gateway Lambda authorizers) helps to validate "C" as the correct answer, not "B".
upvoted 2 times

4fad2f8 2 years, 4 months ago
Selected Answer: B
B. Amazon Kinesis Data Firehose does not save anything
upvoted 2 times

FlyingHawk 1 year, 4 months ago
Amazone Data Firehose can send data to S3: https://docs.aws.amazon.com/firehose/latest/dev/s3-object-name.html
upvoted 1 times

jaswantn 2 years, 4 months ago
option C...Amazon Kinesis Data Firehose that stores the information (that the company receives) in an Amazon S3 bucket.
This answer statement is worded in a complex way. It means to say that Firehose stores the data in S3 ...which company receives from API Gateway.
upvoted 3 times

TariqKipkemei 2 years, 6 months ago
Selected Answer: C
Configure an Amazon API Gateway endpoint in front of an Amazon Kinesis Data Firehose that stores the information that the company receives in ar Amazon S3 bucket. Use an API Gateway Lambda authorizer to resolve authorization.
upvoted 4 times

wsdasdasdqwdaw 2 years, 7 months ago
Using ECS just to stores the information is a overkill. So B or C then, lambda authoriser is the key word => C
upvoted 4 times

Eminenza22 2 years, 9 months ago
Selected Answer: C
https://docs.aws.amazon.com/lambda/latest/dg/services-kinesisfirehose.html
upvoted 3 times

ErnShm 2 years, 9 months ago
C

authorizer is configured for the method. If it is, API Gateway calls the Lambda function. The Lambda function authenticates the caller by means such as the following: Calling out to an OAuth provider to get an OAuth access token
upvoted 3 times

gispankaj 2 years, 9 months ago
Selected Answer: C
lambda authoriser seems to be logical solution.
upvoted 3 times

Question #588 | Topic 1

An ecommerce company wants a disaster recovery solution for its Amazon RDS DB instances that run Microsoft SQL Server Enterprise Edition. The company's current recovery point objective (RPO) and recovery time objective (RTO) are 24 hours.

Which solution will meet these requirements MOST cost-effectively?

A. Create a cross-Region read replica and promote the read replica to the primary instance.

B. Use AWS Database Migration Service (AWS DMS) to create RDS cross-Region replication.

C. Use cross-Region replication every 24 hours to copy native backups to an Amazon S3 bucket.

D. Copy automatic snapshots to another Region every 24 hours. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

TiagueteVital Highly Voted 2 years, 9 months ago

Selected Answer: D

Snapshots are always a cost-efficience way to have a DR plan.

upvoted 5 times

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: D

Cross region data transfer is billable so think of smallest amount of data to transfer every 24 hours

upvoted 5 times

MatAlves Most Recent 1 year, 8 months ago

A, B, C => cross-region $$

D => copy snapshots -> most cost-effectively.

upvoted 2 times

emakid 1 year, 11 months ago

Selected Answer: D

Option D: Copy automatic snapshots to another Region every 24 hours.

Explanation: This option involves copying RDS automatic snapshots to another Region. It is a straightforward way to ensure that snapshots are available in the event of a disaster. Since RDS snapshots are typically incremental and copied periodically, this solution matches the 24-hour RPO requirement effectively and is cost-effective compared to maintaining constant cross-Region replication.

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: D

Amazon RDS creates and saves automated backups of your DB instance or Multi-AZ DB cluster during the backup window of your DB instance. RDS creates a storage volume snapshot of your DB instance, backing up the entire DB instance and not just individual databases. RDS saves the automated backups of your DB instance according to the backup retention period that you specify. If necessary, you can recover your DB instance to any point in time during the backup retention period.

upvoted 3 times

wsdasdasdqwdaw 2 years, 7 months ago

most cost-effective way is just copying the snapshot (24h delta in the storage). => D

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Dddddddddd
upvoted 4 times

Eminenza22 2 years, 9 months ago

Selected Answer: D

This is the most cost-effective solution because it does not require any additional AWS services. Amazon RDS automatically creates snapshots of you DB instances every hour. You can copy these snapshots to another Region every 24 hours to meet your RPO and RTO requirements.

The other solutions are more expensive because they require additional AWS services. For example, AWS DMS is a more expensive service than AWS RDS.

upvoted 4 times

Question #589 *Topic 1*

A company runs a web application on Amazon EC2 instances in an Auto Scaling group behind an Application Load Balancer that has sticky sessions enabled. The web server currently hosts the user session state. The company wants to ensure high availability and avoid user session state loss in the event of a web server outage.

Which solution will meet these requirements?

A. Use an Amazon ElastiCache for Memcached instance to store the session data. Update the application to use ElastiCache for Memcached to store the session state.

B. Use Amazon ElastiCache for Redis to store the session state. Update the application to use ElastiCache for Redis to store the session state. Most Voted

C. Use an AWS Storage Gateway cached volume to store session data. Update the application to use AWS Storage Gateway cached volume to store the session state.

D. Use Amazon RDS to store the session state. Update the application to use Amazon RDS to store the session state.

Correct Answer: B

Community vote distribution

B (91%) 9%

Comments

Guru4Cloud Highly Voted 2 years, 9 months ago

Selected Answer: B

The key points are:

ElastiCache Redis provides in-memory caching that can deliver microsecond latency for session data.
Redis supports replication and multi-AZ which can provide high availability for the cache.
The application can be updated to store session data in ElastiCache Redis rather than locally on the web servers.
If a web server fails, the user can be routed via the load balancer to another web server which can retrieve their session data from the highly available ElastiCache Redis cluster.

upvoted 9 times

czyboi Highly Voted 2 years, 9 months ago

Selected Answer: B

high availability => use redis instead of Elastich memcache

upvoted 5 times

emakid Most Recent 1 year, 11 months ago

Selected Answer: B

Option B: Use Amazon ElastiCache for Redis to store the session state. Update the application to use ElastiCache for Redis to store the session state.

Explanation: Amazon ElastiCache for Redis is suitable for session state storage because Redis provides both in-memory data storage and persistence options. Redis supports features like replication, persistence, and high availability (through Redis Sentinel or clusters). This ensures that session state is preserved and available even if individual web servers fail.

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: B

As Memcached is not HA

upvoted 4 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

A

As cache needs to be distributed as ALB is used.

upvoted 1 times

potomac 2 years, 7 months ago

Selected Answer: B

B is correct

upvoted 3 times

franbarberan 2 years, 8 months ago

Selected Answer: D

Elastic cache is Only for RDS

upvoted 3 times

pentium75 2 years, 5 months ago

Since when?

upvoted 4 times

gispankaj 2 years, 9 months ago

Selected Answer: B

redis is correct since it provides high availability and data persistance

upvoted 5 times

Eminenza22 2 years, 9 months ago

Selected Answer: B

B is the correct answer. It suggests using Amazon ElastiCache for Redis to store the session state. Update the application to use ElastiCache for Redis to store the session state. This solution is cost-effective and requires minimal development effort.

upvoted 4 times

Question #590 *Topic 1*

A company migrated a MySQL database from the company's on-premises data center to an Amazon RDS for MySQL DB instance. The company sized the RDS DB instance to meet the company's average daily workload. Once a month, the database performs slowly when the company runs queries for a report. The company wants to have the ability to run reports and maintain the performance of the daily workloads.

Which solution will meet these requirements?

A. Create a read replica of the database. Direct the queries to the read replica. **Most Voted**

B. Create a backup of the database. Restore the backup to another DB instance. Direct the queries to the new database.

C. Export the data to Amazon S3. Use Amazon Athena to query the S3 bucket.

D. Resize the DB instance to accommodate the additional workload.

Correct Answer: A

Community vote distribution

A (100%)

Comments

TariqKipkemei 1 year, 6 months ago

Selected Answer: A

queries for reports = read replica

upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

Create a read replica of the database. Direct the queries to the read replica.

upvoted 4 times

Eminenza22 1 year, 9 months ago

Selected Answer: A

This is the most cost-effective solution because it does not require any additional AWS services. A read replica is a copy of a database that is synchronized with the primary database. You can direct the queries for the report to the read replica, which will not affect the performance of the daily workloads

upvoted 4 times

TiagueteVital 1 year, 9 months ago

Selected Answer: A

Clearly the right choice, with a read replica all the queries needed for a report are done in the replica, leaving the primary on best perfomance for write

upvoted 3 times

Question #591 *Topic 1*

A company runs a container application by using Amazon Elastic Kubernetes Service (Amazon EKS). The application includes microservices that manage customers and place orders. The company needs to route incoming requests to the appropriate microservices.

Which solution will meet this requirement MOST cost-effectively?

A. Use the AWS Load Balancer Controller to provision a Network Load Balancer.

B. Use the AWS Load Balancer Controller to provision an Application Load Balancer. **Most Voted**

C. Use an AWS Lambda function to connect the requests to Amazon EKS.

D. Use Amazon API Gateway to connect the requests to Amazon EKS.

Correct Answer: B

Community vote distribution

B (76%) D (24%)

Comments

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: B

"The company needs to route incoming requests to the appropriate microservices"
In Kubernetes world, this would be called an Ingress Service so it will need B
https://kubernetes-sigs.github.io/aws-load-balancer-controller/v2.6/
https://kubernetes.io/docs/concepts/services-networking/ingress/

upvoted 7 times

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: B

Not D because
- even with an API gateway you'd need an ALB or ELB (so B+D would work, but D alone does not)
- you would use AWS API Gateway Controller (not "Amazon API Gateway") to create the API Gateway

upvoted 6 times

pentium75 2 years, 5 months ago

https://aws.amazon.com/blogs/containers/microservices-development-using-aws-controllers-for-kubernetes-ack-and-amazon-eks-blueprints/

https://aws.amazon.com/blogs/containers/integrate-amazon-api-gateway-with-amazon-eks/

upvoted 5 times

emakid Most Recent 1 year, 11 months ago

Selected Answer: B

Option B: Use the AWS Load Balancer Controller to provision an Application Load Balancer (ALB).

Explanation: The AWS Load Balancer Controller can provision ALBs, which operate at the application layer (Layer 7). ALBs support advanced routing capabilities such as routing based on HTTP paths or hostnames. This makes ALBs well-suited for routing requests to different microservices based on URL paths or domains. This approach integrates well with Kubernetes and is a common pattern for microservices architectures.

upvoted 3 times

Mikado211 2 years, 6 months ago

Selected Answer: B

ALB is considered as expensive than API Gateway, particularly on higher load.

If you do not need any specific functionalities of API Gateway so you must choose ALB because it will be cheaper.

upvoted 4 times

Mikado211 2 years, 6 months ago

ALB is considered as LESS expensive

upvoted 2 times

riyasara 2 years, 6 months ago

Selected Answer: B

API Gateway has a pricing model that includes a cost per API call, and depending on the volume of requests, this could potentially be more expensive than using an Application Load Balancer.

upvoted 4 times

1rob 2 years, 6 months ago

Selected Answer: B

Routing requests to the appr. microserv. can easily be done with ALB and ingress. The ingress handles routing rules to the micro.serv. With answer D you wil still need ALB or NLB as can be seen in the pics of https://aws.amazon.com/blogs/containers/integrate-amazon-api-gateway-with-amazon-eks/ or https://aws.amazon.com/blogs/containers/microservices-development-using-aws-controllers-for-kubernetes-ack-and-amazon-eks-blueprints/ so that is not the most cost-effectively.

upvoted 3 times

ale_brd_111 2 years, 5 months ago

yeah, I was going with D than checked and seems that you are right to deploy API gateway you still a LB

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: D

Both ALB and API gateway can be used to route traffic to the microservices, but the question seeks the most 'cost effective' option.

You are charged for each hour or partial hour that an Application Load Balancer is running, and the number of Load Balancer Capacity Units (LCU) used per hour.

With Amazon API Gateway, you only pay when your APIs are in use.

I say API gateway is the best option for this case.

upvoted 2 times

pentium75 2 years, 5 months ago

But you still need an ALB or ELB

https://aws.amazon.com/blogs/containers/microservices-development-using-aws-controllers-for-kubernetes-ack-and-amazon-eks-blueprints/

https://aws.amazon.com/blogs/containers/integrate-amazon-api-gateway-with-amazon-eks/

upvoted 2 times

t0nx 2 years, 6 months ago

Selected Answer: B

AWS Load Balancer Controller: The AWS Load Balancer Controller is a Kubernetes controller that makes it easy to set up an Application Load Balance (ALB) or Network Load Balancer (NLB) for your Amazon EKS clusters. It simplifies the process of managing load balancers for applications running on EKS.

Application Load Balancer (ALB): ALB is a Layer 7 load balancer that is capable of routing requests based on content, such as URL paths or hostnames. This makes it suitable for routing requests to different microservices based on specific criteria.

Cost-Effectiveness: ALB is typically more cost-effective than an NLB, and it provides additional features at the application layer, which may be useful for routing requests to microservices based on specific conditions.

Option D: Amazon API Gateway is designed for creating, publishing, and managing APIs. While it can integrate with Amazon EKS, it may be more feature-rich and complex than needed for simple routing to microservices within an EKS cluster.

upvoted 4 times

potomac 2 years, 7 months ago

Selected Answer: D

API Gateway provides an entry point to your microservices.

https://aws.amazon.com/blogs/containers/integrate-amazon-api-gateway-with-amazon-eks/

upvoted 3 times

ccmc 2 years, 7 months ago

B is correct, it is a required before exposing through api gateway

upvoted 2 times

thanhnv142 2 years, 7 months ago

B: is correct.
For EKS, use application load balancer to expose microservices
upvoted 4 times

KhasDenis 2 years, 8 months ago

Selected Answer: B

Routing to ms in k8s -> Ingresses -> Ingress Controller -> AWS Load Balancer Controller https://kubernetes-sigs.github.io/aws-load-balancer-controller/v2.6/
upvoted 4 times

RDM10 2 years, 8 months ago

Microservices--> API--> API GW
upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

D. Use Amazon API Gateway to connect the requests to Amazon EKS.
upvoted 3 times

Mll1975 2 years, 9 months ago

Selected Answer: D

API Gateway is a fully managed service that makes it easy for you to create, publish, maintain, monitor, and secure APIs at any scale. API Gateway provides an entry point to your microservices.
upvoted 1 times

Eminenza22 2 years, 9 months ago

Selected Answer: D

https://aws.amazon.com/blogs/containers/microservices-development-using-aws-controllers-for-kubernetes-ack-and-amazon-eks-blueprints/
upvoted 1 times

ralfj 2 years, 9 months ago

Selected Answer: D

https://aws.amazon.com/blogs/containers/integrate-amazon-api-gateway-with-amazon-eks/
upvoted 1 times

Question #592 Topic 1

A company uses AWS and sells access to copyrighted images. The company's global customer base needs to be able to access these images quickly. The company must deny access to users from specific countries. The company wants to minimize costs as much as possible.

Which solution will meet these requirements?

A. Use Amazon S3 to store the images. Turn on multi-factor authentication (MFA) and public bucket access. Provide customers with a link to the S3 bucket.

B. Use Amazon S3 to store the images. Create an IAM user for each customer. Add the users to a group that has permission to access the S3 bucket.

C. Use Amazon EC2 instances that are behind Application Load Balancers (ALBs) to store the images. Deploy the instances only in the countries the company services. Provide customers with links to the ALBs for their specific country's instances.

D. Use Amazon S3 to store the images. Use Amazon CloudFront to distribute the images with geographic restrictions. Provide a signed URL for each customer to access the data in CloudFront. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

ralfj Highly Voted 1 year, 9 months ago

Selected Answer: D

Use Cloudfront and geographic restriction

upvoted 5 times

TariqKipkemei Most Recent 1 year, 6 months ago

Selected Answer: D

Store images = Amazon S3
global customer base needs to be able to access these images quickly = Amazon CloudFront
deny access to users from specific countries = Amazon CloudFront geographic restrictions, signed URLs

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

D. Use Amazon S3 to store the images. Use Amazon CloudFront to distribute the images with geographic restrictions. Provide a signed URL for each customer to access the data in CloudFront.

upvoted 4 times

Colz 1 year, 9 months ago

Correct answer is D

upvoted 1 times

hubbabubba 1 year, 9 months ago

Selected Answer: D

answer is D

upvoted 2 times

Eminenza22 1 year, 9 months ago

Selected Answer: D

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/georestrictions.html

upvoted 3 times

Question #593 Topic 1

A solutions architect is designing a highly available Amazon ElastiCache for Redis based solution. The solutions architect needs to ensure that failures do not result in performance degradation or loss of data locally and within an AWS Region. The solution needs to provide high availability at the node level and at the Region level.

Which solution will meet these requirements?

A. Use Multi-AZ Redis replication groups with shards that contain multiple nodes. Most Voted

B. Use Redis shards that contain multiple nodes with Redis append only files (AOF) turned on.

C. Use a Multi-AZ Redis cluster with more than one read replica in the replication group.

D. Use Redis shards that contain multiple nodes with Auto Scaling turned on.

Correct Answer: A

Community vote distribution

A (79%) 13% 8%

Comments

pentium75 Highly Voted 1 year, 11 months ago

Selected Answer: A

It seems like "Multi-AZ Redis replication group" (A) and "Multi-AZ Redis cluster" (C) are different wordings for the same configuration. However, "to minimize the impact of a node failure, we recommend that your implementation use multiple nodes in each shard" - and that is mentioned only in A

upvoted 6 times

taustin2 Highly Voted 2 years, 2 months ago

Selected Answer: A

https://docs.aws.amazon.com/AmazonElastiCache/latest/red-ug/Replication.html

upvoted 5 times

Dookie00 Most Recent 5 months, 3 weeks ago

Selected Answer: C

è la c

upvoted 1 times

LocNV 1 year, 11 months ago

Selected Answer: A

high availability at the node level = shard and Multi A-Z = region level

upvoted 5 times

Cyberkayu 1 year, 11 months ago

did client ask for improved performance, unfortunately they didn't, so C is good to have but not part of the business requirement.

My answer A.

upvoted 3 times

SHAAHIBHUSHANAWS 2 years ago

A

Multi-AZ is only option. It is regional service so can use backup to replicate but can not use for failover.

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: A

Multi-AZ is only supported on Redis clusters that have more than one node in each shard (node groups).

https://docs.aws.amazon.com/AmazonElastiCache/latest/red-ug/AutoFailover.html#:~:text=node%20in%20each-,shard.,-Topics

upvoted 5 times

t0nx 2 years ago

Selected Answer: C

C. Use a Multi-AZ Redis cluster with more than one read replica in the replication group.
In summary, option C, using a Multi-AZ Redis cluster with more than one read replica, is designed to provide both node-level and AWS Region-level high availability, making it the most suitable choice for the given requirements.

upvoted 2 times

potomac 2 years, 1 month ago

Selected Answer: A

the replication structure is contained within a shard (called node group in the API/CLI) which is contained within a Redis cluster

A shard (in the API and CLI, a node group) is a hierarchical arrangement of nodes, each wrapped in a cluster. Shards support replication. Within a shard, one node functions as the read/write primary node. All the other nodes in a shard function as read-only replicas of the primary node.

upvoted 1 times

thanhnv142 2 years, 1 month ago

C is correct.
Not A because in replication mode, shard have multiple nodes by default.
B and D not correct because that not an option

upvoted 1 times

iwannabeawsgod 2 years, 1 month ago

Selected Answer: C

its c for me

upvoted 1 times

bsbs1234 2 years, 2 months ago

C:
Cluster mode will create multiple shards, when node level failure, request of shard that not impacted will not has any performance impact. If the issu at AZ level, spread traffic between multiple shards shall also reduce the performance degrade.

upvoted 1 times

loveaws 2 years, 2 months ago

c.
Option A is not ideal because it doesn't mention read replicas, and it's generally better to have read replicas for both performance and high availability.
Option B mentions Redis append-only files (AOF), but AOF alone doesn't provide high availability or fault tolerance.
Option D mentions Auto Scaling, but this doesn't directly address high availability at the Region level or data replication

upvoted 1 times

taustin2 2 years, 2 months ago

Multi-AZ is only supported on Redis clusters that have more than one node in each shard.

upvoted 1 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

Multi-AZ replication groups provide automatic failover between AZs if there is an issue with the primary AZ. This provides high availability at the region level

upvoted 3 times

pentium75 1 year, 11 months ago

What about "the node level"?

upvoted 1 times

xyb 2 years, 3 months ago

Selected Answer: C

Enabling ElastiCache Multi-AZ with automatic failover on your Redis cluster (in the API and CLI, replication group) improves your fault tolerance. This is true particularly in cases where your cluster's read/write primary cluster becomes unreachable or fails for any reason. Multi-AZ with automatic failover is only supported on Redis clusters that support replication

upvoted 1 times

Mll1975 2 years, 3 months ago

Selected Answer: A

I would go with A too

I would go with A, Using AOF can't protect you from all failure scenarios.
For example, if a node fails due to a hardware fault in an underlying physical server, ElastiCache will provision a new node on a different server. In thi case, the AOF is not available and can't be used to recover the data.

upvoted 2 times

Question #594 Topic 1

A company plans to migrate to AWS and use Amazon EC2 On-Demand Instances for its application. During the migration testing phase, a technical team observes that the application takes a long time to launch and load memory to become fully productive.

Which solution will reduce the launch time of the application during the next testing phase?

A. Launch two or more EC2 On-Demand Instances. Turn on auto scaling features and make the EC2 On-Demand Instances available during the next testing phase.

B. Launch EC2 Spot Instances to support the application and to scale the application so it is available during the next testing phase.

C. Launch the EC2 On-Demand Instances with hibernation turned on. Configure EC2 Auto Scaling warm pools during the next testing phase. Most Voted

D. Launch EC2 On-Demand Instances with Capacity Reservations. Start additional EC2 instances during the next testing phase.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Guru4Cloud Highly Voted 2 years, 3 months ago

Selected Answer: C

Using EC2 hibernation and Auto Scaling warm pools will help address this:

Hibernation saves the in-memory state of the EC2 instance to persistent storage and shuts the instance down. When the instance is started again, the in-memory state is restored, which launches much faster than launching a new instance.
Warm pools pre-initialize EC2 instances and keep them ready to fulfill requests, reducing launch time. The hibernated instances can be added to a warm pool.
When auto scaling scales out during the next testing phase, it will be able to launch instances from the warm pool rapidly since they are already initialized

upvoted 8 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: C

https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-warm-pools.html

upvoted 3 times

riyasara 2 years ago

Selected Answer: C

Amazon EC2 hibernation and warm pool

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: C

If an instance or application takes a long time to bootstrap and build a memory footprint in order to become fully productive, you can use hibernation to pre-warm the instance. To pre-warm the instance, you:
Launch it with hibernation enabled.
Bring it to a desired state.
Hibernate it so that it's ready to be resumed to the desired state whenever needed.

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Hibernate.html#:~:text=you%20can%20use-,hibernation,-to%20pre%2Dwarm

upvoted 4 times

potomac 2 years, 1 month ago

Selected Answer: C

With Amazon EC2 hibernation enabled, you can maintain your EC2 instances in a "pre-warmed" state so these can get to a productive state faster.
upvoted 3 times

tabbyDolly 2 years, 2 months ago
C: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Hibernate.html
upvoted 3 times

ralfj 2 years, 3 months ago
Selected Answer: C
just use hibernation option so you won't load the full EC2 Instance
upvoted 2 times

Question #595 *Topic 1*

A company's applications run on Amazon EC2 instances in Auto Scaling groups. The company notices that its applications experience sudden traffic increases on random days of the week. The company wants to maintain application performance during sudden traffic increases.

Which solution will meet these requirements MOST cost-effectively?

A. Use manual scaling to change the size of the Auto Scaling group.

B. Use predictive scaling to change the size of the Auto Scaling group.

C. Use dynamic scaling to change the size of the Auto Scaling group. Most Voted

D. Use schedule scaling to change the size of the Auto Scaling group.

Correct Answer: C

Community vote distribution

C (100%)

Comments

ralfj Highly Voted 2 years, 3 months ago

Selected Answer: C

Dynamic Scaling – This is yet another type of Auto Scaling in which the number of EC2 instances is changed automatically depending on the signals received. Dynamic Scaling is a good choice when there is a high volume of unpredictable traffic.

https://www.developer.com/web-services/aws-auto-scaling-types-best-practices/#:~:text=Dynamic%20Scaling%20%E2%80%93%20This%20is%20yet,high%20volume%20of%20unpredictable%20traffic.

upvoted 6 times

tabbyDolly Highly Voted 2 years, 2 months ago

C - " sudden traffic increases on random days of the week" --> dynamic scaling

upvoted 5 times

awsgeek75 Most Recent 1 year, 10 months ago

Selected Answer: C

random = dynamic
A: Manual is never a solution
B: Predictive is not possible as it's random
D: Cannot schedule random

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: C

Dynamic scaling

upvoted 2 times

dilaaziz 2 years, 1 month ago

Selected Answer: C

https://aws.amazon.com/ec2/autoscaling/faqs/

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

C is the best answer here. Dynamic scaling is the most cost-effective way to automatically scale the Auto Scaling group to maintain performance during random traffic spikes.

upvoted 3 times

Question #596 | Topic 1

An ecommerce application uses a PostgreSQL database that runs on an Amazon EC2 instance. During a monthly sales event, database usage increases and causes database connection issues for the application. The traffic is unpredictable for subsequent monthly sales events, which impacts the sales forecast. The company needs to maintain performance when there is an unpredictable increase in traffic.

Which solution resolves this issue in the MOST cost-effective way?

A. Migrate the PostgreSQL database to Amazon Aurora Serverless v2. **Most Voted**

B. Enable auto scaling for the PostgreSQL database on the EC2 instance to accommodate increased usage.

C. Migrate the PostgreSQL database to Amazon RDS for PostgreSQL with a larger instance type.

D. Migrate the PostgreSQL database to Amazon Redshift to accommodate increased usage.

Correct Answer: A

Community vote distribution

A (95%) | 5%

Comments

Guru4Cloud Highly Voted 2 years, 3 months ago

Selected Answer: A

Answer is A.
Aurora Serverless v2 got autoscaling, highly available and cheaper when compared to the other options.

upvoted 8 times

pentium75 Most Recent 1 year, 11 months ago

Selected Answer: A

Not B - we can auto-scale the EC2 instance, but not "the [self-managed] PostgreSQL database ON the EC2 instance"
Not C - This does not mention scaling, so it would incur high cost and still it might not be able to keep up with the "unpredictable" spikes
Not D - Redshift is OLAP Data Warehouse

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: A

Amazon Aurora Serverless is an on-demand, auto-scaling configuration for Aurora where the database automatically starts up, shuts down, and scales capacity up or down based on your application's needs. This is the least costly option for unpredictable traffic.

upvoted 3 times

tabbyDolly 2 years, 2 months ago

A: "he traffic is unpredictable for subsequent monthly sales events" --> serverless

upvoted 3 times

Wayne23Fang 2 years, 3 months ago

Selected Answer: C

A is probably more expensive than C. Aurora is serverless and fast. But nevertheless it needs DB migration service. Not sure DMS may not be free.

upvoted 1 times

danielmakita 2 years, 1 month ago

C is more expensive if you think the scenario where the traffic is low. You are paying for a larger hardware but not using it. That's why I think A is correct.

upvoted 4 times

TiagueteVital 2 years, 3 months ago

Selected Answer: A

A to autoscaling

upvoted 3 times

manOfThePeople 2 years, 3 months ago

Answer is A.

Aurora Serverless v2 got autoscaling, highly available and cheaper when compared to the other options.

upvoted 2 times

anikety123 2 years, 3 months ago

Selected Answer: A

The correct answer is A

upvoted 2 times

Question #597 Topic 1

A company hosts an internal serverless application on AWS by using Amazon API Gateway and AWS Lambda. The company's employees report issues with high latency when they begin using the application each day. The company wants to reduce latency.

Which solution will meet these requirements?

A. Increase the API Gateway throttling limit.

B. Set up a scheduled scaling to increase Lambda provisioned concurrency before employees begin to use the application each day. Most Voted

C. Create an Amazon CloudWatch alarm to initiate a Lambda function as a target for the alarm at the beginning of each day.

D. Increase the Lambda function memory.

Correct Answer: B

Community vote distribution

B (100%)

Comments

oayoade Highly Voted 2 years, 9 months ago

Selected Answer: B

https://aws.amazon.com/blogs/compute/scheduling-aws-lambda-provisioned-concurrency-for-recurring-peak-usage/

upvoted 5 times

Guru4Cloud Highly Voted 2 years, 9 months ago

Selected Answer: B

Set up a scheduled scaling to increase Lambda provisioned concurrency before employees begin to use the application each day.

upvoted 5 times

emakid Most Recent 1 year, 11 months ago

Selected Answer: B

Option B: Set up a scheduled scaling to increase Lambda provisioned concurrency before employees begin to use the application each day.

Explanation: Provisioned concurrency ensures that a specified number of Lambda instances are initialized and ready to handle requests. By scheduling this scaling, you can pre-warm Lambda functions before peak usage times, reducing cold start latency. This solution directly addresses th latency issue caused by cold starts.

upvoted 4 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

Provisioned concurrency pre-initializes execution environments for your functions. These execution environments are prepared to respond immediately to incoming function requests at start of day.

upvoted 3 times

potomac 2 years, 7 months ago

Selected Answer: B

A is wrong
API Gateway throttling limit is for better throughput, not for latency

upvoted 2 times

Mll1975 2 years, 9 months ago

Selected Answer: B

Provisioned Concurrency incurs additional costs, so it is cost-efficient to use it only when necessary. For example, early in the morning when activity starts, or to handle recurring peak usage.

upvoted 4 times

Eminenza22 2 years, 9 months ago

Selected Answer: B

B option setting up a scheduled scaling to increase Lambda provisioned concurrency before employees begin to use the application each day. This solution is cost-effective and requires minimal development effort.

upvoted 2 times

Question #598 Topic 1

A research company uses on-premises devices to generate data for analysis. The company wants to use the AWS Cloud to analyze the data. The devices generate .csv files and support writing the data to an SMB file share. Company analysts must be able to use SQL commands to query the data. The analysts will run queries periodically throughout the day.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose three.)

A. Deploy an AWS Storage Gateway on premises in Amazon S3 File Gateway mode. Most Voted

B. Deploy an AWS Storage Gateway on premises in Amazon FSx File Gateway made.

C. Set up an AWS Glue crawler to create a table based on the data that is in Amazon S3. Most Voted

D. Set up an Amazon EMR cluster with EMR File System (EMRFS) to query the data that is in Amazon S3. Provide access to analysts.

E. Set up an Amazon Redshift cluster to query the data that is in Amazon S3. Provide access to analysts.

F. Setup Amazon Athena to query the data that is in Amazon S3. Provide access to analysts. Most Voted

Correct Answer: ACF

Community vote distribution

ACF (96%) 4%

Comments

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: ACF

SQL Queries is Athena so DE are wrong and we are now dependant on S3
A to get files into S3
C Glue to convert CSV to S3 table data
B irrelevant as we don't have anything to consume data from FSx in other options

upvoted 6 times

JA2018 1 year, 6 months ago

err,,, pardon my ignorance but I did not recall seeing any mention of S3 in the stem

upvoted 1 times

awsgeek75 2 years, 4 months ago

My only reservation with this answer is C.
CSV is technically a table and Athena can query multiple csv from S3. Glue just seems overengineering over here

upvoted 3 times

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: ACF

A to upload the files to S3 via SMB
C to convert the data from CSV format
F to query with SQL

Not B (we need the data in S3, not in FSx)
Not D or E (we should provide the ability to run SQL queries)

upvoted 5 times

TariqKipkemei Most Recent 2 years, 6 months ago

Selected Answer: ACF

SMB + use SQL commands to query the data = Amazon S3 File Gateway mode + Amazon Athena

upvoted 1 times

wsdasdasdqwdaw 2 years, 7 months ago https://aws.amazon.com/storagegateway/file/s3/#:~:text=Amazon%20S3%20File%20Gateway%20provides,Amazon%20S3%20with%20local%20caching.

"Amazon S3 File Gateway provides a seamless way to connect to the cloud in order to store application data files and backup images as durable objects in Amazon S3 cloud storage. Amazon S3 File Gateway offers SMB or NFS-based access to data in Amazon S3 with local caching"

=> SMB and NFS is supported in Amazon S3 File Gateway => ACF

upvoted 2 times

iwannabeawsgod 2 years, 7 months ago

Selected Answer: ACF

ACF 100% sure

upvoted 3 times

Ramdi1 2 years, 8 months ago

Selected Answer: ACF

I thought the correct answer was BCF however I have changed my mind to BCF
FSx does support SMB protocol. However so does s3 file gateway which is version 2 and 3 of the SMB protocol. Hence using it with athena ACF should be correct

upvoted 4 times

RDM10 2 years, 8 months ago

SMB file share- is B incorrect?

upvoted 1 times

pentium75 2 years, 5 months ago

Yes because "FSx File Gateway" uploads the files to FSx, but we need it in S3.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: BCE

BCF is the correct

upvoted 1 times

pentium75 2 years, 5 months ago

No because FSx File Gateway (B) uploads it to FSx while we we need it in S3. S3 File Gateway provides access to S3 via SMB.

upvoted 3 times

Eminenza22 2 years, 9 months ago

Selected Answer: ACF

https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-etl-format-csv-home.html
https://aws.amazon.com/blogs/aws/amazon-athena-interactive-sql-queries-for-data-in-amazon-s3/
https://aws.amazon.com/storagegateway/faqs/

upvoted 3 times

anikety123 2 years, 9 months ago

Selected Answer: ACF

It should be ACF

upvoted 2 times

ralfj 2 years, 9 months ago

Selected Answer: ACF

ACF use S3 File Gateway, Use Glue and Use Athena

upvoted 3 times

Question #599 Topic 1

A company wants to use Amazon Elastic Container Service (Amazon ECS) clusters and Amazon RDS DB instances to build and run a payment processing application. The company will run the application in its on-premises data center for compliance purposes.

A solutions architect wants to use AWS Outposts as part of the solution. The solutions architect is working with the company's operational team to build the application.

Which activities are the responsibility of the company's operational team? (Choose three.)

A. Providing resilient power and network connectivity to the Outposts racks **Most Voted**

B. Managing the virtualization hypervisor, storage systems, and the AWS services that run on Outposts

C. Physical security and access controls of the data center environment **Most Voted**

D. Availability of the Outposts infrastructure including the power supplies, servers, and networking equipment within the Outposts racks

E. Physical maintenance of Outposts components

F. Providing extra capacity for Amazon ECS clusters to mitigate server failures and maintenance events **Most Voted**

Correct Answer: ACF

Community vote distribution

ACF (55%) ACE (26%) Other

Comments

taustin2 Highly Voted 2 years, 8 months ago

Selected Answer: ACF

From https://docs.aws.amazon.com/whitepapers/latest/aws-outposts-high-availability-design/aws-outposts-high-availability-design.html

With Outposts, you are responsible for providing resilient power and network connectivity to the Outpost racks to meet your availability requirements for workloads running on Outposts. You are responsible for the physical security and access controls of the data center environment. You must provide sufficient power, space, and cooling to keep the Outpost operational and network connections to connect the Outpost back to the Region. Since Outpost capacity is finite and determined by the size and number of racks AWS installs at your site, you must decide how much EC2, EBS, and S3 on Outposts capacity you need to run your initial workloads, accommodate future growth, and to provide extra capacity to mitigate server failures and maintenance events.

upvoted 28 times

ibu007 Highly Voted 2 years, 9 months ago

Selected Answer: ACE

My exam is tomorrow. thank you all for the answers and links.

upvoted 18 times

pentium75 2 years, 5 months ago

"Physical maintenance" such as replacing faulty disks is NOT your responsibility.

upvoted 7 times

FlyingHawk Most Recent 1 year, 4 months ago

Selected Answer: ACF

this is a tricky question. I selected ACE initially, but after reading the AWS doc:
https://docs.aws.amazon.com/whitepapers/latest/aws-outposts-high-availability-design/aws-outposts-high-availability-design.html
Based on it , capacity is customer's responsibility. E is AWS responsibility

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Since Outpost capacity is finite and determined by the size and number of racks AWS installs at your site, you must decide how much EC2, EBS, and S3 on Outposts capacity you need to run your initial workloads, accommodate future growth, and to provide extra capacity to mitigate server failures and maintenance events.(F)
AWS is responsible for the availability of the Outposts infrastructure including the power supplies, servers, and networking equipment within the AWS Outposts racks. (D, E) AWS also manages the virtualization hypervisor, storage systems, and the AWS services that run on Outposts.(B)
upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: ACF

F: "If there is no additional capacity on the Outpost, the instance remains in the stopped state. The Outpost owner can try to free up used capacity or request additional capacity for the Outpost so that the migration can complete."

Not D: "Equipment within the Outposts rack" is AWS' responsibility, you're not supposed to touch that

Not E: "When the AWS installation team arrives on site, they will replace the unhealthy hosts, switches, or rack elements"
upvoted 6 times

1rob 2 years, 6 months ago

Selected Answer: ACF

From <https://aws.amazon.com/outposts/rack/faqs/ : Your site must support the basic power, networking and space requirements to host an Outpost ===> A
From <https://docs.aws.amazon.com/whitepapers/latest/applying-security-practices-to-network-workload-for-csps/the-shared-responsibility-model.html : In AWS Outposts, the customer takes the responsibility of securing the physical infrastructure to host the AWS Outposts equipment in their own data centers. ===> C
upvoted 2 times

1rob 2 years, 6 months ago

and From <https://docs.aws.amazon.com/whitepapers/latest/aws-outposts-high-availability-design/aws-outposts-high-availability-design.html :
Since Outpost capacity is finite and determined by the size and number of racks AWS installs at your site, you must decide how much EC2, EBS, and S3 on Outposts capacity you need to run your initial workloads, accommodate future growth, and to provide extra capacity to mitigate server failures and maintenance events. ===> F
upvoted 3 times

1rob 2 years, 6 months ago

From <https://docs.aws.amazon.com/whitepapers/latest/aws-outposts-high-availability-design/aws-outposts-high-availability-design.html>
AWS is responsible for the availability of the Outposts infrastructure including the power supplies, servers, and networking equipment within the AWS Outposts racks. AWS also manages the virtualization hypervisor, storage systems, and the AWS services that run on Outposts. So The customer isn't so not D.
upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: AC

Only A and C are correct.
AWS is responsible for the hardware and software that run on AWS Outposts. This is a fully managed infrastructure service. AWS manages security patches, updates firmware, and maintains the Outpost equipment. AWS also monitors the performance, health, and metrics for your Outpost and determines whether any maintenance is required.

https://docs.aws.amazon.com/outposts/latest/userguide/outpost-maintenance.html
upvoted 1 times

pentium75 2 years, 5 months ago

"Choose three".

You're missing F, you must order the Outposts rack with excess capacity
upvoted 2 times

[Removed] 2 years, 6 months ago

Selected Answer: A다E

The role that physical companies will play is ACE.
upvoted 1 times

potomac 2 years, 7 months ago

Selected Answer: ACD

E is wrong
If there is a need to perform physical maintenance, AWS will reach out to schedule a time to visit your site.
https://aws.amazon.com/outposts/rack/faqs/#:~:text=As%20AWS%20Outposts%20rack%20runs,the%20Outpost%20for%20compliance%20certification.
upvoted 1 times

pentium75 2 years, 5 months ago

So is D, "equipment WITHIN the Outposts rack" is something that your Infra team should stay away from.

upvoted 2 times

beast2091 2 years, 7 months ago

ACE

AWS is responsible for the availability of the Outposts infrastructure including the power supplies, servers, and networking equipment within the AWS Outposts racks. AWS also manages the virtualization hypervisor, storage systems, and the AWS services that run on Outposts.

https://d1.awsstatic.com/whitepapers/aws-outposts-high-availability-design-and-architecture-considerations.pdf

upvoted 1 times

dilaaziz 2 years, 7 months ago

Selected Answer: ACF

https://docs.aws.amazon.com/whitepapers/latest/aws-outposts-high-availability-design/aws-outposts-high-availability-design.html

upvoted 2 times

canonlycontainletters1 2 years, 7 months ago

Selected Answer: ACD

I choose ACD

upvoted 1 times

danielmakita 2 years, 7 months ago

Selected Answer: ACD

I think ACD is correct

upvoted 1 times

chris0975 2 years, 7 months ago

Selected Answer: ACF

You get to choose the capacity. F

upvoted 2 times

thanhnv142 2 years, 7 months ago

A, C and D

upvoted 1 times

aleksand41 2 years, 8 months ago

ACD https://docs.aws.amazon.com/outposts/latest/userguide/outpost-maintenance.html

upvoted 1 times

Ramdi1 2 years, 8 months ago

Selected Answer: ACD

I think because of the shared responsibility model it is ACD

upvoted 3 times

taustin2 2 years, 8 months ago

Selected Answer: ACF

A and C are obviously right. D is wrong because "within the Outpost racks". Between E and F, E is wrong because (https://aws.amazon.com/outposts/rack/faqs/) says "If there is a need to perform physical maintenance, AWS will reach out to schedule a time to visi your site. AWS may replace a given module as appropriate but will not perform any host or network switch servicing on customer premises." So, choosing F.

upvoted 2 times

Question #600 *Topic 1*

A company is planning to migrate a TCP-based application into the company's VPC. The application is publicly accessible on a nonstandard TCP port through a hardware appliance in the company's data center. This public endpoint can process up to 3 million requests per second with low latency. The company requires the same level of performance for the new public endpoint in AWS.

What should a solutions architect recommend to meet this requirement?

A. Deploy a Network Load Balancer (NLB). Configure the NLB to be publicly accessible over the TCP port that the application requires. Most Voted

B. Deploy an Application Load Balancer (ALB). Configure the ALB to be publicly accessible over the TCP port that the application requires.

C. Deploy an Amazon CloudFront distribution that listens on the TCP port that the application requires. Use an Application Load Balancer as the origin.

D. Deploy an Amazon API Gateway API that is configured with the TCP port that the application requires. Configure AWS Lambda functions with provisioned concurrency to process the requests.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Sugarbear_01 Highly Voted 2 years, 2 months ago

Selected Answer: A

Since the company requires the same level of performance for the new public endpoint in AWS.

A Network Load Balancer functions at the fourth layer of the Open Systems Interconnection (OSI) model. It can handle millions of requests per second. After the load balancer receives a connection request, it selects a target from the target group for the default rule. It attempts to open a TCP connection to the selected target on the port specified in the listener configuration.

Link;
https://docs.aws.amazon.com/elasticloadbalancing/latest/network/introduction.html

upvoted 11 times

TariqKipkemei Highly Voted 2 years ago

Selected Answer: A

TCP = NLB

upvoted 7 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: A

B: Is wrong as ALB is not going to help with TCP traffic
C: CloudFront is CDN. There is no content here
D: API Gateway is for HTTP web/API stuff, not custom TCP port applicationns

upvoted 3 times

taustin2 2 years, 2 months ago

Selected Answer: A

NLBs handle millions of requests per second. NLBs can handle general TCP traffic.

upvoted 4 times

Question #601 *Topic 1*

A company runs its critical database on an Amazon RDS for PostgreSQL DB instance. The company wants to migrate to Amazon Aurora PostgreSQL with minimal downtime and data loss.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create a DB snapshot of the RDS for PostgreSQL DB instance to populate a new Aurora PostgreSQL DB cluster.

B. Create an Aurora read replica of the RDS for PostgreSQL DB instance. Promote the Aurora read replicate to a new Aurora PostgreSQL DB cluster. **Most Voted**

C. Use data import from Amazon S3 to migrate the database to an Aurora PostgreSQL DB cluster.

D. Use the pg_dump utility to back up the RDS for PostgreSQL database. Restore the backup to a new Aurora PostgreSQL DB cluster.

Correct Answer: B

Community vote distribution

B (88%) 12%

Comments

Guru4Cloud Highly Voted 2 years, 8 months ago

Selected Answer: B

The key reasons are:

Aurora read replicas allow setting up replication from RDS PostgreSQL to Aurora PostgreSQL with minimal downtime.
Once replication is set up, the read replica can be promoted to a full standalone Aurora DB cluster with little to no downtime.
This approach leverages AWS's managed replication between the source RDS PostgreSQL instance and Aurora. It avoids having to manually create backups and restore data.
Using DB snapshots or pg_dump backups requires manually restoring data which increases downtime and operational overhead.
Data import from S3 would require exporting, uploading and then importing data which adds overhead.

upvoted 7 times

JA2018 1 year, 6 months ago

can share the link? thanks

upvoted 1 times

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: B

Not A: Would work but have some (though minor) downtime
B: "The Aurora read replica option minimizes downtime during a migration"
Not C: "If your data is stored using Amazon Simple Storage Service (Amazon S3)" ... in this case it is not
Not D: "If ... you don't have downtime considerations, you can use this option"
https://repost.aws/knowledge-center/aurora-postgresql-migrate-from-rds

upvoted 5 times

Firdous586 Most Recent 2 years, 4 months ago

B is correct as the question says least down time and data loss

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

"Use an RDS for PostgreSQL DB instance as the basis for a new Aurora PostgreSQL DB cluster by using an Aurora read replica. The Aurora read replica is available for migrating only within the same AWS Region and account. The Aurora read replica option minimizes downtime during a migration. You can promote the new cluster when you have zero (0) replication lag between the primary RDS instance and the Aurora read replica."
https://repost.aws/knowledge-center/aurora-postgresql-migrate-from-rds

upvoted 3 times

Cyberkayu 2 years, 5 months ago

Selected Answer: B

ACD will have delta changes issue. Which means, RDS snapshot/export at 2pm, upload/import the table into Aurora, configure and populated completed by 6pm. This created a 4-hour gap of delta changes

upvoted 2 times

aws94 2 years, 6 months ago

Selected Answer: A

please focus, we have RDS not Aurora, I don't know how you vote to create an Aurora read replica to migrate an RDS to Aurora.

upvoted 1 times

pentium75 2 years, 5 months ago

I thought that too but B is correct: https://repost.aws/knowledge-center/aurora-postgresql-migrate-from-rds

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

LEAST operational overhead = read replica

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: B

A,B,C are all valid option.
But B: The Aurora read replica option minimizes downtime during a migration.

upvoted 2 times

thanhnv142 2 years, 7 months ago

B is correct guys. Lets see what we got here:
C and D is not correct of course. We have to consider A and B.
A: migration using a snapshot: this would, of course, introduce heavy data loss and down time
B: migration using read replica: nearly no dataloss and downtime.

upvoted 4 times

RRya 2 years, 7 months ago

Selected Answer: A

RDS PostgreSQL to Aurora PostgreSQL:
• Option 1: DB Snapshots from RDS PostgreSQL restored as PostgreSQL Aurora DB
• Option 2: Create an Aurora Read Replica from your RDS PostgreSQL, and when the replication lag is 0, promote it as its own DB cluster (can take time and cost $)

upvoted 1 times

pentium75 2 years, 5 months ago

"The Aurora read replica option minimizes downtime during a migration"
https://repost.aws/knowledge-center/aurora-postgresql-migrate-from-rds

upvoted 2 times

Jay2k23 2 years, 8 months ago

Selected Answer: A

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraPostgreSQL.Migrating.html

upvoted 1 times

Sugarbear_01 2 years, 8 months ago

Answer [B]

There are five options for migrating data from your existing Amazon RDS for PostgreSQL database to an Amazon Aurora PostgreSQL-Compatible DB cluster.
1-Using a snapshot
2-Using an Aurora read replica
3-Using a pg_dump utility
4-Using logical replication
5-Using a data import from Amazon S3

(2-Using an Aurora read replica)
The Aurora read replica option minimizes downtime during a migration. Which is what the question demand so answer B; is the correct ;
https://repost.aws/knowledge-center/aurora-postgresql-migrate-from-rds

upvoted 5 times

Sugarbear_01 2 years, 8 months ago

Using (4 - using logical replication) RDS for PostgreSQL and Aurora PostgreSQL instance to migrate data off minimal downtime. But is not part of the option in the answer. Which makes answer B the best solution.

upvoted 2 times

taustin2 2 years, 8 months ago

Selected Answer: B

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraPostgreSQL.Migrating.html

upvoted 2 times

Question #602 Topic 1

A company's infrastructure consists of hundreds of Amazon EC2 instances that use Amazon Elastic Block Store (Amazon EBS) storage. A solutions architect must ensure that every EC2 instance can be recovered after a disaster.

What should the solutions architect do to meet this requirement with the LEAST amount of effort?

A. Take a snapshot of the EBS storage that is attached to each EC2 instance. Create an AWS CloudFormation template to launch new EC2 instances from the EBS storage.

B. Take a snapshot of the EBS storage that is attached to each EC2 instance. Use AWS Elastic Beanstalk to set the environment based on the EC2 template and attach the EBS storage.

C. Use AWS Backup to set up a backup plan for the entire group of EC2 instances. Use the AWS Backup API or the AWS CLI to speed up the restore process for multiple EC2 instances. Most Voted

D. Create an AWS Lambda function to take a snapshot of the EBS storage that is attached to each EC2 instance and copy the Amazon Machine Images (AMIs). Create another Lambda function to perform the restores with the copied AMIs and attach the EBS storage.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Guru4Cloud Highly Voted 1 year, 8 months ago

Selected Answer: C

The key reasons are:

AWS Backup automates backup of resources like EBS volumes. It allows defining backup policies for groups of resources. This removes the need to manually create backups for each resource.
The AWS Backup API and CLI allow programmatic control of backup plans and restores. This enables restoring hundreds of EC2 instances programmatically after a disaster instead of manually.
AWS Backup handles cleanup of old backups based on policies to minimize storage costs.

upvoted 14 times

LeonSauveterre Most Recent 1 year, 6 months ago

Selected Answer: C

A - Manually managing templates for hundreds of EC2 instances is TOO MUCH WORK.
B - Beanstalk is designed for managing application environments, not backups.
C - AWS Backup API or CLI simplifies bulk restore operations. YES.
D - Never create a lambda function when there's already to tool to do that and you need minimal effort. This is true for almost all the questions in this list.

upvoted 1 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: C

LEAST amount of effort = AWS Backup

upvoted 3 times

Chiquitabandita 1 year, 6 months ago

for the question, I would choose C as well, AWS Backup of the EC2, but design, why would anything of importance be on the Ec2 that would need to be restored? Shouldn't any critical or important data be on the EBS volumes in this example or similar location?

upvoted 3 times

LeonSauveterre 1 year, 6 months ago

There are actually scenarios where an EC2 instance itself might still need restoration, even when EBS is being used for storage.

1. Some workloads may have specific configurations or dependencies baked into the EC2 instance itself, like custom softwares and OS-level configs
2. Some applications may be running on EC2 relies on temporary but critical data stored on the instance's root volume.
3. Some stateful applications might store temporary state information locally on the EC2 instance. If this state isn't externalized (like DynamoDB, Redis, or EFS), then recovery is needed.

upvoted 1 times

taustin2 1 year, 8 months ago

Selected Answer: C

Going with Backup. Can restore programmatically using Backup API.

upvoted 3 times

Question #603 *Topic 1*

A company recently migrated to the AWS Cloud. The company wants a serverless solution for large-scale parallel on-demand processing of a semistructured dataset. The data consists of logs, media files, sales transactions, and IoT sensor data that is stored in Amazon S3. The company wants the solution to process thousands of items in the dataset in parallel.

Which solution will meet these requirements with the MOST operational efficiency?

A. Use the AWS Step Functions Map state in Inline mode to process the data in parallel.

B. Use the AWS Step Functions Map state in Distributed mode to process the data in parallel. **Most Voted**

C. Use AWS Glue to process the data in parallel.

D. Use several AWS Lambda functions to process the data in parallel.

Correct Answer: B

Community vote distribution

B (96%) 4%

Comments

Guru4Cloud Highly Voted 2 years, 8 months ago

Selected Answer: B

AWS Step Functions allows you to orchestrate and scale distributed processing using the Map state. The Map state can process items in a large dataset in parallel by distributing the work across multiple resources.
Using the Map state in Distributed mode will automatically handle the parallel processing and scaling. Step Functions will add more workers to process the data as needed.
Step Functions is serverless so there are no servers to manage. It will scale up and down automatically based on demand.

upvoted 9 times

FlyingHawk Most Recent 1 year, 4 months ago

Selected Answer: B

https://docs.aws.amazon.com/step-functions/latest/dg/sample-dist-map-s3data-process.html
With Step Functions, you can orchestrate large-scale parallel workloads to perform tasks, such as on-demand processing of semi-structured data. These parallel workloads let you concurrently process large-scale data sources stored in Amazon S3. For example, you might process a single JSON or CSV file that contains large amounts of data. Or you might process a large set of Amazon S3 objects.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

A - AWS Step Function Map State in inline mode accepts only a JSON array as input and supports up to 40 concurrent iterations.

upvoted 1 times

Sandy1254 1 year, 11 months ago

Selected Answer: B

https://docs.aws.amazon.com/step-functions/latest/dg/use-dist-map-orchestrate-large-scale-parallel-workloads.html

upvoted 2 times

bogdannb 1 year, 11 months ago

Selected Answer: C

Using step functions will be overwill from my point of view. I would use Glue, it's serverless and purposely designed for such use case

upvoted 1 times

Lin878 1 year, 11 months ago

Selected Answer: B

Simple - user Lambda / Complex - user Step Functions

upvoted 3 times

Lx016 2 years, 4 months ago

A Map in Inline mode can support concurrency of 40 parallel branches and execution history limits of 25,000 events or approximately 6,500 state transitions in a workflow. With the Distributed mode, you can run at concurrency of up to 10,000 parallel branches. So I believe if it has to process thousands of items in parallel Distributed Mode is more appropriate

upvoted 4 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

https://aws.amazon.com/blogs/aws/step-functions-distributed-map-a-serverless-solution-for-large-scale-parallel-data-processing/
https://docs.aws.amazon.com/step-functions/latest/dg/sample-dist-map-s3data-process.html

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

The Distributed Map has been optimized for Amazon S3.,helping you more easily iterate over objects in an S3 bucket. With the Distributed mode, you can run at concurrency of up to 10,000 parallel branches.

https://aws.amazon.com/step-functions/faqs/#:~:text=A%20Map%20in%20Inline%20mode,up%20to%2010%2C000%20parallel%20branches.

upvoted 3 times

Sugarbear_01 2 years, 8 months ago

Selected Answer: B

https://docs.aws.amazon.com/step-functions/latest/dg/concepts-orchestrate-large-scale-parallel-workloads.html

upvoted 2 times

taustin2 2 years, 8 months ago

Selected Answer: B

With Step Functions, you can orchestrate large-scale parallel workloads to perform tasks, such as on-demand processing of semi-structured data. These parallel workloads let you concurrently process large-scale data sources stored in Amazon S3. https://docs.aws.amazon.com/step-functions/latest/dg/concepts-orchestrate-large-scale-parallel-workloads.html

upvoted 3 times

Sugarbear_01 2 years, 8 months ago

After going through the link I confirmed the answer is B

upvoted 2 times

Question #604 *Topic 1*

A company will migrate 10 PB of data to Amazon S3 in 6 weeks. The current data center has a 500 Mbps uplink to the internet. Other on-premises applications share the uplink. The company can use 80% of the internet bandwidth for this one-time migration task.

Which solution will meet these requirements?

A. Configure AWS DataSync to migrate the data to Amazon S3 and to automatically verify the data.

B. Use rsync to transfer the data directly to Amazon S3.

C. Use the AWS CLI and multiple copy processes to send the data directly to Amazon S3.

D. Order multiple AWS Snowball devices. Copy the data to the devices. Send the devices to AWS to copy the data to Amazon S3. **Most Voted**

Correct Answer: D

Community vote distribution

D (93%) 7%

Comments

Cyberkayu Highly Voted 2 years, 5 months ago

7 Years, 5 Months, 3 Weeks, 5 Days required to transfer 10PB on 400 Mbps. Finger cross the upload don't drop or timeout on year 7.

upvoted 14 times

zits88 1 year, 9 months ago

As a data engineer, this comment made both laugh and shudder at the same time -- hits too close to home

upvoted 2 times

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: D

PB = snowball

upvoted 7 times

Ravan Most Recent 2 years, 3 months ago

Selected Answer: D

To calculate the total time required in weeks, we can use the result we obtained earlier, which was approximately
6.26
×
1
0
10
6.26×10
10
weeks.

So, the total time required to transfer 10 PB of data to Amazon S3, given a 500 Mbps uplink, would be approximately
6.26
×
1
0
10
6.26×10
10
weeks. However, this is an extremely large value and not practically feasible.

It's important to note that the result obtained might not accurately reflect real-world scenarios due to various factors such as network limitations, bandwidth constraints, and other practical considerations. Additionally, this calculation assumes a constant transfer rate and does not consider potential optimizations or parallelization techniques that could be employed to expedite the data transfer process.

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

10PB on 80% of 500Mbps (Megabits not Megabytes) will take 6.5 years. But for the sake of exam when you cannot use calculators etc, just use snowball for petabytes of transfer if it is an option!

upvoted 2 times

awsgeek75 2 years, 5 months ago

Answer is D! not A! Fiddly fingers!

upvoted 7 times

wsdasdasdqwdaw 2 years, 7 months ago

D, but even if you do not know, all 3 option (A,B and C) have the same nature (transfer via bandwidth) and we know that there is only one correct answer => D.

upvoted 4 times

iwannabeawsgod 2 years, 7 months ago

Selected Answer: D

snowball for sure

upvoted 3 times

joshik 2 years, 8 months ago

Selected Answer: D

1Gbps will roughly do 7 TB in 24 hours. This means 400Mbps will only do 3x42TB.

upvoted 3 times

Xin123 2 years, 8 months ago

D

1Gbps will roughly do 7 TB in 24 hours. This means 400Mbps will only do 3x42TB.

upvoted 3 times

Sugarbear_01 2 years, 8 months ago

Selected Answer: D

D

1Gbps will roughly do 7 TB in 24 hours. This means 400Mbps will only do 3x42TB.

upvoted 3 times

Devsin2000 2 years, 8 months ago

D

1Gbps will roughly do 7 TB in 24 hours. This means 400Mbps will only do 3x42TB.

upvoted 2 times

Guru4Cloud 2 years, 8 months ago

Selected Answer: D

D. Order multiple AWS Snowball devices. Copy the data to the devices. Send the devices to AWS to copy the data to Amazon S3.

upvoted 2 times

taustin2 2 years, 8 months ago

Selected Answer: D

10 PB = It's Snowballs.

upvoted 5 times

kambarami 2 years, 8 months ago

Answer is DDDDD

upvoted 3 times

Question #605 Topic 1

A company has several on-premises Internet Small Computer Systems Interface (ISCSI) network storage servers. The company wants to reduce the number of these servers by moving to the AWS Cloud. A solutions architect must provide low-latency access to frequently used data and reduce the dependency on on-premises servers with a minimal number of infrastructure changes.

Which solution will meet these requirements?

A. Deploy an Amazon S3 File Gateway.

B. Deploy Amazon Elastic Block Store (Amazon EBS) storage with backups to Amazon S3.

C. Deploy an AWS Storage Gateway volume gateway that is configured with stored volumes.

D. Deploy an AWS Storage Gateway volume gateway that is configured with cached volumes. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

Guru4Cloud Highly Voted 2 years, 2 months ago

Selected Answer: D

The key reasons are:

The Storage Gateway volume gateway provides iSCSI block storage using cached volumes. This allows replacing the on-premises iSCSI servers with minimal changes.
Cached volumes store frequently accessed data locally for low latency access, while storing less frequently accessed data in S3.
This reduces the number of on-premises servers while still providing low latency access to hot data.
EBS does not provide iSCSI support to replace the existing servers.
S3 File Gateway is for file storage, not block storage.
Stored volumes would store all data on-premises, not in S3.

upvoted 10 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: D

Low latency = always look for cache or local storage.
A: Doesn't address low latency
B: Don't think this is possible
CD are both low latency but D is better:
https://aws.amazon.com/storagegateway/faqs/#:~:text=In%20the%20cached%20mode%2C%20your,asynchronously%20backed%20up%20to%20AWS.

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: D

low-latency access to frequently used data = cached volumes

upvoted 4 times

Sugarbear_01 2 years, 2 months ago

Answer D

Here is the link ;
https://docs.aws.amazon.com/storagegateway/latest/vgw/WhatIsStorageGateway.html

upvoted 2 times

taustin2 2 years, 2 months ago

Selected Answer: D

ISCI=Volume Gateway.
low-latency access to frequently used data = cached volumes

upvoted 4 times

nnecode 2 years, 2 months ago

Selected Answer: D

An AWS Storage Gateway volume gateway is a hybrid storage solution that connects your on-premises applications to your cloud storage. It provide low-latency access to frequently used data while storing your entire dataset in the cloud.

When you configure an AWS Storage Gateway volume gateway with cached volumes, the gateway stores a copy of frequently accessed data locally. This allows you to provide low-latency access to your frequently accessed data while reducing your dependency on on-premises servers.

upvoted 3 times

Question #606 Topic 1

A solutions architect is designing an application that will allow business users to upload objects to Amazon S3. The solution needs to maximize object durability. Objects also must be readily available at any time and for any length of time. Users will access objects frequently within the first 30 days after the objects are uploaded, but users are much less likely to access objects that are older than 30 days.

Which solution meets these requirements MOST cost-effectively?

A. Store all the objects in S3 Standard with an S3 Lifecycle rule to transition the objects to S3 Glacier after 30 days.

B. Store all the objects in S3 Standard with an S3 Lifecycle rule to transition the objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days. Most Voted

C. Store all the objects in S3 Standard with an S3 Lifecycle rule to transition the objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.

D. Store all the objects in S3 Intelligent-Tiering with an S3 Lifecycle rule to transition the objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days.

Correct Answer: B

Community vote distribution

B (74%) C (26%)

Comments

TheLaPlanta Highly Voted 2 years, 2 months ago

Selected Answer: C

I believe it's C. The following link mentions One Zone-IA offers 99.999999999% durability. Questions says nothing about HA

upvoted 5 times

Sugarbear_01 Highly Voted 2 years, 8 months ago

Selected Answer: B

Minimum Days for Transition to S3 Standard-IA or S3 One Zone-IA

Before you transition objects to S3 Standard-IA or S3 One Zone-IA, you must store them for at least 30 days in Amazon S3. For example, you cannot create a Lifecycle rule to transition objects to the S3 Standard-IA storage class one day after you create them. Amazon S3 doesn't support this transition within the first 30 days because newer objects are often accessed more frequently or deleted sooner than is suitable for S3 Standard-IA or S3 One Zone-IA storage.

Similarly, if you are transitioning noncurrent objects (in versioned buckets), you can transition only objects that are at least 30 days noncurrent to S3 Standard-IA or S3 One Zone-IA storage.

https://docs.aws.amazon.com/AmazonS3/latest/userguide/lifecycle-transition-general-considerations.html

upvoted 5 times

abriggy Most Recent 1 year, 10 months ago

Answer is B.

C is wrong because one-zone doesn't maximize durability, it compromises it.

upvoted 3 times

MatAlves 1 year, 8 months ago

No, both Standard-IA and OZ-IA have SAME durability. What differs it the availability.

Durability is calculated based on how many times the data is copied and made redundant, inherently by the service itself. But that redundancy can be across one building (AZ) or multiple availability zones.

upvoted 2 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

B
Intelligent tiering will automatically transition to S3 One Zone-IA which is not needed for durability.

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

'Objects also must be readily available at any time and for any length of time'...definitely option B.

upvoted 3 times

potomac 2 years, 7 months ago

Selected Answer: B

B is correct

upvoted 2 times

thanhnv142 2 years, 7 months ago

B is correct
C is not correct because data must be durable. C is only for data that can be regenerated.

upvoted 4 times

Xin123 2 years, 8 months ago

Selected Answer: B

Durability. Available any time for any duration => B

upvoted 2 times

Devsin2000 2 years, 8 months ago

A
S3 Glacier is most cost effective

upvoted 4 times

awsgeek75 2 years, 5 months ago

Between A & B, this is the tie-breaker:
"Objects also must be readily available at any time and for any length of time"

While Glacier IS more cost effective but it won't make the objects readily available at any time for any duration.... this is only possible with IA.

upvoted 2 times

taustin2 2 years, 8 months ago

Selected Answer: B

B meets the requirements. No need for intelligent Tiering because of 30 days.

upvoted 2 times

Question #607

Topic 1

A company has migrated a two-tier application from its on-premises data center to the AWS Cloud. The data tier is a Multi-AZ deployment of Amazon RDS for Oracle with 12 TB of General Purpose SSD Amazon Elastic Block Store (Amazon EBS) storage. The application is designed to process and store documents in the database as binary large objects (blobs) with an average document size of 6 MB.

The database size has grown over time, reducing the performance and increasing the cost of storage. The company must improve the database performance and needs a solution that is highly available and resilient.

Which solution will meet these requirements MOST cost-effectively?

A. Reduce the RDS DB instance size. Increase the storage capacity to 24 TiB. Change the storage type to Magnetic.

B. Increase the RDS DB instance size. Increase the storage capacity to 24 TiChange the storage type to Provisioned IOPS.

C. Create an Amazon S3 bucket. Update the application to store documents in the S3 bucket. Store the object metadata in the existing database. Most Voted

D. Create an Amazon DynamoDB table. Update the application to use DynamoDB. Use AWS Database Migration Service (AWS DMS) to migrate data from the Oracle database to DynamoDB.

Correct Answer: C

Community vote distribution

C (100%)

Comments

awsgeek75 Highly Voted 1 year, 11 months ago

Selected Answer: C

When using BLOB, always try to pick a solution with S3.

upvoted 6 times

ferdzcruz Most Recent 1 year, 10 months ago

process and store documents as objects. S3 is known for object storage.

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: C

MOST cost-effectively = store the objects in S3, and object metadata in the existing DB.

upvoted 4 times

taustin2 2 years, 2 months ago

DynamoDB's limit on the size of each record is 400KB, so D is wrong.

upvoted 3 times

Guru4Cloud 2 years, 2 months ago

Selected Answer: C

C. Create an Amazon S3 bucket. Update the application to store documents in the S3 bucket. Store the object metadata in the existing database.

upvoted 4 times

taustin2 2 years, 2 months ago

Selected Answer: C

Storing the blobs in the db is more expensive than s3 with references in the db.

upvoted 4 times

Question #608 *Topic 1*

A company has an application that serves clients that are deployed in more than 20.000 retail storefront locations around the world. The application consists of backend web services that are exposed over HTTPS on port 443. The application is hosted on Amazon EC2 instances behind an Application Load Balancer (ALB). The retail locations communicate with the web application over the public internet. The company allows each retail location to register the IP address that the retail location has been allocated by its local ISP.

The company's security team recommends to increase the security of the application endpoint by restricting access to only the IP addresses registered by the retail locations.

What should a solutions architect do to meet these requirements?

A. Associate an AWS WAF web ACL with the ALB. Use IP rule sets on the ALB to filter traffic. Update the IP addresses in the rule to include the registered IP addresses. **Most Voted**

B. Deploy AWS Firewall Manager to manage the ALConfigure firewall rules to restrict traffic to the ALModify the firewall rules to include the registered IP addresses.

C. Store the IP addresses in an Amazon DynamoDB table. Configure an AWS Lambda authorization function on the ALB to validate that incoming requests are from the registered IP addresses.

D. Configure the network ACL on the subnet that contains the public interface of the ALB. Update the ingress rules on the network ACL with entries for each of the registered IP addresses.

Correct Answer: A

Community vote distribution

A (91%) 9%

Comments

pentium75 Highly Voted 1 year, 11 months ago

Selected Answer: A

WAF, you can have 100 "rule sets" per account, each with up to 10,000 IP addresses.

https://docs.aws.amazon.com/waf/latest/developerguide/limits.html

upvoted 8 times

Karls Most Recent 1 year, 8 months ago

Selected Answer: C

AWS Lambda and DynamoDB to dynamically manage and validate incoming requests based on registered IP addresses.

https://docs.aws.amazon.com/lambda/latest/dg/services-alb.html

upvoted 1 times

ferdzcruz 1 year, 10 months ago

web services and HTTPS = WAF

upvoted 4 times

awsgeek75 1 year, 11 months ago

Selected Answer: A

B: Looks like an incomplete solution for something different

C: Not workable as Lambda for IP filtering means you have already allowed the request to pass through

D NACL with entries for each registered IP is not possible.

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: A

endpoint restriction by IP addresses = AWS WAF
upvoted 4 times

Passexam4sure_com 2 years, 1 month ago

Selected Answer: A

Associate an AWS WAF web ACL with the ALB. Use IP rule sets on the ALB to filter traffic. Update the IP addresses in the rule to include the registerec IP addresses.
upvoted 4 times

Sugarbear_01 2 years, 2 months ago

Selected Answer: A

AWS WAF cannot be directly associated with a Web Application. But, can only be associated with Application Load Balancer, CloudFront and API Gateway.
upvoted 4 times

taustin2 2 years, 2 months ago

Selected Answer: C

Changing answer to C because of "20000" IP addresses. Use Lambda with ALB.
upvoted 3 times

bsbs1234 2 years, 2 months ago
I will choose this answer if it is API Gateway. But I cannot figure out how to do lambda authentication on ALB. I will go A
upvoted 2 times

taustin2 2 years, 1 month ago
You are right. I don't know of a way to use Lambda with ALB in this way. Answer is A.
upvoted 2 times

potomac 2 years, 1 month ago
ALB invokes Lambda function, sending the incoming data in JSON format. Lambda function performs task, returns HTTP response to ALB.
upvoted 1 times

potomac 2 years, 1 month ago
WAF seems still better
upvoted 3 times

pentium75 1 year, 11 months ago
WAF allows 100 rule sets, each with up to 10,000 IP addresses, per account.
upvoted 2 times

potomac 2 years, 1 month ago
10,000 IP addresses
For the latest version of AWS WAF, see AWS WAF. If you want to allow or block web requests based on the IP addresses that the requests originate from, create one or more IP match conditions. An IP match condition lists up to 10,000 IP addresses or IP address ranges that your requests originate from.
upvoted 2 times

Guru4Cloud 2 years, 2 months ago

Selected Answer: A

A. Associate an AWS WAF web ACL with the ALB. Use IP rule sets on the ALB to filter traffic. Update the IP addresses in the rule to include the registered IP addresses.
upvoted 3 times

taustin2 2 years, 2 months ago

Selected Answer: A

WAF meets the requirements.
upvoted 3 times

Question #609 Topic 1

A company is building a data analysis platform on AWS by using AWS Lake Formation. The platform will ingest data from different sources such as Amazon S3 and Amazon RDS. The company needs a secure solution to prevent access to portions of the data that contain sensitive information.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an IAM role that includes permissions to access Lake Formation tables.

B. Create data filters to implement row-level security and cell-level security. Most Voted

C. Create an AWS Lambda function that removes sensitive information before Lake Formation ingests the data.

D. Create an AWS Lambda function that periodically queries and removes sensitive information from Lake Formation tables.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Guru4Cloud Highly Voted 2 years, 8 months ago

Selected Answer: B

The key reasons are:

Lake Formation data filters allow restricting access to rows or cells in data tables based on conditions. This allows preventing access to sensitive data
Data filters are implemented within Lake Formation and do not require additional coding or Lambda functions.
Lambda functions to pre-process data or purge tables would require ongoing development and maintenance.
IAM roles only provide user-level permissions, not row or cell level security.
Data filters give granular access control over Lake Formation data with minimal configuration, avoiding complex custom code.

upvoted 10 times

awsgeek75 2 years, 5 months ago

https://docs.aws.amazon.com/lake-formation/latest/dg/data-filters-about.html

upvoted 2 times

emakid Most Recent 1 year, 11 months ago

Selected Answer: B

B. Create data filters to implement row-level security and cell-level security.
Explanation:

Row-Level and Cell-Level Security: AWS Lake Formation provides built-in support for row-level and cell-level security. By using data filters, you can define policies that control access to specific rows and cells within your tables. This allows you to restrict access to sensitive information without needing to manually filter or remove data.

Least Operational Overhead: This solution leverages built-in Lake Formation capabilities, reducing the need for additional infrastructure or custom code. Once the data filters are set up, they automatically enforce the security policies, minimizing ongoing operational overhead.

upvoted 2 times

KennethNg923 1 year, 12 months ago

Selected Answer: B

As it said “prevent access to portions of the data that contain sensitive information”, not the access to S3, so data filter is enough

upvoted 3 times

wizcloudifa 2 years, 1 month ago

Selected Answer: B

Focus on the exact wordings: "to prevent access to portions of the data that contain sensitive information."
Only option B restricts the platform to access sensitive data, option A restrict users to restrict access that doesn't serve the req here, C and D are talking about removing the sensitive data which is not the ask here

upvoted 2 times

ferdzcruz 2 years, 4 months ago

portions of the data that contain sensitive information = Filtered data.

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

A is possible but it does not secure the data properly and only provides table level access control (if any).
CD are too much overhead
B is exactly for this purpose and is a built-in feature of Lake formation

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: B

https://docs.aws.amazon.com/lake-formation/latest/dg/data-filters-about.html

upvoted 3 times

taustin2 2 years, 8 months ago

Selected Answer: B

You can create data filters based on the values of columns in a Lake Formation table. Easy. Lowest operational overhead.

upvoted 3 times

nnecode 2 years, 8 months ago

Selected Answer: B

The best solution to meet the requirements with the least operational overhead is to create data filters to implement row-level security and cell-level security.

Data filters are a feature of Lake Formation that allow you to restrict access to data based on row and column values. This can be used to implement row-level security and cell-level security.

To implement row-level security, you would create a data filter that only allows users to access rows where the values in certain columns meet certair criteria. For example, you could create a data filter that only allows users to access rows where the value in the customer_id column matches the user's own customer ID.

upvoted 3 times

Question #610 *Topic 1*

A company deploys Amazon EC2 instances that run in a VPC. The EC2 instances load source data into Amazon S3 buckets so that the data can be processed in the future. According to compliance laws, the data must not be transmitted over the public internet. Servers in the company's on-premises data center will consume the output from an application that runs on the EC2 instances.

Which solution will meet these requirements?

A. Deploy an interface VPC endpoint for Amazon EC2. Create an AWS Site-to-Site VPN connection between the company and the VPC.

B. Deploy a gateway VPC endpoint for Amazon S3. Set up an AWS Direct Connect connection between the on-premises network and the VPC. **Most Voted**

C. Set up an AWS Transit Gateway connection from the VPC to the S3 buckets. Create an AWS Site-to-Site VPN connection between the company and the VPC.

D. Set up proxy EC2 instances that have routes to NAT gateways. Configure the proxy EC2 instances to fetch S3 data and feed the application instances.

Correct Answer: B

Community vote distribution

B (79%) A (21%)

Comments

taustin2 Highly Voted 2 years, 8 months ago

Selected Answer: B

Gateway VPC Endpoint = no internet to access S3. Direct Connect = secure access to VPC.

upvoted 11 times

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: B

No public internet != encrypted public internet (VPN)
Direct connect is the only option.

upvoted 6 times

zdi561 Most Recent 1 year, 4 months ago

Selected Answer: A

While not as ideal for high-traffic scenarios, a properly configured Site-to-Site VPN can also create a private connection between your on-premise network and AWS. Gateway endpoint is not working for on-premise

upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: B

Deploy a gateway VPC endpoint for Amazon S3 = so traffic between EC2 and S3 doesn't live AWS private network.

Set up an AWS Direct Connect connection between the on-premises network and the VPC = servers on-premises can consume the output from ec2 instances via private connection.

upvoted 2 times

OSHOAIB 2 years, 5 months ago

Selected Answer: B

A gateway VPC endpoint for Amazon S3 allows the EC2 instances within the VPC to access Amazon S3 buckets without using the public internet. The traffic between the VPC and S3 is routed within the AWS network.
AWS Direct Connect establishes a private connection between the on-premises data center and AWS infrastructure, avoiding data transfer over the public internet and ensuring compliance with the specified requirements. It provides a dedicated network link with higher bandwidth options and potentially more consistent network performance than internet-based connections.

Whereas Option A uses Site-to-Site VPN connection which is secure. However it typically runs over the public internet, which would not meet the company's requirement of avoiding public internet data transit.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: B

I think the last sentence ("Servers in the company's on-premises data center will consume the output from an application that runs on the EC2 instances") refers to a different application. Purely from the wording, it does NOT seem to refer to the data 'loaded into S3 buckets so that it can be processed in the future' before. So the EC2 instances could write to S3, the on-premises servers can talk to the EC2 application, and data would not be transmitted over the public internet.

Not A: There's no such thing as a "VPC endpoint for Amazon EC2 (!)"
Not C: Transit Gateway is not for EC2->S3, VPN is over public internet
Not D: Would address only the first part and use public Internet

upvoted 2 times

wizcloudifa 2 years, 1 month ago

Interface endpoint is a thing, the only reason A is not true is because of the presence of site-to-site vpn which is essentially accessing public interne

upvoted 2 times

ale_brd_111 2 years, 5 months ago

Selected Answer: A

I would go for A, for two reasons:
1) "S3 gateway endpoints do not currently support access from resources in a different Region, different VPC, or from an on-premises (non-AWS) environment.
2) we tryna access an output from an application hosted in e2 instances and not to access the s3 stored data so ideally we should use Interface Endpoints for the applications running in ec2.

upvoted 3 times

MatAlves 1 year, 8 months ago

You forgot the traffic from EC2 to S3. Without the Gateway Endpoint, that would go via public internet.

1. Deploy a gateway VPC endpoint for Amazon S3 = so traffic between EC2 and S3 doesn't live AWS private network.

2. Set up an AWS Direct Connect connection between the on-premises network and the VPC = servers on-premises can consume the output from ec2 instances via private connection.

upvoted 2 times

pentium75 2 years, 5 months ago

Plus, in A you deploy a VPC endpoint "for EC2" (!) which doesn't exist

upvoted 3 times

elmyth 1 year, 9 months ago

exists, check the docs, interface VPS endpoint != gateway VPC endpoint, they have different range of services

upvoted 2 times

pentium75 2 years, 5 months ago

"Data must not be transmitted over the public internet", as it would with A (VPN).

upvoted 3 times

ftaws 2 years, 5 months ago

I standhood answer is B, but why not A?

upvoted 1 times

pentium75 2 years, 5 months ago

there's no such things a 'VPC endpoint for EC2', and it uses public Internet

upvoted 2 times

achechen 2 years, 6 months ago

Selected Answer: A

https://aws.amazon.com/blogs/architecture/choosing-your-vpc-endpoint-strategy-for-amazon-s3/ According to this document, " S3 gateway endpoints do not currently support access from resources in a different Region, different VPC, or from an on-premises (non-AWS) environment. However, if you're willing to manage a complex custom architecture, you can use proxies. In all those scenarios, where access is from resources external to VPC, S3 interface endpoints access S3 in a secure way." so, the answer is A.

upvoted 4 times

pentium75 2 years, 5 months ago

A uses a VPC endpoint "for Amazon EC2", not S3. Also it uses public Internet.

upvoted 3 times

elmyth 1 year, 9 months ago

interface VPC endpoint works with PrivateLink, so it can be connected to huge amount of services, and to EC2. Gateway VPC endpoint can't work for on-prem

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

data must not be transmitted over the public internet = gateway VPC endpoint for Amazon S3 and AWS Direct Connect connection between the on-premises network and the VPC.

upvoted 2 times

Guru4Cloud 2 years, 8 months ago

Selected Answer: B

Gateway VPC Endpoint = no internet to access S3. Direct Connect = secure access to VPC
I agree with you @taustin2- Happy Learning all

upvoted 5 times

Question #611 Topic 1

A company has an application with a REST-based interface that allows data to be received in near-real time from a third-party vendor. Once received, the application processes and stores the data for further analysis. The application is running on Amazon EC2 instances.

The third-party vendor has received many 503 Service Unavailable Errors when sending data to the application. When the data volume spikes, the compute capacity reaches its maximum limit and the application is unable to process all requests.

Which design should a solutions architect recommend to provide a more scalable solution?

A. Use Amazon Kinesis Data Streams to ingest the data. Process the data using AWS Lambda functions. Most Voted

B. Use Amazon API Gateway on top of the existing application. Create a usage plan with a quota limit for the third-party vendor.

C. Use Amazon Simple Notification Service (Amazon SNS) to ingest the data. Put the EC2 instances in an Auto Scaling group behind an Application Load Balancer.

D. Repackage the application as a container. Deploy the application using Amazon Elastic Container Service (Amazon ECS) using the EC2 launch type with an Auto Scaling group.

Correct Answer: A

Community vote distribution

A (94%) 6%

Comments

Guru4Cloud Highly Voted 2 years, 2 months ago

Selected Answer: A

The key reasons are:

Kinesis Data Streams provides an auto-scaling stream that can handle large amounts of streaming data ingestion and throughput. This removes the bottlenecks around receiving the data.
AWS Lambda can process and store the data in a scalable serverless manner, avoiding EC2 capacity limits.
API Gateway adds API management capabilities but does not improve the underlying scalability of the EC2 application.
SNS is for event publishing/notifications, not large scale data ingestion. ECS still relies on EC2 capacity.

upvoted 7 times

taustin2 Highly Voted 2 years, 2 months ago

Selected Answer: A

For near-real time data ingest and processing, Kinesis and Lambda are most scalable choice.

upvoted 5 times

zdi561 Most Recent 1 year, 4 months ago

Selected Answer: B

A change the whole architect, B adding a API gateway to make the app more scalable not perfect as in A it may not be possible within budget, deadline, Yes, AWS API Gateway significantly contributes to making an application more scalable by handling a large volume of concurrent API requests, automatically managing traffic distribution, it can cache response and allow asyn call

upvoted 1 times

ferdzcruz 1 year, 10 months ago

A.
Kinesis Data Streams = near realtime and scalable
AWS Lambda functions = scalable

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: A
more scalable solution? = serverless = Amazon Kinesis Data Streams and AWS Lambda functions
upvoted 3 times

wsdasdasdqwdaw 2 years, 1 month ago
Only A is pure serverless which means scale. A for sure.
upvoted 2 times

Question #612 Topic 1

A company has an application that runs on Amazon EC2 instances in a private subnet. The application needs to process sensitive information from an Amazon S3 bucket. The application must not use the internet to connect to the S3 bucket.

Which solution will meet these requirements?

A. Configure an internet gateway. Update the S3 bucket policy to allow access from the internet gateway. Update the application to use the new internet gateway.

B. Configure a VPN connection. Update the S3 bucket policy to allow access from the VPN connection. Update the application to use the new VPN connection.

C. Configure a NAT gateway. Update the S3 bucket policy to allow access from the NAT gateway. Update the application to use the new NAT gateway.

D. Configure a VPC endpoint. Update the S3 bucket policy to allow access from the VPC endpoint. Update the application to use the new VPC endpoint. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

Guru4Cloud Highly Voted 2 years, 2 months ago

Selected Answer: D

The solution that will meet these requirements is to:

Configure a VPC endpoint for Amazon S3
Update the S3 bucket policy to allow access from the VPC endpoint
Update the application to use the new VPC endpoint
The key reasons are:

VPC endpoints allow private connectivity from VPCs to AWS services like S3 without using an internet gateway.
The application can connect to S3 through the VPC endpoint while remaining in the private subnet, without internet access.

upvoted 8 times

ferdzcruz Most Recent 1 year, 10 months ago

D.
VPC endpoint = not internet, direct access from VPC to S3

upvoted 3 times

awsgeek75 1 year, 11 months ago

Selected Answer: D

https://docs.aws.amazon.com/whitepapers/latest/aws-privatelink/what-are-vpc-endpoints.html

upvoted 3 times

achechen 2 years ago

Selected Answer: D

Answer is D

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: D

application must not use the internet to connect to the S3 bucket = VPC endpoint

upvoted 4 times

taustin2 2 years, 2 months ago

Selected Answer: D

VPC Endpoint for S3.

upvoted 2 times

aleariva 2 years, 2 months ago

D is the correct...https://docs.aws.amazon.com/whitepapers/latest/aws-privatelink/what-are-vpc-endpoints.html

upvoted 1 times

awslearnerin2022 2 years, 2 months ago

Selected Answer: D

VPC endpoint enables communication between VPC subnet and S3 bucket.

upvoted 1 times

nnecode 2 years, 2 months ago

Selected Answer: D

A VPC endpoint is a managed endpoint in your VPC that is connected to a public AWS service. It provides a private connection between your VPC and the service, and it does not require an internet gateway or a NAT device.

Option A (internet gateway) would involve exposing the S3 bucket to the internet, which is not recommended for security reasons.

Option B (VPN connection) would require additional setup and would still involve traffic going over the internet.

Option C (NAT gateway) is used for outbound internet access from private subnets, not for accessing S3 without the internet.

upvoted 4 times

Question #613 Topic 1

A company uses Amazon Elastic Kubernetes Service (Amazon EKS) to run a container application. The EKS cluster stores sensitive information in the Kubernetes secrets object. The company wants to ensure that the information is encrypted.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use the container application to encrypt the information by using AWS Key Management Service (AWS KMS).

B. Enable secrets encryption in the EKS cluster by using AWS Key Management Service (AWS KMS). Most Voted

C. Implement an AWS Lambda function to encrypt the information by using AWS Key Management Service (AWS KMS).

D. Use AWS Systems Manager Parameter Store to encrypt the information by using AWS Key Management Service (AWS KMS).

Correct Answer: B

Community vote distribution

B (100%)

Comments

Guru4Cloud Highly Voted 2 years, 8 months ago

Selected Answer: B

EKS supports encrypting Kubernetes secrets at the cluster level using AWS KMS keys. This provides an automated way to encrypt secrets.
Enabling this feature requires minimal configuration changes to the EKS cluster and no code changes.
Other options like using Lambda functions or modifying the application code to encrypt secrets require additional development effort and overhead
Systems Manager Parameter Store could store encrypted parameters but does not natively integrate with EKS to encrypt Kubernetes secrets.
The EKS secrets encryption feature leverages AWS KMS without the need to directly call KMS APIs from the application.

upvoted 8 times

KennethNg923 Most Recent 1 year, 12 months ago

Selected Answer: B

System manager: irrelevant
Lambda or application: operational overhead
So it will be B secret encryption

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

LEAST operational overhead? = Enable secrets encryption in the EKS cluster

upvoted 3 times

potomac 2 years, 7 months ago

Selected Answer: B

https://aws.amazon.com/about-aws/whats-new/2020/03/amazon-eks-adds-envelope-encryption-for-secrets-with-aws-kms/

upvoted 3 times

dilaaziz 2 years, 7 months ago

Selected Answer: B

https://aws.amazon.com/about-aws/whats-new/2020/03/amazon-eks-adds-envelope-encryption-for-secrets-with-aws-kms/

upvoted 2 times

iwannabeawsgod 2 years, 7 months ago

BBBBBBB

upvoted 2 times

taustin2 2 years, 8 months ago

Selected Answer: B

Use KMS. Enable secrets encryption in KMS.

upvoted 3 times

nnecode 2 years, 8 months ago

Selected Answer: B

Enabling secrets encryption in the EKS cluster by using AWS Key Management Service (AWS KMS) is the least operationally overhead way to encrypt the sensitive information in the Kubernetes secrets object.

When you enable secrets encryption in the EKS cluster, AWS KMS encrypts the secrets before they are stored in the EKS cluster. You do not need to make any changes to your container application or implement any additional Lambda functions.

upvoted 3 times

Question #614 | *Topic 1*

A company is designing a new multi-tier web application that consists of the following components:

• Web and application servers that run on Amazon EC2 instances as part of Auto Scaling groups
• An Amazon RDS DB instance for data storage

A solutions architect needs to limit access to the application servers so that only the web servers can access them.

Which solution will meet these requirements?

A. Deploy AWS PrivateLink in front of the application servers. Configure the network ACL to allow only the web servers to access the application servers.

B. Deploy a VPC endpoint in front of the application servers. Configure the security group to allow only the web servers to access the application servers.

C. Deploy a Network Load Balancer with a target group that contains the application servers' Auto Scaling group. Configure the network ACL to allow only the web servers to access the application servers.

D. Deploy an Application Load Balancer with a target group that contains the application servers' Auto Scaling group. Configure the security group to allow only the web servers to access the application servers. **Most Voted**

Correct Answer: D

Community vote distribution

D (87%) | 13%

Comments

Guru4Cloud Highly Voted 2 years, 2 months ago

Selected Answer: D

The key reasons are:

An Application Load Balancer (ALB) allows directing traffic to the application servers and provides access control via security groups.
Security groups act as a firewall at the instance level and can control access to the application servers from the web servers.
Network ACLs work at the subnet level and are less flexible for security groups for instance-level access control.
VPC endpoints are used to provide private access to AWS services, not for access between EC2 instances.
AWS PrivateLink provides private connectivity between VPCs, which is not required in this single VPC scenario.

upvoted 21 times

Dz1990 Most Recent 1 year ago

Selected Answer: D

Option D is correct because:

ALB is the right tool to put in front of application servers
Security groups are the right tool to control access (not network ACLs)
This creates a proper barrier where only web servers can reach the application servers

Option B is wrong because VPC endpoints are for AWS services, not your own EC2 application servers.
Think of it like this: ALB is like a security guard at a building entrance - only people (web servers) with the right ID can get through to reach the workers (application servers) inside.

upvoted 1 times

Ravan 1 year, 9 months ago

Selected Answer: B

A VPC endpoint is a managed endpoint in your VPC that is connected to a public AWS service. It provides a private connection between your VPC and the service, and it does not require an internet gateway or a NAT device.
The other options do not meet all of the requirements:

Option A: AWS PrivateLink is a service that allows you to connect your VPC to private services that are owned by AWS or by other AWS customers. It is not designed to be used to limit access to resources within the same VPC.
Option C: A Network Load Balancer can be used to distribute traffic across multiple application servers, but it does not provide a way to limit access to the application servers.
Option D: An Application Load Balancer can be used to distribute traffic across multiple application servers, but it does not provide a way to limit access to the application servers.

upvoted 2 times

awsgeek75 1 year, 10 months ago

Selected Answer: D

"limit access to the application servers so that only the web servers can access them"
Can be done via NACL or SG
A: Irrelevant as everything is inside the same VPC
B: VPC endpoint are attached to VPC and if you deploy a VPC endpoint like this it will be in front of both app and web server. Language is weird here
C: Potentially a good solution but NACL is allowing on web to app traffic and no response will reach to web servers as NACL have to be configured in both directions
D: ALB in front of ASG will give an internal endpoint which can be secured by SG as recommended. ASG itself is not an endpoint that can be used with SG which is why we need ALB here.

Hence D is correct

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: D

Deploy an Application Load Balancer with a target group that contains the application servers' Auto Scaling group. Configure the security group to allow only the web servers to access the application servers

upvoted 4 times

Tekk97 2 years ago

Selected Answer: D

I think B also working. but A company has Auto Scaling groups. D has strategy for Auto Scaling. D is correct

upvoted 2 times

pentium75 1 year, 11 months ago

How do you want to "deploy a VPC endpoint" for a group of EC2 instances that are inside your VPC?

upvoted 2 times

potomac 2 years, 1 month ago

Selected Answer: D

D is correct

upvoted 2 times

iwannabeawsgod 2 years, 1 month ago

Selected Answer: D

Scaling group to Scaling group.

upvoted 3 times

Devsin2000 2 years, 2 months ago

C - ALB is for Web applications only. NLB can be internal / not public

upvoted 1 times

pentium75 1 year, 11 months ago

Both ALB and NLB can be internal or public. NLB works on layer 3 while ALB works on layer 7.

Both ALB and NLB could help here, but C uses a network ACL that's missing the outbound traffic.

upvoted 3 times

taustin2 2 years, 2 months ago

Selected Answer: D

ALB with Security Group is simplest solution.

upvoted 4 times

nnecode 2 years, 2 months ago

Selected Answer: B

A VPC endpoint is a managed endpoint in your VPC that is connected to a public AWS service. It provides a private connection between your VPC and the service, and it does not require an internet gateway or a NAT device.
The other options do not meet all of the requirements:

Option A: AWS PrivateLink is a service that allows you to connect your VPC to private services that are owned by AWS or by other AWS customers. It

is not designed to be used to limit access to resources within the same VPC.
Option C: A Network Load Balancer can be used to distribute traffic across multiple application servers, but it does not provide a way to limit access to the application servers.
Option D: An Application Load Balancer can be used to distribute traffic across multiple application servers, but it does not provide a way to limit access to the application servers.

upvoted 4 times

pentium75 1 year, 11 months ago

We don't want to connect "to a public AWS service" but to EC2 instances.

"An Application Load Balancer can be used to distribute traffic across multiple application servers, but it does not provide a way to limit access to the application servers" but the Security Group of the web servers does.

upvoted 2 times

Question #615 *Topic 1*

A company runs a critical, customer-facing application on Amazon Elastic Kubernetes Service (Amazon EKS). The application has a microservices architecture. The company needs to implement a solution that collects, aggregates, and summarizes metrics and logs from the application in a centralized location.

Which solution meets these requirements?

A. Run the Amazon CloudWatch agent in the existing EKS cluster. View the metrics and logs in the CloudWatch console.

B. Run AWS App Mesh in the existing EKS cluster. View the metrics and logs in the App Mesh console.

C. Configure AWS CloudTrail to capture data events. Query CloudTrail by using Amazon OpenSearch Service.

D. Configure Amazon CloudWatch Container Insights in the existing EKS cluster. View the metrics and logs in the CloudWatch console. **Most Voted**

Correct Answer: D

Community vote distribution

D (94%) 6%

Comments

pentium75 Highly Voted 1 year, 11 months ago

Selected Answer: D

'Running the Amazon CloudWatch agent in the existing EKS cluster' is called Amazon CloudWatch Container Insights: https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/Container-Insights-setup-metrics.html

upvoted 7 times

potomac Highly Voted 2 years, 1 month ago

Selected Answer: D

Use CloudWatch Container Insights to collect, aggregate, and summarize metrics and logs from your containerized applications and microservices. Container Insights is available for Amazon Elastic Container Service (Amazon ECS), Amazon Elastic Kubernetes Service (Amazon EKS), and Kubernetes platforms on Amazon EC2. Container Insights supports collecting metrics from clusters deployed on AWS Fargate for both Amazon ECS and Amazon EKS.

upvoted 6 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: D

https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/ContainerInsights.html
"Use CloudWatch Container Insights to collect, aggregate, and summarize metrics and logs from your containerized applications and microservices."

upvoted 5 times

SHAAHIBHUSHANAWS 2 years ago

Selected Answer: D
https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/ContainerInsights.html

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: D

EKS monitoring = Amazon CloudWatch Container Insights

upvoted 3 times

Examanier1217 2 years, 1 month ago

Selected Answer: A

I have worked on it. A is the right answer

upvoted 1 times

pentium75 1 year, 11 months ago
But 'running the Amazon CloudWatch agent in the existing EKS cluster' is called Container Insights, thus D.

https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/Container-Insights-setup-metrics.html
upvoted 3 times

dilaaziz 2 years, 1 month ago
Selected Answer: D
https://aws.amazon.com/cloudwatch/features/
upvoted 2 times

Guru4Cloud 2 years, 2 months ago
Selected Answer: D
The key reasons are:

CloudWatch Container Insights automatically collects metrics and logs from containers running in EKS clusters. This provides visibility into resource utilization, application performance, and microservice interactions.
The metrics and logs are stored in CloudWatch Logs and CloudWatch metrics for central access.
The CloudWatch console allows querying, filtering, and visualizing the metrics and logs in one centralized place.
upvoted 3 times

ErnShm 2 years, 2 months ago
D

Amazon CloudWatch Application Insights facilitates observability for your applications and underlying AWS resources. It helps you set up the best monitors for your application resources to continuously analyze data for signs of problems with your applications.
upvoted 3 times

taustin2 2 years, 2 months ago
Selected Answer: D
What Cloudwatch Container Insights is for.
upvoted 2 times

kambarami 2 years, 2 months ago
https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/deploy-container-insights-EKS.html
upvoted 2 times

awslearnerin2022 2 years, 2 months ago
Selected Answer: A
Cloudwatch monitors applications and provides metrics. Cloudtrail is used for API activities in the account.
upvoted 1 times

nnecode 2 years, 2 months ago
Selected Answer: D
Amazon CloudWatch Container Insights is a service that collects, aggregates, and summarizes metrics and logs from containerized applications. It is designed to work with Amazon EKS and Kubernetes.
upvoted 2 times

Question #616 *Topic 1*

A company has deployed its newest product on AWS. The product runs in an Auto Scaling group behind a Network Load Balancer. The company stores the product's objects in an Amazon S3 bucket.

The company recently experienced malicious attacks against its systems. The company needs a solution that continuously monitors for malicious activity in the AWS account, workloads, and access patterns to the S3 bucket. The solution must also report suspicious activity and display the information on a dashboard.

Which solution will meet these requirements?

A. Configure Amazon Macie to monitor and report findings to AWS Config.

B. Configure Amazon Inspector to monitor and report findings to AWS CloudTrail.

C. Configure Amazon GuardDuty to monitor and report findings to AWS Security Hub. **Most Voted**

D. Configure AWS Config to monitor and report findings to Amazon EventBridge.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Guru4Cloud **Highly Voted** 2 years, 8 months ago

Selected Answer: C

The key reasons are:

Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior. It analyzes AWS CloudTrail, VPC Flow Logs, and DNS logs.
GuardDuty can detect threats like instance or S3 bucket compromise, malicious IP addresses, or unusual API calls.
Findings can be sent to AWS Security Hub which provides a centralized security dashboard and alerts.
Amazon Macie and Amazon Inspector do not monitor the breadth of activity that GuardDuty does. They focus more on data security and application vulnerabilities respectively.
AWS Config monitors for resource configuration changes, not malicious activity.

upvoted 13 times

MatAlves **Most Recent** 1 year, 8 months ago

Selected Answer: C

- Amazon Inspector = automated vulnerability management service

- Amazon GuardDuty = threat detection service that monitors for malicious activity and anomalous behavior to protect AWS accounts, workloads, and data.

upvoted 4 times

KennethNg923 1 year, 12 months ago

Selected Answer: C

" continuously monitors for malicious activity in the AWS account, workloads, and access patterns to the S3 bucket" only guard duty for this purpose in the options

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

Amazon Inspector provides you with security assessments of your applications settings and configurations on your EC2 instances while Amazon GuardDuty helps with analyzing your entire AWS environment for potential threats.
AWS Security Hub is a cloud security posture management service that aggregates alerts, and enables automated remediation.

upvoted 4 times

dilaaziz 2 years, 7 months ago

Selected Answer: C

Guardduty

upvoted 3 times

taustin2 2 years, 8 months ago

Selected Answer: C

What Guard Duty is for.

upvoted 3 times

Guru4Cloud 2 years, 8 months ago

The key reasons are:

Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior. It analyzes AWS CloudTrail, VPC Flow Logs, and DNS logs.
GuardDuty can detect threats like instance or S3 bucket compromise, malicious IP addresses, or unusual API calls.
Findings can be sent to AWS Security Hub which provides a centralized security dashboard and alerts.
Amazon Macie and Amazon Inspector do not monitor the breadth of activity that GuardDuty does. They focus more on data security and application vulnerabilities respectively.
AWS Config monitors for resource configuration changes, not malicious activity.

upvoted 3 times

kambarami 2 years, 8 months ago

Answer is C.

upvoted 2 times

aleariva 2 years, 8 months ago

C is the correct. https://aws.amazon.com/guardduty/

upvoted 2 times

brownie23 2 years, 8 months ago

Answer is C Since Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior to protect your AWS accounts, Amazon Elastic Compute Cloud (EC2) workloads, container applications, Amazon Aurora databases, and data stored in Amazon Simple Storage Service (S3).

upvoted 3 times

awslearnerin2022 2 years, 8 months ago

Selected Answer: C

Gaurd duty is a threat detection service for accounts and workloads.

upvoted 2 times

Question #617 *Topic 1*

A company wants to migrate an on-premises data center to AWS. The data center hosts a storage server that stores data in an NFS-based file system. The storage server holds 200 GB of data. The company needs to migrate the data without interruption to existing services. Multiple resources in AWS must be able to access the data by using the NFS protocol.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

A. Create an Amazon FSx for Lustre file system.

B. Create an Amazon Elastic File System (Amazon EFS) file system. Most Voted

C. Create an Amazon S3 bucket to receive the data.

D. Manually use an operating system copy command to push the data into the AWS destination.

E. Install an AWS DataSync agent in the on-premises data center. Use a DataSync task between the on-premises location and AWS. Most Voted

Correct Answer: BE

Community vote distribution

BE (100%)

Comments

Guru4Cloud Highly Voted 2 years, 2 months ago

Selected Answer: BE

Amazon EFS provides a scalable, high performance NFS file system that can be accessed from multiple resources in AWS.
AWS DataSync can perform the migration from the on-prem NFS server to EFS without interruption to existing services.
This avoids having to manually move the data which could cause downtime. DataSync incrementally syncs changed data.
EFS and DataSync together provide a cost-optimized approach compared to using S3 or FSx, while still meeting the requirements.
Manually copying 200 GB of data to AWS would be slow and risky compared to using DataSync.

upvoted 11 times

awslearnerin2022 Highly Voted 2 years, 2 months ago

Selected Answer: BE

EFS can be accessed by multiple AWS resources.
Datasync allowes NFS migrations.

upvoted 5 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: BE

A: FSX Lustre is for parallel high performance file storage not NFS
C: S3 is a blob storage, not a file system
D: Too much to copy with a lot of overhead
A: NFS maps to EFS and allows NFS protocol for access
E: DataSync solves copy problems without interruptions

upvoted 5 times

dilaaziz 2 years, 1 month ago

Selected Answer: BE

https://aws.amazon.com/compare/the-difference-between-nfs-smb/

upvoted 2 times

taustin2 2 years, 2 months ago

Selected Answer: BE

NFS file system = EFS, Use DataSync for the migration with NFS support.

upvoted 5 times

Question #618 *Topic 1*

A company wants to use Amazon FSx for Windows File Server for its Amazon EC2 instances that have an SMB file share mounted as a volume in the us-east-1 Region. The company has a recovery point objective (RPO) of 5 minutes for planned system maintenance or unplanned service disruptions. The company needs to replicate the file system to the us-west-2 Region. The replicated data must not be deleted by any user for 5 years.

Which solution will meet these requirements?

A. Create an FSx for Windows File Server file system in us-east-1 that has a Single-AZ 2 deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in compliance mode for a target vault in us-west-2. Configure a minimum duration of 5 years.

B. Create an FSx for Windows File Server file system in us-east-1 that has a Multi-AZ deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in governance mode for a target vault in us-west-2. Configure a minimum duration of 5 years.

C. Create an FSx for Windows File Server file system in us-east-1 that has a Multi-AZ deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in compliance mode for a target vault in us-west-2. Configure a minimum duration of 5 years. **Most Voted**

D. Create an FSx for Windows File Server file system in us-east-1 that has a Single-AZ 2 deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in governance mode for a target vault in us-west-2. Configure a minimum duration of 5 years.

Correct Answer: C

Community vote distribution

C (100%)

Comments

taustin2 Highly Voted 2 years, 8 months ago

Selected Answer: C

Need to use Compliance Mode, so it's either A or C. RPO leads to Multi-AZ so C.

upvoted 13 times

KennethNg923 1 year, 11 months ago

Agree Compliance Mode with Multi-AZ

upvoted 2 times

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: C

high availability = multi AZ
data must be retained for 5 years = compliance mode

upvoted 5 times

TheLaPlanta 2 years, 2 months ago

No HA was mentioned though. But RPO leads to that, so IDK

upvoted 2 times

dilaaziz Most Recent 2 years, 7 months ago

Selected Answer: C

https://docs.aws.amazon.com/aws-backup/latest/devguide/vault-lock.html

upvoted 3 times

thanhnv142 2 years, 7 months ago

C is correct.
A and C is potential answer because they mention compliance mode. But single AZ is recommended for test and development only. For production workloads, we need multi AZ, which is C

upvoted 3 times

Xin123 2 years, 8 months ago

Selected Answer: C

Trust me bro

upvoted 4 times

Question #619 *Topic 1*

A solutions architect is designing a security solution for a company that wants to provide developers with individual AWS accounts through AWS Organizations, while also maintaining standard security controls. Because the individual developers will have AWS account root user-level access to their own accounts, the solutions architect wants to ensure that the mandatory AWS CloudTrail configuration that is applied to new developer accounts is not modified.

Which action meets these requirements?

A. Create an IAM policy that prohibits changes to CloudTrail. and attach it to the root user.

B. Create a new trail in CloudTrail from within the developer accounts with the organization trails option enabled.

C. Create a service control policy (SCP) that prohibits changes to CloudTrail, and attach it the developer accounts. **Most Voted**

D. Create a service-linked role for CloudTrail with a policy condition that allows changes only from an Amazon Resource Name (ARN) in the management account.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Xin123 Highly Voted 2 years, 2 months ago

Selected Answer: C

Organizations + Restricts = SCP

upvoted 7 times

taustin2 Highly Voted 2 years, 2 months ago

Selected Answer: C

For Organizations to restrict users in accounts, use an SCP.

upvoted 7 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: C

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps.html

upvoted 4 times

awsgeek75 1 year, 10 months ago

C is correct but for my sanity I want to know what D is talking about as it makes no sense to me. Can someone explain?

upvoted 2 times

LeonSauveterre 1 year, 6 months ago

The "policy condition" mentioned in option D can never be created. A service-linked role is a type of IAM role that is predefined by AWS and tightly integrated with AWS services, allowing AWS services to perform actions on your behalf. However, service-linked roles are not meant to control or restrict user actions, but simply designed for services to function correctly.

upvoted 1 times

TariqKipkemei 2 years ago

Selected Answer: C

Guardrails = service control policy

upvoted 1 times

Ramdi1 2 years, 2 months ago

Selected Answer: C

C - Use SCP best way

upvoted 4 times

Question #620 *Topic 1*

A company is planning to deploy a business-critical application in the AWS Cloud. The application requires durable storage with consistent, low-latency performance.

Which type of storage should a solutions architect recommend to meet these requirements?

A. Instance store volume

B. Amazon ElastiCache for Memcached cluster

C. Provisioned IOPS SSD Amazon Elastic Block Store (Amazon EBS) volume **Most Voted**

D. Throughput Optimized HDD Amazon Elastic Block Store (Amazon EBS) volume

Correct Answer: C

Community vote distribution

C (100%)

Comments

taustin2 Highly Voted 2 years, 2 months ago

Selected Answer: C

Durable storage excludes A and B. Low-latency excludes D. Choose C.

upvoted 12 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: C

AB are not storage or this purpose
D is HDD so slow by nature
C best fit

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: C

durable storage, low-latency performance = Provisioned IOPS SSD Amazon EBS volume

upvoted 3 times

potomac 2 years, 1 month ago

Selected Answer: C

Provisioned IOPS SSD — Provides high performance for mission-critical, low-latency, or high-throughput workloads. Throughput Optimized HDD — A low-cost HDD designed for frequently accessed, throughput-intensive workloads.

upvoted 3 times

dilaaziz 2 years, 1 month ago

Selected Answer: C

https://aws.amazon.com/ebs/volume-types/

upvoted 3 times

Question #621 *Topic 1*

An online photo-sharing company stores its photos in an Amazon S3 bucket that exists in the us-west-1 Region. The company needs to store a copy of all new photos in the us-east-1 Region.

Which solution will meet this requirement with the LEAST operational effort?

A. Create a second S3 bucket in us-east-1. Use S3 Cross-Region Replication to copy photos from the existing S3 bucket to the second S3 bucket. **Most Voted**

B. Create a cross-origin resource sharing (CORS) configuration of the existing S3 bucket. Specify us-east-1 in the CORS rule's AllowedOrigin element.

C. Create a second S3 bucket in us-east-1 across multiple Availability Zones. Create an S3 Lifecycle rule to save photos into the second S3 bucket.

D. Create a second S3 bucket in us-east-1. Configure S3 event notifications on object creation and update events to invoke an AWS Lambda function to copy photos from the existing S3 bucket to the second S3 bucket.

Correct Answer: A

Community vote distribution

A (86%) 14%

Comments

Guru4Cloud Highly Voted 2 years, 8 months ago

Selected Answer: A

S3 Cross-Region Replication handles automatically copying new objects added to the source bucket to the destination bucket in a different region.
It continuously replicates new photos without needing to manually copy files or set up Lambda triggers.
CORS only enables cross-origin access, it does not copy objects.
Using Lifecycle rules or Lambda functions requires custom code and logic to handle the copying.
S3 Cross-Region Replication provides automated replication that minimizes operational overhead.

upvoted 8 times

xBUGx Most Recent 2 years, 3 months ago

Selected Answer: D

All NEW photo, not all photo.
We dont want to copy existing photos

upvoted 3 times

MatAlves 1 year, 8 months ago

"There are two types of replication: live replication and on-demand replication.

Live replication – To automatically replicate new and updated objects as they are written to the source bucket, use live replication. Live replication doesn't replicate any objects that existed in the bucket before you set up replication. To replicate objects that existed before you set up replication, use on-demand replication."

https://docs.aws.amazon.com/AmazonS3/latest/userguide/replication.html

upvoted 3 times

TheLaPlanta 2 years, 2 months ago

A does exactly that

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

https://docs.aws.amazon.com/AmazonS3/latest/userguide/replication.html

To automatically replicate new objects as they are written to the bucket, use live replication, such as Cross-Region Replication (CRR). To replicate existing objects to a different bucket on demand, use S3 Batch Replication.

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

LEAST operational effort = Cross-Region Replication

upvoted 2 times

dilaaziz 2 years, 7 months ago

Selected Answer: A

https://aws.amazon.com/about-aws/whats-new/2015/03/amazon-s3-introduces-cross-region-replication/

upvoted 3 times

taustin2 2 years, 8 months ago

Selected Answer: A

S3 Cross-Region Replication is least operational overhead.

upvoted 3 times

Question #622 Topic 1

A company is creating a new web application for its subscribers. The application will consist of a static single page and a persistent database layer. The application will have millions of users for 4 hours in the morning, but the application will have only a few thousand users during the rest of the day. The company's data architects have requested the ability to rapidly evolve their schema.

Which solutions will meet these requirements and provide the MOST scalability? (Choose two.)

A. Deploy Amazon DynamoDB as the database solution. Provision on-demand capacity. **Most Voted**

B. Deploy Amazon Aurora as the database solution. Choose the serverless DB engine mode.

C. Deploy Amazon DynamoDB as the database solution. Ensure that DynamoDB auto scaling is enabled.

D. Deploy the static content into an Amazon S3 bucket. Provision an Amazon CloudFront distribution with the S3 bucket as the origin. **Most Voted**

E. Deploy the web servers for static content across a fleet of Amazon EC2 instances in Auto Scaling groups. Configure the instances to periodically refresh the content from an Amazon Elastic File System (Amazon EFS) volume.

Correct Answer: AD

Community vote distribution

AD (58%) CD (29%) 6%

Comments

taustin2 Highly Voted 2 years, 8 months ago

Selected Answer: AD

Changing answer to A,D. DynamoDB on-demand is more scalable than DynamoDB auto-scaling.

upvoted 12 times

bogobob Highly Voted 2 years, 6 months ago

Selected Answer: CD

For those answering A over C, the question asks about scalability, but the text says the traffic patterns are known and don't state they will change. Both auto-scaling and on-demand can "scale", but auto-scaling is for known, on-demand is better for unknown traffic patterns. Its likely the "scalability" is more to do with the file hosting (EC2 wouldn't scale well at all vs S3)

upvoted 8 times

pentium75 2 years, 5 months ago

"Most scalability" = A. Might be more expensive, but cost is irrelevant in the question.

upvoted 4 times

ln2718 Most Recent 10 months, 3 weeks ago

Selected Answer: CD

I agree with the comments for C and D. I would go with autoscaling as a tighter fit than on-demand.

upvoted 1 times

bora4motion 1 year ago

Selected Answer: BD

I'm going with BD.

"the company's data architects have requested the ability to rapidly evolve their schema"

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: BD

Amazon Aurora Serverless for a highly fluctuating DB workload, and S3 for static webpage

upvoted 1 times

RamanAgarwal 1 year, 6 months ago
Cant change schema in Aurora
upvoted 1 times

marcosviniciuscb 1 year, 2 months ago
Aurora is slower than DynamoDB, Dynamo scales to millions of requests, which is what the company asks for
upvoted 1 times

Ravee_L 1 year, 10 months ago
B and D B. Deploy Amazon Aurora as the database solution. Choose the serverless DB engine mode: Aurora Serverless is an on-demand, auto-scaling configuration for Amazon Aurora.
upvoted 3 times

a7md0 1 year, 11 months ago
Selected Answer: AD
on-demand capacity for DynamoDB since traffic is not consistent, and S3 & CloudFront for static files
upvoted 2 times

KennethNg923 1 year, 11 months ago
Selected Answer: AD
MOST scalability - DynamoDB On-Demand > Auto Scaling + Static Content host in S3
upvoted 2 times

NSA_Poker 2 years ago
Selected Answer: CD
(A) is incorrect. "On-demand capacity mode is probably best when you have new tables with unknown workloads, unpredictable application traffic." We know the workload upper limit is the total amount of subscribers & we know it's busy in the morning & not so much in the afternoon. Nothing in the question says it bursts in the morning.

(C) is correct. The traffic pattern is known, so prep for the morning & the exact upper boundary is the # of subscribers.
upvoted 2 times

jaswantn 2 years, 4 months ago
For autoscaling we need to know the lower and upper limits. Anh the question says....application will have millions of users for 4 hours in the morning....how many millions , how much upper limit we need to set for to handle this much request?
here we can't have exact estimation for the upper limit in autoscaling. Thus, better option is (A)
upvoted 3 times

jaswantn 2 years, 4 months ago
With autoscaling we can face throttling initially, when there is surge of requests and the load is greater than the scaling upper limit. We can gradually increase the upper limit of autoscaling and would be then able to handle the load in subsequent requests preventing ourself from using OnDemand.
Thus Option (C) is more scalable as it can handle the both types of load(high & low) in efficient manner.
upvoted 1 times

JA2018 1 year, 6 months ago
why not Amazon Aurora Serverless?
upvoted 1 times

1Alpha1 2 years, 4 months ago
Selected Answer: AD
AD vs CD ?
1) Please read the final sentence. Which solutions will meet these requirements and provide the "MOST" scalability?
2) It is not possible to predict an exact boundary based on the number of "millions of users".
So I would choose "AD".
upvoted 4 times

NSA_Poker 2 years ago
The exact boundary is known, the application's subscribers. Leaning towards AC now.
upvoted 1 times

06042022 2 years, 4 months ago
Selected Answer: CD
The traffic pattern is known here.
upvoted 2 times

awsgeek75 2 years, 5 months ago
Selected Answer: AD

A: On-demand scaling because the demand changes drastically (millions to thousands)
D: S3 for static page is perfect

B: Aurora is RDMS so not much rapid schema changes (it's subjective and DBA will argue but better options on the table are DynamoDB)
E: Too much work and overhead

upvoted 2 times

awsgeek75 2 years, 4 months ago

To be fair, 4 hours is a strange time duration for burst traffic. 20 hours of low traffic may benefit from auto-scaling's as it is long enough to be callec a "depressed" traffic mode in autoscaling config. 4 hours in the morning can also be termed as "sustained period" of burst in autoscaling.

This question is not theoretical, someone who has scaled Dynamo in similar scenarios will be able to answer correctly.

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: AD

Question asks for "most scalability", not cost optimization. "DynamoDB auto scaling ... modifies provisioned throughput settings only when the actua workload stays elevated or depressed for a sustained period of several minutes. ... This means that provisioned capacity is probably best for you if you have relatively predictable application traffic, run applications whose traffic is consistent, and ramps up or down gradually."

upvoted 5 times

pentium75 2 years, 5 months ago

"The on-demand pricing model is ideal for bursty, new, or unpredictable workloads whose traffic can spike in seconds or minutes, and when under-provisioned capacity would impact the user experience."

Whereas on-demand capacity mode is probably best when you have new tables with unknown workloads, unpredictable application traffic and alsc if you only want to pay exactly for what you use. The on-demand pricing model is ideal for bursty, new, or unpredictable workloads whose traffic can spike in seconds or minutes, and when under-provisioned capacity would impact the user experience.

https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html

upvoted 2 times

Salilgen 1 year, 5 months ago

This support C not A

upvoted 1 times

Ashhher 2 years, 5 months ago

Selected Answer: BD

I understand the argument between A and C, but why not B?

upvoted 2 times

pentium75 2 years, 5 months ago

"Ability to rapidly evolve their schema" -> NoSQL database, schema changes in transactional databases like RDS are difficult

upvoted 5 times

Derek_G 2 years, 5 months ago

Selected Answer: AD

Provisioned on-demand capacity:
Manual: Requires manual setup and management of capacity.
Cost-Effectiveness: Requires manual estimation of workload, which can result in either excess or insufficient capacity.
Use Case: Suitable for relatively stable workloads with predictable capacity needs.

predictable capacity needs.： 4 hours in the morning,a few thousand users during the rest of the day.

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: AD

rapidly evolve their schema, MOST scalability for data layer = DynamoDB with on-demand capacity. on-demand capacity mode automatically enables autoscaling.
MOST scalability for single page app = Amazon CloudFront distribution with the S3 bucket as the origin.

upvoted 4 times

Question #623 *Topic 1*

A company uses Amazon API Gateway to manage its REST APIs that third-party service providers access. The company must protect the REST APIs from SQL injection and cross-site scripting attacks.

What is the MOST operationally efficient solution that meets these requirements?

A. Configure AWS Shield.

B. Configure AWS WAF. **Most Voted**

C. Set up API Gateway with an Amazon CloudFront distribution. Configure AWS Shield in CloudFront.

D. Set up API Gateway with an Amazon CloudFront distribution. Configure AWS WAF in CloudFront.

Correct Answer: B

Community vote distribution

B (88%) 13%

Comments

taustin2 Highly Voted 2 years, 2 months ago

Selected Answer: B

SQL Injection and Cross-Site Scripting = WAF so Either B or D. Both B and D are valid options but the question doesn't indicate a real need for CloudFront, so just use WAF with the API Gateway. Answer is B.

upvoted 14 times

awslearnerin2022 Highly Voted 2 years, 2 months ago

Selected Answer: B

WAF helps with layer 7 attacks like SQL injection and XSS. Shield is helpful for DDOS attacks.

upvoted 8 times

marcosviniciuscb Most Recent 1 year, 2 months ago

Selected Answer: D

CloudFront no se elige por su caché

✅ No se elige por mejorar la latencia

🔐 Se elige porque es el único sitio donde puedes aplicar AWS WAF si usas REST API (v1)

Entonces:

Aunque CloudFront tenga caché y latencia optimizada, eso no es lo relevante aquí.
Lo importante es que es el único recurso donde puedes colgar el WAF para proteger tu REST API.

upvoted 1 times

marcosviniciuscb 1 year, 2 months ago

Option B is technically sound and more straightforward if you only want to protect a specific API.

Option D is more scalable and operationally efficient, especially if you're building a modern architecture with multiple services and want protection from the edge.

The question uses the keyword "MOST operationally efficient," so option D is the best.

upvoted 2 times

Salilgen 1 year, 5 months ago

Selected Answer: B

https://aws.amazon.com/blogs/compute/amazon-api-gateway-adds-support-for-aws-waf/

upvoted 1 times

awsgeek75 1 year, 11 months ago

Selected Answer: B

WAF is good enough for SQL Injection and Cross Site scripting so A is good
A: AWS Shield (basic) is not for SQL injection
C: Same as A
D: Good solution and will work but it provides extra DDoS protection and caching which is not needed (as we don't know much about the API also)

upvoted 3 times

pentium75 1 year, 11 months ago

Selected Answer: B

Question asks for protection against SQL injection and XSS, both is provided by WAF (option B). D would work too, but it would add another layer (CloudFront) with benefits that nobody asked for (and that would cost money), thus it would IMO be less 'operationally efficient'.

upvoted 3 times

Naijaboy99 1 year, 11 months ago

Selected Answer: D

D. Set up API Gateway with an Amazon CloudFront distribution. Configure AWS WAF in CloudFront.

Option A (Configure AWS Shield) is a DDoS protection service but doesn't specifically address SQL injection and cross-site scripting attacks.

Option B (Configure AWS WAF) alone is a valid option, but integrating it with CloudFront (Option D) provides additional benefits like improved performance through caching.

Option C (Set up API Gateway with CloudFront and configure AWS Shield in CloudFront) might provide DDoS protection, but for SQL injection and cross-site scripting, AWS WAF is the more appropriate service.

upvoted 5 times

LeonSauveterre 1 year, 6 months ago

In questions like this that mentioned "operationally efficient" or "operationally effective", aim for the indicated jobs only. We don't need additional benefits, so CloudFront is not a necessity here.

upvoted 1 times

TariqKipkemei 2 years ago

Selected Answer: B

SQL injection and cross-site scripting attacks = AWS WAF

upvoted 4 times

potomac 2 years, 1 month ago

Selected Answer: B

B or D
But no need for CloudFront

upvoted 2 times

Sugarbear_01 2 years, 1 month ago

Selected Answer: B

AWS WAF protect agains :
Presence of SQL code that is likely to be malicious (known as SQL injection).
Presence of a script that is likely to be malicious (known as cross-site scripting).

AWS Shield provides protection against distributed denial of service (DDoS) attacks for AWS resources, at the network and transport layers (layer 3 and 4) and the application layer (layer 7).

https://docs.aws.amazon.com/waf/latest/developerguide/what-is-aws-waf.html

upvoted 2 times

thanhnv142 2 years, 1 month ago

Finally, I am here at the end. Thank you guys for your support!

upvoted 5 times

Guru4Cloud 2 years, 2 months ago

Selected Answer: B

B. Configure AWS WAF.

upvoted 5 times

aleariva 2 years, 2 months ago

B is the correct. https://docs.aws.amazon.com/waf/latest/developerguide/classic-web-acl-xss-conditions.html

upvoted 5 times

Question #624 *Topic 1*

A company wants to provide users with access to AWS resources. The company has 1,500 users and manages their access to on-premises resources through Active Directory user groups on the corporate network. However, the company does not want users to have to maintain another identity to access the resources. A solutions architect must manage user access to the AWS resources while preserving access to the on-premises resources.

What should the solutions architect do to meet these requirements?

A. Create an IAM user for each user in the company. Attach the appropriate policies to each user.

B. Use Amazon Cognito with an Active Directory user pool. Create roles with the appropriate policies attached.

C. Define cross-account roles with the appropriate policies attached. Map the roles to the Active Directory groups.

D. Configure Security Assertion Markup Language (SAML) 2 0-based federation. Create roles with the appropriate policies attached Map the roles to the Active Directory groups. Most Voted

Correct Answer: D

Community vote distribution

D (89%) 11%

Comments

pentium75 Highly Voted 1 year, 11 months ago

Selected Answer: D

Though you can federate Cognito with Active Directory, Cognito is for providing access to your own applications, NOT to AWS Resources.

upvoted 11 times

tsdsmth Highly Voted 1 year, 11 months ago

Selected Answer: D

While Amazon Cognito can integrate with Active Directory, it is more focused on providing identity management for mobile and web applications. In this scenario, where the primary concern is integrating with existing on-premises resources, using SAML-based federation with IAM roles is more appropriate.

upvoted 8 times

Salilgen Most Recent 1 year, 5 months ago

Selected Answer: D

https://aws.amazon.com/blogs/security/aws-federated-authentication-with-active-directory-federation-services-ad-fs/
B is wrong because user pool is in Cognito not in Active Directory
https://aws.amazon.com/blogs/security/how-to-set-up-amazon-cognito-for-federated-authentication-using-azure-ad/

upvoted 1 times

sangavi_vijay 1 year, 11 months ago

Selected Answer: B

why its not b?

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: D

Use Amazon Cognito via SAML integration. (SAML) is an open federation standard that allows an identity provider (for this case on-prem AD) to authenticate users and pass identity and security information about them to a service provider (for this case AWS).

I will settle for D, because this is definitely required for this to work.

upvoted 5 times

NickGordon 2 years, 1 month ago

Selected Answer: D

D.

An Amazon Cognito user pool is a user directory for WEB and MOBILE app authentication and authorization. So it is not a best option for corporate users.

upvoted 3 times

potomac 2 years, 1 month ago

Selected Answer: D

I think it is D

upvoted 2 times

ahlofan 2 years, 1 month ago

Selected Answer: B

Access to Aws resource -> cognito, then use iam role
SAML or AD -> identity pool

upvoted 2 times

pentium75 1 year, 11 months ago

Cognito is for app users, to authenticate users accessing your apps. Cognito is NOT for granting access to AWS resources.

upvoted 3 times

dilaaziz 2 years, 1 month ago

Selected Answer: D

https://aws.amazon.com/identity/saml/

upvoted 2 times

Question #625 *Topic 1*

A company is hosting a website behind multiple Application Load Balancers. The company has different distribution rights for its content around the world. A solutions architect needs to ensure that users are served the correct content without violating distribution rights.

Which configuration should the solutions architect choose to meet these requirements?

A. Configure Amazon CloudFront with AWS WAF.

B. Configure Application Load Balancers with AWS WAF

C. Configure Amazon Route 53 with a geolocation policy **Most Voted**

D. Configure Amazon Route 53 with a geoproximity routing policy

Correct Answer: C

Community vote distribution

C (77%) A (23%)

Comments

potomac Highly Voted 2 years, 7 months ago

Selected Answer: C

Geolocation routing policy — Use when you want to route traffic based on the location of users.

Geo-proximity routing policy — Use when you want to route traffic based on the location of your resources and optionally switch resource traffic at one location to resources elsewhere.

upvoted 14 times

MatAlves 1 year, 8 months ago

"Restrict the geographic distribution of your content
You can use geographic restrictions, sometimes known as geo blocking, to prevent users in specific geographic locations from accessing content that you're distributing through an Amazon CloudFront distribution."

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/georestrictions.html

upvoted 2 times

MatAlves 1 year, 8 months ago

"Violation to distribution rights" is something you want to handle seriously by actually blocking the access. This can only be accomplished by "OPTION A".

upvoted 1 times

pentium75 2 years, 5 months ago

DNS routing can be easily bypassed, and just routing traffic from different countries to different endpoints does still not restrict what each country can see. It's clearly A.

upvoted 4 times

loverduck 1 year, 9 months ago

They're not talking about detail like your case

upvoted 1 times

upliftinghut Highly Voted 2 years, 4 months ago

Selected Answer: C

"You can also use geolocation routing to restrict distribution of content to only the locations in which you have distribution rights"
Link: https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-geo.html

upvoted 8 times

EllenLiu Most Recent 1 year, 5 months ago

Selected Answer: C

A is incorrect, what is config cloudfront with waf? should be WAF geo match statement only

upvoted 1 times

bujuman 1 year, 7 months ago

Selected Answer: C

When you use geolocation routing, you can localize your content and present some or all of your website in the language of your users. You can alsc use geolocation routing to restrict distribution of content to only the locations in which you have distribution rights. Another possible use is for balancing load across endpoints in a predictable, easy-to-manage way, so that each user location is consistently routed to the same endpoint.

upvoted 2 times

MatAlves 1 year, 8 months ago

Selected Answer: A

"Restrict the geographic distribution of your content
You can use geographic restrictions, sometimes known as geo blocking, to prevent users in specific geographic locations from accessing content tha you're distributing through an Amazon CloudFront distribution."

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/georestrictions.html

upvoted 1 times

MatAlves 1 year, 8 months ago

Actually, it also says:

"You can also use geolocation routing to restrict distribution of content to only the locations in which you have distribution rights."

It can be either C or A, but C seems to address the question wording better.

upvoted 2 times

JA2018 1 year, 6 months ago

Agreed

upvoted 1 times

Johnoppong101 1 year, 10 months ago

Selected Answer: C

Check line 6 on this documentation.
...You can also use geolocation routing to restrict distribution of content to only the locations in which you have distribution rights.
https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-geo.html

upvoted 2 times

Lin878 1 year, 11 months ago

Selected Answer: A

Route 53 can only restrict for geolocation users in this case, it's not for contents. I vote for "A".

upvoted 2 times

FZBianco 2 years ago

Answer is A.

upvoted 1 times

xBUGx 2 years, 3 months ago

Selected Answer: A

I vote for A

upvoted 1 times

Ravan 2 years, 3 months ago

Selected Answer: C

. Configure Amazon Route 53 with a geolocation policy.

By configuring Amazon Route 53 with a geolocation policy, the solutions architect can direct users to different Application Load Balancers based on their geographical location. This allows the company to serve the correct content to users in different regions without violating distribution rights. Geolocation routing policies enable you to route traffic based on the geographic location of your users, ensuring that users are directed to the nearest or most appropriate endpoint based on their location. This solution is suitable for scenarios where content distribution rights vary by region and need to be enforced accordingly.

upvoted 4 times

Pics00094 2 years, 3 months ago

Selected Answer: A

I think it's A

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

WAF for filtering web traffic based on rules. In this case it may be IP address, geolocation, region. CloudFront for global distribution.

B: Just balances and does not filter
CD: Connects the user to the NEAREST server which is not same as AUTHORISED content

upvoted 1 times

awsgeek75 2 years, 4 months ago

WAF for geo filtering can be configured like this:
https://docs.aws.amazon.com/waf/latest/developerguide/waf-rule-statement-type-geo-match.html

How CloudFront integrates with your WAF rules:
https://docs.aws.amazon.com/waf/latest/developerguide/cloudfront-features.html

upvoted 2 times

Marco_St 2 years, 5 months ago

Selected Answer: A

distributions + restriction of content deleivery target = A

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: A

We want to restrict access by country. People in Spain are allowed to access certain content while people in Portugal are not. A Route 53 geolocation policy that returns the "nearest" endpoint will not help, because a) the "nearest" endpoint could be identical for multiple countries with different distribution rights and b) it could easily be bypassed.

upvoted 3 times

Salilgen 1 year, 5 months ago

A doesn't talk about geographic restrictions

upvoted 1 times

master9 2 years, 5 months ago

Selected Answer: A

AWS CloudFront supports geographic restrictions, also known as geo-blocking, which can be used to control the distribution of your content based on the geographic location of your viewers.

You can use the CloudFront geographic restrictions feature to either grant permission to your users to access your content only if they're in one of the approved countries on your allowlist, or prevent your users from accessing your content if they're in one of the banned countries on your denylist.

For example, if a request comes from a country where you are not authorized to distribute your content, you can use CloudFront geographic restrictions to block the request.

upvoted 1 times

pentium75 2 years, 5 months ago

Edit: Even though you can specify DNS targets by country, this will not help.

upvoted 2 times

Murtadhaceit 2 years, 5 months ago

Selected Answer: C

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-geo.html

upvoted 4 times

ekisako 2 years, 6 months ago

Selected Answer: A

https://repost.aws/knowledge-center/cloudfront-geo-restriction

upvoted 1 times

Question #626 *Topic 1*

A company stores its data on premises. The amount of data is growing beyond the company's available capacity.

The company wants to migrate its data from the on-premises location to an Amazon S3 bucket. The company needs a solution that will automatically validate the integrity of the data after the transfer.

Which solution will meet these requirements?

A. Order an AWS Snowball Edge device. Configure the Snowball Edge device to perform the online data transfer to an S3 bucket

B. Deploy an AWS DataSync agent on premises. Configure the DataSync agent to perform the online data transfer to an S3 bucket. **Most Voted**

C. Create an Amazon S3 File Gateway on premises Configure the S3 File Gateway to perform the online data transfer to an S3 bucket

D. Configure an accelerator in Amazon S3 Transfer Acceleration on premises. Configure the accelerator to perform the online data transfer to an S3 bucket.

Correct Answer: B

Community vote distribution

B (100%)

Comments

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: B

During a transfer, AWS DataSync always checks the integrity of your data.

https://docs.aws.amazon.com/datasync/latest/userguide/configure-data-verification-options.html

upvoted 12 times

potomac Highly Voted 2 years, 7 months ago

Selected Answer: B

During a transfer, AWS DataSync always checks the integrity of your data, but you can specify how and when this verification happens with the following options: Verify only the data transferred (recommended) – DataSync calculates the checksum of transferred files and metadata at the source location.

https://docs.aws.amazon.com/datasync/latest/userguide/configure-data-verification-options.html

upvoted 6 times

KennethNg923 Most Recent 1 year, 11 months ago

Selected Answer: B

"automatically validate the integrity of the data after the transfer" -> so it should be datasync

upvoted 2 times

dilaaziz 2 years, 7 months ago

Selected Answer: B

https://aws.amazon.com/datasync/faqs/

upvoted 2 times

Question #627 Topic 1

A company wants to migrate two DNS servers to AWS. The servers host a total of approximately 200 zones and receive 1 million requests each day on average. The company wants to maximize availability while minimizing the operational overhead that is related to the management of the two servers.

What should a solutions architect recommend to meet these requirements?

A. Create 200 new hosted zones in the Amazon Route 53 console Import zone files. Most Voted

B. Launch a single large Amazon EC2 instance Import zone tiles. Configure Amazon CloudWatch alarms and notifications to alert the company about any downtime.

C. Migrate the servers to AWS by using AWS Server Migration Service (AWS SMS). Configure Amazon CloudWatch alarms and notifications to alert the company about any downtime.

D. Launch an Amazon EC2 instance in an Auto Scaling group across two Availability Zones. Import zone files. Set the desired capacity to 1 and the maximum capacity to 3 for the Auto Scaling group. Configure scaling alarms to scale based on CPU utilization.

Correct Answer: A

Community vote distribution

A (95%) 5%

Comments

awsgeek75 Highly Voted 1 year, 11 months ago

Selected Answer: A

Key requirement it "maximize availability while minimizing the operational overhead" of 200 zones to process million requests

R53 is designed exactly to do this and supports zone import functionality so literally does the job of their EC2 servers but much better so BCD become "overhead" by default. I doubt D will work.

upvoted 5 times

pentium75 Most Recent 1 year, 11 months ago

Selected Answer: A

B, C and D would not "maximize availability" (not HA) and also not minimize the operational overhead.

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: A

'maximize availability while minimizing the operational overhead' = severless = Amazon Route 53

upvoted 4 times

EdenWang 2 years ago

Selected Answer: A

Only A makes sense

upvoted 3 times

NickGordon 2 years, 1 month ago

Selected Answer: A

Should be A

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/migrate-dns-domain-in-use.html

upvoted 4 times

potomac 2 years, 1 month ago

Selected Answer: D

D makes more sense to me

upvoted 1 times

awsgeek75 1 year, 10 months ago

1 EC2 server for millions of requests?

upvoted 2 times

pentium75 1 year, 11 months ago

No, "Desired capacity 1" meaning that usually only 1 server would run, but they want to "maximize availability". And operating EC2 servers would not be "minimizing the operational overhead that is related to the management of the two servers."

upvoted 3 times

Question #628 Topic 1

A global company runs its applications in multiple AWS accounts in AWS Organizations. The company's applications use multipart uploads to upload data to multiple Amazon S3 buckets across AWS Regions. The company wants to report on incomplete multipart uploads for cost compliance purposes.

Which solution will meet these requirements with the LEAST operational overhead?

A. Configure AWS Config with a rule to report the incomplete multipart upload object count.

B. Create a service control policy (SCP) to report the incomplete multipart upload object count.

C. Configure S3 Storage Lens to report the incomplete multipart upload object count. Most Voted

D. Create an S3 Multi-Region Access Point to report the incomplete multipart upload object count.

Correct Answer: C

Community vote distribution

C (100%)

Comments

warp Highly Voted 2 years, 1 month ago

Selected Answer: C

S3 storage lenses can be used to find incomplete multipart uploads: https://aws.amazon.com/blogs/aws-cloud-financial-management/discovering-and-deleting-incomplete-multipart-uploads-to-lower-amazon-s3-costs/

upvoted 7 times

LocNV Highly Voted 1 year, 11 months ago

Selected Answer: C

S3 Storage Lens provides four Cost Efficiency metrics for analyzing incomplete multipart uploads in your S3 buckets. These metrics are free of charge and automatically configured for all S3 Storage Lens dashboards.

Incomplete Multipart Upload Storage Bytes – The total bytes in scope with incomplete multipart uploads
% Incomplete MPU Bytes – The percentage of bytes in scope that are results of incomplete multipart uploads
Incomplete Multipart Upload Object Count – The number of objects in scope that are incomplete multipart uploads
% Incomplete MPU Objects – The percentage of objects in scope that are incomplete multipart uploads
https://aws.amazon.com/blogs/aws-cloud-financial-management/discovering-and-deleting-incomplete-multipart-uploads-to-lower-amazon-s3-costs/

upvoted 5 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: C

ABD cannot do any of this so C is the right product for this use case

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: C

Amazon S3 Storage Lens is a cloud storage analytics solution with support for AWS Organizations to give you organization-wide visibility into object storage, with point-in-time metrics and trend lines as well as actionable recommendations.

upvoted 4 times

potomac 2 years, 1 month ago

Selected Answer: C

C for sure

upvoted 2 times

Question #629 Topic 1

A company runs a production database on Amazon RDS for MySQL. The company wants to upgrade the database version for security compliance reasons. Because the database contains critical data, the company wants a quick solution to upgrade and test functionality without losing any data.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an RDS manual snapshot. Upgrade to the new version of Amazon RDS for MySQL.

B. Use native backup and restore. Restore the data to the upgraded new version of Amazon RDS for MySQL.

C. Use AWS Database Migration Service (AWS DMS) to replicate the data to the upgraded new version of Amazon RDS for MySQL.

D. Use Amazon RDS Blue/Green Deployments to deploy and test production changes. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: D

A blue/green deployment copies a production database environment to a separate, synchronized staging environment. You can make changes to the database in the staging environment without affecting the production environment. When you are ready, you can promote the staging environment to be the new production database environment, with downtime typically under one minute.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/blue-green-deployments.html

upvoted 11 times

warp Highly Voted 2 years, 7 months ago

Selected Answer: D

You can make changes to the RDS DB instances in the green environment without affecting production workloads. For example, you can upgrade the major or minor DB engine version, upgrade the underlying file system configuration, or change database parameters in the staging environment. You can thoroughly test changes in the green environment.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/blue-green-deployments-overview.html

upvoted 6 times

cjace Most Recent 1 year, 12 months ago

Option A (Create an RDS manual snapshot and upgrade) is the most straightforward and least operationally intensive method to upgrade your Amazon RDS for MySQL instance while ensuring data safety and allowing thorough testing of application functionality post-upgrade. This approach leverages RDS's snapshot capabilities to provide a reliable rollback mechanism if needed, making it the recommended choice for your scenario.

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

Least overhead, only CD qualify and D is actually a managed solution for what is being proposed (hopefully) in C so it's better.

upvoted 4 times

foha2012 2 years, 5 months ago

C works for me

upvoted 1 times

potomac 2 years, 7 months ago

Selected Answer: D

D is the answer

upvoted 1 times

Question #630 *Topic 1*

A solutions architect is creating a data processing job that runs once daily and can take up to 2 hours to complete. If the job is interrupted, it has to restart from the beginning.

How should the solutions architect address this issue in the MOST cost-effective manner?

A. Create a script that runs locally on an Amazon EC2 Reserved Instance that is triggered by a cron job.

B. Create an AWS Lambda function triggered by an Amazon EventBridge scheduled event.

C. Use an Amazon Elastic Container Service (Amazon ECS) Fargate task triggered by an Amazon EventBridge scheduled event. **Most Voted**

D. Use an Amazon Elastic Container Service (Amazon ECS) task running on Amazon EC2 triggered by an Amazon EventBridge scheduled event.

Correct Answer: C

Community vote distribution

C (91%) 9%

Comments

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: C

A: Nonsense
B: Lambda max running time is 15 mins
D: EC2 is more expensive than Fargate for 2 hours duration as EC2 instance will be billed.

upvoted 11 times

awsgeek75 2 years, 4 months ago

A is also nonsense because an EC2 reserved instance will cost the most for the period when the 2 hour job is not running!

upvoted 4 times

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: C

Not B because of running time

upvoted 5 times

KennethNg923 Most Recent 1 year, 11 months ago

Selected Answer: C

EC2 expensive than Fargate or Lambda, but Lambda has 15 mins limit, so only could choose Fargate for micro service which is C.
I think no one will create script for that by the way

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

AWS Fargate will bill you based on the amount of vCPU, RAM, OS, CPU architecture, and storage that your containerized apps consume while running on EKS or ECS.

upvoted 2 times

cevin93 2 years, 6 months ago

Selected Answer: C

should be C

upvoted 3 times

Alex1atd 2 years, 6 months ago

Selected Answer: C

Lambda function have a limit timeout about 15 minutes, so cannot be B.
Answer is C

upvoted 3 times

hungta 2 years, 6 months ago

Selected Answer: C

Lamda function have a limit timeout about 15 minutes

upvoted 2 times

cciesam 2 years, 7 months ago

Selected Answer: B

I think it should be B. Considering the Cost.

upvoted 3 times

zhdetn 2 years, 6 months ago

Lambda Maximum execution time: 900 seconds (15 minutes)

upvoted 6 times

Murtadhaceit 2 years, 5 months ago

Lambda times out after 15 minutes. This job requires a 2-hour time without interruption block. So, definitely not B.

upvoted 5 times

potomac 2 years, 7 months ago

Selected Answer: C

I guess it is C

upvoted 3 times

Question #631 *Topic 1*

A social media company wants to store its database of user profiles, relationships, and interactions in the AWS Cloud. The company needs an application to monitor any changes in the database. The application needs to analyze the relationships between the data entities and to provide recommendations to users.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Neptune to store the information. Use Amazon Kinesis Data Streams to process changes in the database.

B. Use Amazon Neptune to store the information. Use Neptune Streams to process changes in the database. **Most Voted**

C. Use Amazon Quantum Ledger Database (Amazon QLDB) to store the information. Use Amazon Kinesis Data Streams to process changes in the database.

D. Use Amazon Quantum Ledger Database (Amazon QLDB) to store the information. Use Neptune Streams to process changes in the database.

Correct Answer: B

Community vote distribution

B (88%) 12%

Comments

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: B

Relationships between entities = Graph data = Neptune

upvoted 8 times

awsgeek75 2 years, 5 months ago

Also, Neptune Streams can monitor changes in the data and create a changelog

upvoted 7 times

MatAlves 1 year, 8 months ago

I come looking for one of you guys comments. Nice.

upvoted 4 times

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: B

Amazon Neptune Database is a serverless graph database designed for superior scalability and availability. Neptune Database provides built-in security, continuous backups, and integrations with other AWS services. Suitable for social media. With the Neptune Streams feature, you can generate a complete sequence of change-log entries that record every change made to your graph data as it happens.

upvoted 7 times

FlyingHawk Most Recent 1 year, 4 months ago

Selected Answer: B

Neptune is optimized for storing billions of relationships and querying the graph with milliseconds latence. Neptune Streams logs every change to your graph as it happens, in the order that it is made, in a fully managed way.

Amazon Quantum Ledger Database (QLDB) is a purpose-built ledger database that provides a complete and cryptographically verifiable history of al changes made to your application data.

upvoted 1 times

KennethNg923 1 year, 11 months ago

Selected Answer: B

Normally Amazon Quantum Ledger Database use in blockchain DB more. So i will go for B using Neptune and Neptune Stream for relationship between entities.

upvoted 2 times

haci 2 years, 3 months ago

Selected Answer: C

Amazon QLDB tracks and maintains a sequential history of every application data change using an immutable and transparent log. It trusts the integrity of your data. Built-in cryptographic authentication provides third-party verification of data changes. QLDB ACID transactions can create accurate, event-driven systems with support for real-time streaming to Amazon Kinesis.

upvoted 1 times

NickGordon 2 years, 7 months ago

Selected Answer: B

B

Social network -> Graph Structure -> Neptune

upvoted 3 times

ekisako 2 years, 7 months ago

Selected Answer: B

Keyword: analyze the relationships
With Amazon Neptune, you can create sophisticated, interactive graph applications that can query billions of relationships in milliseconds.

https://aws.amazon.com/neptune/features/

upvoted 5 times

potomac 2 years, 7 months ago

Selected Answer: C

Amazon Neptune is primarily used for managing highly connected graph data, making it well-suited for graph-based applications.

In contrast, Amazon QLDB is designed for applications that require an immutable and auditable transaction history to ensure data integrity.

upvoted 3 times

pentium75 2 years, 5 months ago

Exactly, thus B. "Relationships between the data entities" is "graph data".

upvoted 2 times

warp 2 years, 7 months ago

Selected Answer: B

Neptune is a graph type database and Neptune streams provides view on changes into the database:
https://docs.aws.amazon.com/neptune/latest/userguide/streams.html

upvoted 3 times

AF_1221 2 years, 7 months ago

C is the correct answer
provides a well-suited, managed, and scalable solution for storing and monitoring the database with the least operational overhead, meeting the requirements of the social media company.

upvoted 2 times

awsgeek75 2 years, 4 months ago

AQLB is like a blockchain database. Are you sure this is the correct option for graph data?

upvoted 2 times

Question #632 *Topic 1*

A company is creating a new application that will store a large amount of data. The data will be analyzed hourly and will be modified by several Amazon EC2 Linux instances that are deployed across multiple Availability Zones. The needed amount of storage space will continue to grow for the next 6 months.

Which storage solution should a solutions architect recommend to meet these requirements?

A. Store the data in Amazon S3 Glacier. Update the S3 Glacier vault policy to allow access to the application instances.

B. Store the data in an Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume on the application instances.

C. Store the data in an Amazon Elastic File System (Amazon EFS) file system. Mount the file system on the application instances. **Most Voted**

D. Store the data in an Amazon Elastic Block Store (Amazon EBS) Provisioned IOPS volume shared between the application instances.

Correct Answer: C

Community vote distribution

C (100%)

Comments

TariqKipkemei Highly Voted 1 year, 6 months ago

Selected Answer: C

Multiple Linux instances = Amazon Elastic File System (Amazon EFS) with multiple mount targets.

upvoted 6 times

AF_1221 Highly Voted 1 year, 7 months ago

C is correct

Shared File System: Amazon EFS allows multiple Amazon EC2 instances to mount the same file system simultaneously, making it easy for multiple instances to access and modify the data concurrently.

upvoted 5 times

Salilgen Most Recent 1 year, 5 months ago

Selected Answer: C

S3 Glacier (flexible retrieval) is not compatible with hourly job

EBS can be shared only in same AZ (B and D are out)

upvoted 2 times

potomac 1 year, 7 months ago

Selected Answer: C

C is correct

upvoted 4 times

Question #633 *Topic 1*

A company manages an application that stores data on an Amazon RDS for PostgreSQL Multi-AZ DB instance. Increases in traffic are causing performance problems. The company determines that database queries are the primary reason for the slow performance.

What should a solutions architect do to improve the application's performance?

A. Serve read traffic from the Multi-AZ standby replica.

B. Configure the DB instance to use Transfer Acceleration.

C. Create a read replica from the source DB instance. Serve read traffic from the read replica. **Most Voted**

D. Use Amazon Kinesis Data Firehose between the application and Amazon RDS to increase the concurrency of database requests.

Correct Answer: C

Community vote distribution

C (100%)

Comments

TariqKipkemei Highly Voted 2 years ago

Selected Answer: C

A Multi-AZ DB instance Creates a primary DB instance with one standby DB instance in a different Availability Zone. Using a Multi-AZ DB instance provides high availability, but the standby DB instance doesn't support connections for read workloads.
Therefore you will need to create a read replica from the source DB instance then serve read traffic from the read replica.

upvoted 7 times

awsgeek75 Highly Voted 1 year, 11 months ago

Selected Answer: C

Read replica split for read traffic will distribute the overall load and improve the performance.

A: Standby replica cannot serve traffic (Correct me if I am wrong here)
B: Transfer Accelerator is to speed up S3 traffic. Not the case here
D: Kiensis will increase concurrency but won't solve the DB performance issues

upvoted 5 times

potomac Most Recent 2 years, 1 month ago

Selected Answer: C

you can't read from the standby DB instance. If applications require more read capacity, you should create or add additional read replicas.

upvoted 3 times

warp 2 years, 1 month ago

Selected Answer: C

After you create a read replica from a source DB instance, the source becomes the primary DB instance. When you make updates to the primary DB instance, Amazon RDS copies them asynchronously to the read replica.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReadRepl.html

upvoted 4 times

Question #634 *Topic 1*

A company collects 10 GB of telemetry data daily from various machines. The company stores the data in an Amazon S3 bucket in a source data account.

The company has hired several consulting agencies to use this data for analysis. Each agency needs read access to the data for its analysts. The company must share the data from the source data account by choosing a solution that maximizes security and operational efficiency.

Which solution will meet these requirements?

A. Configure S3 global tables to replicate data for each agency.

B. Make the S3 bucket public for a limited time. Inform only the agencies.

C. Configure cross-account access for the S3 bucket to the accounts that the agencies own. Most Voted

D. Set up an IAM user for each analyst in the source data account. Grant each user access to the S3 bucket.

Correct Answer: C

Community vote distribution

C (88%) 12%

Comments

pentium75 Highly Voted 1 year, 11 months ago

Selected Answer: C

A doesn't exist
B is a big "hell no"
D is a bad practice, even with IAM you'd use groups

upvoted 9 times

xBUGx Highly Voted 1 year, 9 months ago

What if other agencies don't have an aws account?

upvoted 8 times

chickenmf 1 year, 9 months ago

then we politely tell them "no."

upvoted 10 times

In2718 Most Recent 10 months, 3 weeks ago

Selected Answer: D

The question does not specify if the agencies have AWS accounts. This does not mean that C is the correct answer. A terribly written question, when all they had to say was agencies need access from their AWS accounts, otherwise the real correct answer is D>

upvoted 2 times

awsgeek75 1 year, 11 months ago

Selected Answer: C

Others have given reason by ABD are wrong. In case you need it, here is an AWS example exercise of understanding option C

https://docs.aws.amazon.com/AmazonS3/latest/userguide/example-walkthroughs-managing-access-example2.html

upvoted 5 times

TariqKipkemei 2 years ago

Selected Answer: C

With cross-account bucket permissions Account A—can grant another AWS account, Account B, permission to access its resources such as buckets and objects. Account B can then delegate those permissions to users in its account.

https://docs.aws.amazon.com/AmazonS3/latest/userguide/example-walkthroughs-managing-access-example2.html#:~:text=4%3A%20Clean%20up-,An%20AWS%20account,-%E2%80%94for%20example%2C%20Account

upvoted 4 times

NickGordon 2 years, 1 month ago

Selected Answer: C

C is the best answer

upvoted 3 times

cciesam 2 years, 1 month ago

Selected Answer: D

C may not correct as it's doesn't say if the analyst are using AWS services

upvoted 1 times

NickGordon 2 years, 1 month ago

in that case, an analyst user group should be created and the access should be assigned to the group. So C is better

upvoted 3 times

awsgeek75 1 year, 11 months ago

"consulting agencies" means some companies which may have one or more analysts each. Making IAM users for each individual to manage permissions is not well-architected. You would at least create groups and assign it to users.
D will work as it is possible but it won't minimize "operational efficiency"

upvoted 2 times

potomac 2 years, 1 month ago

Selected Answer: C

I think it is C

upvoted 2 times

Question #635 Topic 1

A company uses Amazon FSx for NetApp ONTAP in its primary AWS Region for CIFS and NFS file shares. Applications that run on Amazon EC2 instances access the file shares. The company needs a storage disaster recovery (DR) solution in a secondary Region. The data that is replicated in the secondary Region needs to be accessed by using the same protocols as the primary Region.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an AWS Lambda function to copy the data to an Amazon S3 bucket. Replicate the S3 bucket to the secondary Region.

B. Create a backup of the FSx for ONTAP volumes by using AWS Backup. Copy the volumes to the secondary Region. Create a new FSx for ONTAP instance from the backup.

C. Create an FSx for ONTAP instance in the secondary Region. Use NetApp SnapMirror to replicate data from the primary Region to the secondary Region. Most Voted

D. Create an Amazon Elastic File System (Amazon EFS) volume. Migrate the current data to the volume. Replicate the volume to the secondary Region.

Correct Answer: C

Community vote distribution

C (100%)

Comments

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: C

This is a very rare usage scenario so here are the docs related to the product:
https://docs.aws.amazon.com/fsx/latest/ONTAPGuide/scheduled-replication.html

AD: Not compatible solutions
B: Either wrongly worded or missing something but if I read it correctly, it means just take a backup and restore whereas the question is about continuous replication. If B was scheduled then it would have made sense
C is correct as SnapMirror is a managed solution to replicate the data

upvoted 9 times

KennethNg923 1 year, 11 months ago

Agree, thus it said "replicated in the secondary Region" only, not create a backup infrastructure in other region for failover, i don't think we need to use Backup feature

upvoted 2 times

pentium75 Most Recent 2 years, 5 months ago

Selected Answer: C

Not A, no access with CIFS (SMB) or NFS
Not B, one-time copy
Not D, EFS does not offer SMB

upvoted 3 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

C
https://aws.amazon.com/blogs/storage/cross-region-disaster-recovery-with-amazon-fsx-for-netapp-ontap/

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

Amazon FSx for NetApp ONTAP supports NetApp SnapMirror, a replication technology that you can use to replicate data between two ONTAP file systems. You can configure automatic NetApp SnapMirror replication of your data to another Amazon FSx for NetApp ONTAP file system, including a file system in another AWS Region. If needed, you can fail over your applications and users to use the other Amazon FSx for NetApp ONTAP file system. With SnapMirror, you can configure replication with a Recovery Point Objective (RPO) of as low as 5 minutes, and a Recovery Time Objective (RTO) in single-digit minutes. You can configure SnapMirror using the ONTAP CLI or REST API.

upvoted 3 times

Oblako 2 years, 6 months ago

Selected Answer: C

SnapMirror enables you to configure replication with an RPO of as low as five minutes, and an RTO in single digit minutes. It is the recommended solution for DR when using FSx for ONTAP: https://aws.amazon.com/blogs/storage/cross-region-disaster-recovery-with-amazon-fsx-for-netapp-ontap/

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: C

You can use NetApp SnapMirror to schedule periodic replication of your FSx for ONTAP file system to or from a second file system. This capability is available for both in-Region and cross-Region deployments.

https://docs.aws.amazon.com/fsx/latest/ONTAPGuide/scheduled-replication.html

upvoted 3 times

Question #636 *Topic 1*

A development team is creating an event-based application that uses AWS Lambda functions. Events will be generated when files are added to an Amazon S3 bucket. The development team currently has Amazon Simple Notification Service (Amazon SNS) configured as the event target from Amazon S3.

What should a solutions architect do to process the events from Amazon S3 in a scalable way?

A. Create an SNS subscription that processes the event in Amazon Elastic Container Service (Amazon ECS) before the event runs in Lambda.

B. Create an SNS subscription that processes the event in Amazon Elastic Kubernetes Service (Amazon EKS) before the event runs in Lambda

C. Create an SNS subscription that sends the event to Amazon Simple Queue Service (Amazon SQS). Configure the SOS queue to trigger a Lambda function. Most Voted

D. Create an SNS subscription that sends the event to AWS Server Migration Service (AWS SMS). Configure the Lambda function to poll from the SMS event.

Correct Answer: C

Community vote distribution

C (100%)

Comments

potomac Highly Voted 2 years, 7 months ago

Selected Answer: C

Amazon SQS is designed for event-driven and scalable message processing. It can handle large volumes of messages and automatically scales based on the incoming workload. This allows for better load distribution and scaling as compared to direct Lambda invocation.

upvoted 6 times

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: C

scalable service = serverless = Amazon SQS implemented with FAN-OUT.
However SQS is a pull based event distribution service, it does not trigger other services.
C is the closest option.

upvoted 5 times

KennethNg923 Most Recent 1 year, 11 months ago

Selected Answer: C

SQS + SNS standard solution

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

AB are way too complicated to scale without more specifics (no idea about number of events)
D SMS is not for this, it's for server migrations
C SNS is notified on file creation in S3. SNS publishes to SQS which can scale according to the input load automatically. Lambda execution can scale a lot when attached to SQS.

ABC have scaling limits each but C's scaling limit is much better than AB

upvoted 3 times

Question #637 Topic 1

A solutions architect is designing a new service behind Amazon API Gateway. The request patterns for the service will be unpredictable and can change suddenly from 0 requests to over 500 per second. The total size of the data that needs to be persisted in a backend database is currently less than 1 GB with unpredictable future growth. Data can be queried using simple key-value requests.

Which combination ofAWS services would meet these requirements? (Choose two.)

A. AWS Fargate
B. AWS Lambda Most Voted
C. Amazon DynamoDB Most Voted
D. Amazon EC2 Auto Scaling
E. MySQL-compatible Amazon Aurora

Correct Answer: BC

Community vote distribution

BC (100%)

Comments

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: BC

Scalable, unpredictable request patterns = AWS Lambda
Scalable, key-value data = Amazon DynamoDB

upvoted 11 times

potomac Highly Voted 2 years, 7 months ago

Selected Answer: BC

B and C

upvoted 11 times

KennethNg923 Most Recent 1 year, 11 months ago

Selected Answer: BC

Auto Scaling cannot handle "suddenly from 0 requests to over 500 per second", use Lambda and Dynamo which for Key-value pair.

upvoted 2 times

Phi143 2 years, 2 months ago

Why not AC? Size of the data has unpredictable future growth and Lambda may not be able to handle it.

upvoted 2 times

JA2018 1 year, 6 months ago

Lambda is fully managed by AWS, fully scalable to meet actual spikes in demand

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: BC

Unpredictable scaling of API load = Lambda + SPI Gateway
Unpredictable growth of key/value DB = DynamoDB

Fargate behind API requires EKS/ECS setup which is not suitable for 0-500 varying load. Same with EC2 autoscaling.
Aurora MySQL is not ideal for key/value and is better suited for relational databases

upvoted 4 times

Ashhher 2 years, 5 months ago

Selected Answer: BC

why not Fargate?

upvoted 3 times

Question #638 *Topic 1*

A company collects and shares research data with the company's employees all over the world. The company wants to collect and store the data in an Amazon S3 bucket and process the data in the AWS Cloud. The company will share the data with the company's employees. The company needs a secure solution in the AWS Cloud that minimizes operational overhead.

Which solution will meet these requirements?

A. Use an AWS Lambda function to create an S3 presigned URL. Instruct employees to use the URL. **Most Voted**

B. Create an IAM user for each employee. Create an IAM policy for each employee to allow S3 access. Instruct employees to use the AWS Management Console.

C. Create an S3 File Gateway. Create a share for uploading and a share for downloading. Allow employees to mount shares on their local computers to use S3 File Gateway.

D. Configure AWS Transfer Family SFTP endpoints. Select the custom identity provider options. Use AWS Secrets Manager to manage the user credentials Instruct employees to use Transfer Family.

Correct Answer: A

Community vote distribution

A (37%) C (37%) D (25%)

Comments

t0nx Highly Voted 2 years, 6 months ago

Selected Answer: D

AWS Transfer Family (Option D)

By configuring AWS Transfer Family SFTP endpoints, you can provide a secure and convenient way for employees to access and transfer data to and from the S3 bucket.
Using custom identity provider options allows you to integrate with existing identity systems, and AWS Secrets Manager can be used to manage use credentials securely.

A suggests using an AWS Lambda function to create an S3 presigned URL. While this can work, it involves manual generation of URLs and sharing them, which may not be as scalable or user-friendly.

B suggests creating an IAM user for each employee with IAM policies for S3 access. This involves more operational overhead, as managing IAM users for each employee can be cumbersome and less scalable.

C suggests using an S3 File Gateway. While this can work, it introduces additional components and may not be as straightforward or as efficient as using AWS Transfer Family for SFTP access.

upvoted 16 times

pentium75 2 years, 5 months ago

"Use AWS Secrets Manager to manage the user credentials", so manage separate credentials for every user in Secrets Manager? And "instruct employees to use Transfer Family", actually Transfer Family is the server component, employees would use an SFTP client.

upvoted 8 times

xxichlas 1 year, 11 months ago

https://docs.aws.amazon.com/secretsmanager/latest/userguide/integrating_FTPlong.html

upvoted 2 times

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: C

Not A - S3 presigned URLs are temporary (max. 7 days); you'd need to create a new URL at least every 7 days and "instruct employees" to use it. Definitely NOT 'minimizing operational overhead'.

Not B - "Instruct employees to use the AWS Management Console", using Management console to up- and download files is complex

Not D - Secrets Manager is not for managing user credentials, and employees would not "use Transfer Family", they would use an (S)FTP client to access the files.

C grants simple access for up/downloading, no operational overhead.

upvoted 13 times

awsgeek75 2 years, 5 months ago

Glad that someone else also sees what I see in this question!

upvoted 4 times

KennethNg923 1 year, 11 months ago

Agree, Use an AWS Lambda function to create an S3 presigned URL for 7 days limits, create URL every 7 days have operational overhead more than use Secret Manager

upvoted 5 times

bora4motion 1 year, 1 month ago

but then you will have to manage the S3 File Gateway in the long run, which could be a VM, patching etc. I am tempted to go with D.

upvoted 1 times

KarlosRC Most Recent 3 months ago

Selected Answer: D

A: Presigned URLs are typically used:
- for temporary access
- for a specific object
This is inconvenient for permanent file sharing.
B: Administrative overhead
D: Best choice
C: for hybrid scenarios, so that on-premises servers can see S3 as a network drive.

upvoted 1 times

In2718 10 months, 3 weeks ago

Selected Answer: D

I agree with t0nx - good explanation, good job. Answer D has the security, access both internal an internet, if some employees do not have AWS access, and is more secure than A, and less overhead than B. Not sure if D would be the answer on the test, but it would be more in line with doing this in real life.

upvoted 1 times

ODRAMIREZ 1 year ago

Selected Answer: C

I think is C

upvoted 1 times

Dz1990 1 year ago

Selected Answer: A

Option A is correct because:

Lowest operational overhead - just one Lambda function vs managing many users/systems
Secure - presigned URLs expire automatically
Simple for employees - just click a link, no special software needed
Works globally - URLs work from anywhere
Scalable - handles any number of employees easily

Think of it like this: Instead of giving everyone a key to your house (IAM users), you give them a temporary visitor pass (presigned URL) that expires after a set time.

Option D: Transfer Family SFTP

How it works: SFTP service with custom identity provider
Security: ✅ Secure
Operational overhead: ❌ MEDIUM-HIGH - managing SFTP endpoints, user credentials in Secrets Manager
Global access: ✅ Works globally

upvoted 1 times

srcntpc 1 year ago

Selected Answer: A

Why A is Correct:
S3 presigned URLs allow secure, time-limited access to S3 objects without needing to manage user credentials or IAM permissions for each employee.
AWS Lambda function can dynamically generate presigned URLs based on request parameters or business logic.
Simple, secure, and minimizes operational overhead.

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: A

Sharing S3 access with one who has no amazon credential is to provide presigned URL. C is not right becasue S3 File Gateway is for on-premise access(corporate) to S3.

upvoted 1 times

FlyingHawk 1 year, 5 months ago

Selected Answer: C

I will vote for C. For A, the S3 presigned URL is expired in 7 days, you need to implement a process to allow user send an access request, then validate the user, then trigger the lamdba to generate the resigned URL and send it back to the user. For D, using AWS secret manager to manage the user credentials does not sound practical unless it uses Active Directory.

upvoted 1 times

bora4motion 1 year, 1 month ago

With C, in the long run you will have to manage the s3 gateway which is more for on prem use. So this means you have to patch it, update it, reboo it etc. I am tempted to go with D. With pre-signed url you will have to lean on users to regenerate the links, D - yes, AD for auth and that's it.

upvoted 1 times

EllenLiu 1 year, 5 months ago

Selected Answer: A

S3 File Gateway is designed for hybrid cloud use cases where on-premises applications need to interact with S3. It introduces additional infrastructure complexity and costs, making it less suitable for the described requirements.

upvoted 1 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: C

A - Definitely not ideal. Presigned URLs have a maximum expiration time of 7 days, which might be limiting for ongoing or long-term data sharing. If employees require frequent or dynamic access to multiple files (which is likely the case because this is a research-type company), they'd need a new URL each time.

B - This is cumbersome and tiresome.

C - This is better, but still not ideal for global employees unless the gateway is highly distributed.

D - Secrets Manager stores sensitive information like passwords or API keys but is not a user directory, so they cannot "manage" credentials.

upvoted 2 times

Rhydian25 1 year, 11 months ago

Selected Answer: C

It is not operationally eficient to manage, for example, 1000 signed URLs or user credentials. In addition, it is sometimes dificult to instruct that many people.

It's easier to create an S3 File Gateway and allow the users to mount it locally to access the bucket.

It could be D if the answer said to use IAM roles instead of managing user credentials in Secrets Manager

upvoted 5 times

MandAsh 1 year, 11 months ago

Selected Answer: C

but they didnt mention access in for daily use of occasional.
If its occasional A works well but its permanant thing them mouting drive is solution.

upvoted 4 times

stalk98 2 years ago

Selected Answer: D

i think is d

upvoted 1 times

TwinSpark 2 years, 1 month ago

Selected Answer: C

Less operational overheade is C
https://docs.aws.amazon.com/filegateway/latest/files3/GettingStartedAccessFileShare.html
on client pc is easily mounted. I remain with some doubts but i will go for C

upvoted 3 times

alawada 2 years, 2 months ago

i would go with A

upvoted 1 times

seetpt 2 years, 3 months ago

Selected Answer: D

D seems right

upvoted 1 times

Question #639 Topic 1

A company is building a new furniture inventory application. The company has deployed the application on a fleet ofAmazon EC2 instances across multiple Availability Zones. The EC2 instances run behind an Application Load Balancer (ALB) in their VPC.

A solutions architect has observed that incoming traffic seems to favor one EC2 instance, resulting in latency for some requests.

What should the solutions architect do to resolve this issue?

A. Disable session affinity (sticky sessions) on the ALB **Most Voted**

B. Replace the ALB with a Network Load Balancer

C. Increase the number of EC2 instances in each Availability Zone

D. Adjust the frequency of the health checks on the ALB's target group

Correct Answer: A

Community vote distribution

A (88%) 13%

Comments

In2718 10 months, 3 weeks ago

Selected Answer: A

This is a terrible question. You don't just remove sticky sessions because of this issue, especially if the application relies on sticky sessions in the first place! Unfortunately no other answer is adequate, and you would not replace a network load balancer because you would lose some functionality that only the application load balancer has. This may be one of the worst questions so far.

upvoted 1 times

KennethNg923 1 year, 11 months ago

Selected Answer: A

"favor one EC2 instance" it is because you enable the sticky session feature, so you have to disable it

upvoted 2 times

1Alpha1 2 years, 4 months ago

Selected Answer: A

Answer: *A*
Enabling stickiness may bring imbalance to the load over the backend EC2 instances since sticky sessions help the same client to always redirect to the same instance behind a load balancer.

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

The question is too vague. Doesn't say much about the application or EC2 instance setup. So:
If you assume that application uses session management then A is correct.
If you think application is crashing then D is correct for health checks
If you don't assume anything about the application then B is also correct
SMH, I'll go with B... happy to discuss

upvoted 2 times

awsgeek75 2 years, 5 months ago

I'm not entirely happy with any choice but since others have chosen A, I'm just throwing B for discussion.

upvoted 2 times

MikeSWA 2 years, 5 months ago

what about c?
it actually helps distribute traffic equally across instances in all enabled AZs.

upvoted 2 times

mr123dd 2 years, 4 months ago

nope, if the sticky season is on, no matter how many instances you have in AZ or region, it will only send traffic to you favor session

upvoted 2 times

evelynsun 2 years, 5 months ago

it's A!!

Session affinity is a feature of the Application Load Balancer that keeps client requests on the same EC2 instance for the duration of the session. This can cause latency issues if one EC2 instance is overloaded while others are not, as the overloaded instance will handle all subsequent requests until it is taken offline.

To resolve this issue, the solutions architect should disable session affinity on the ALB. This can be done by setting the "Session affinity" parameter to "Off" in the ALB's configuration.

Disabling session affinity will cause the ALB to distribute requests across all EC2 instances in the target group, rather than keeping them on a single instance. This will help to balance the load and reduce latency for all requests.

upvoted 3 times

awsgeek75 2 years, 5 months ago

I agree with A but it assumes that ALB has session affinity enabled and app doesn't require it. What if the EC2 instances are running an application that requires session affinity? I think the question is missing some important context

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

Disable session affinity (sticky sessions) on the ALB

upvoted 2 times

NickGordon 2 years, 7 months ago

Selected Answer: A

A

https://repost.aws/knowledge-center/elb-fix-unequal-traffic-routing

upvoted 3 times

awsgeek75 2 years, 5 months ago

The same article says to check health of instances. This makes D as a good candidate too.
"Available healthy instances aren’t evenly distributed across Availability Zones."

upvoted 4 times

potomac 2 years, 7 months ago

Selected Answer: A

A makes more sense than others

upvoted 3 times

Question #640 Topic 1

A company has an application workflow that uses an AWS Lambda function to download and decrypt files from Amazon S3. These files are encrypted using AWS Key Management Service (AWS KMS) keys. A solutions architect needs to design a solution that will ensure the required permissions are set correctly.

Which combination of actions accomplish this? (Choose two.)

A. Attach the kms:decrypt permission to the Lambda function's resource policy

B. Grant the decrypt permission for the Lambda IAM role in the KMS key's policy **Most Voted**

C. Grant the decrypt permission for the Lambda resource policy in the KMS key's policy.

D. Create a new IAM policy with the kms:decrypt permission and attach the policy to the Lambda function.

E. Create a new IAM role with the kms:decrypt permission and attach the execution role to the Lambda function. **Most Voted**

Correct Answer: BE

Community vote distribution

BE (81%) | Other

Comments

NickGordon Highly Voted 2 years, 7 months ago

Selected Answer: BE

BE is right.

The key policy has to be modified to give lambda execution role access. You can't set another resource policy as principle. So C is not right

upvoted 9 times

ln2718 Most Recent 10 months, 3 weeks ago

Selected Answer: BD

I will agree with the explanations for B and D as the solution from the responses below.

upvoted 1 times

vaibhavianup09 1 year, 1 month ago

Selected Answer: BD

B. Grant the decrypt permission for the Lambda IAM role in the KMS key's policy

KMS key policies control who can use the key. The Lambda role must be explicitly allowed in the key policy for decryption to work.

D. Create a new IAM policy with the kms:decrypt permission and attach the policy to the Lambda function

The Lambda function (more precisely, its execution role) must have the kms:Decrypt permission in its IAM policy to invoke the KMS service.

upvoted 1 times

MaxMingxing 1 year, 5 months ago

BE, IAM role act as the agent between lambda function and KMS

upvoted 1 times

1166ae3 1 year, 11 months ago

Selected Answer: BD

E is wrong, AWS Lambda function can hold only one IAM role. This role is known as the execution role. What we should do is: creating an IAM policy that allows the kms:Decrypt action and attach it to the Lambda function's execution role.

upvoted 2 times

cjace 1 year, 12 months ago

B D - The combination of Option B (Grant the decrypt permission for the Lambda IAM role in the KMS key's policy) and Option D (Create a new IAM policy with the kms permission and attach the policy to the Lambda function) ensures that both the IAM role used by the Lambda function and the KMS key policy are correctly configured to allow decryption of the files. This setup meets the security requirements and ensures the Lambda function can perform its tasks without issues.

upvoted 1 times

wizcloudifa 2 years ago

Selected Answer: BE

when it comes to permissions look for the "IAM ROLE" word, lambda would need a role to decrypt the s3 object, only roles can be attached to a function not policies

upvoted 3 times

1Alpha1 2 years, 4 months ago

Selected Answer: BE

B. Grant the decrypt permission for the Lambda ***IAM ROLE*** in the KMS key's policy
E. Create a new ***IAM ROLE*** with the kms:decrypt permission and attach the execution role to the Lambda function.

upvoted 4 times

awsgeek75 2 years, 5 months ago

Selected Answer: BE

AC are resource policy, i.e. who can use lambda.
https://docs.aws.amazon.com/lambda/latest/dg/access-control-resource-based.html
D: The wording is confusing so it sort of sounds as if it is correct but you cannot attach a policy to a function.

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: BE

Not A and C because they are about function's "resource policy" which controls who can manage the function, NOT what the function can do.
Not D because you attach an IAM policy to an IAM principal, not to a Lambda function.

upvoted 4 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: BE

Create a new IAM role with the kms:decrypt permission and attach the execution role to the Lambda function then grant the decrypt permission for the Lambda IAM role in the KMS key's policy

upvoted 3 times

louisaok 2 years, 7 months ago

Selected Answer: CE

CE is right

upvoted 1 times

pentium75 2 years, 5 months ago

No, the "Lambda resource policy" is about who can manage the Lambda function

upvoted 2 times

potomac 2 years, 7 months ago

Selected Answer: DE

DE?
Create an IAM role for the Lambda function that also grants decryption permission to the S3 bucket.
Configure the IAM role as the Lambda functions execution role.

To use an IAM policy to control access to a KMS key, the key policy for the KMS key must give the account permission to use IAM policies.

https://repost.aws/knowledge-center/lambda-execution-role-s3-bucket
https://docs.aws.amazon.com/kms/latest/developerguide/iam-policies.html

upvoted 1 times

potomac 2 years, 7 months ago

change to CE
C. Grant the decrypt permission for the Lambda resource policy in the KMS key's policy.
E. Create a new IAM role with the kms:decrypt permission and attach the execution role to the Lambda function.

https://docs.aws.amazon.com/lambda/latest/dg/access-control-resource-based.html
https://docs.aws.amazon.com/kms/latest/developerguide/key-policies.html

upvoted 2 times

pentium75 2 years, 5 months ago

C is about the "Lambda resource policy", who can manage the function.

upvoted 2 times

Question #641 | *Topic 1*

A company wants to monitor its AWS costs for financial review. The cloud operations team is designing an architecture in the AWS Organizations management account to query AWS Cost and Usage Reports for all member accounts. The team must run this query once a month and provide a detailed analysis of the bill.

Which solution is the MOST scalable and cost-effective way to meet these requirements?

A. Enable Cost and Usage Reports in the management account. Deliver reports to Amazon Kinesis. Use Amazon EMR for analysis.

B. Enable Cost and Usage Reports in the management account. Deliver the reports to Amazon S3 Use Amazon Athena for analysis. Most Voted

C. Enable Cost and Usage Reports for member accounts. Deliver the reports to Amazon S3 Use Amazon Redshift for analysis.

D. Enable Cost and Usage Reports for member accounts. Deliver the reports to Amazon Kinesis. Use Amazon QuickSight tor analysis.

Correct Answer: B

Community vote distribution

B (100%)

Comments

NickGordon Highly Voted 1 year, 7 months ago

Selected Answer: B

B

https://aws.amazon.com/blogs/big-data/analyze-amazon-s3-storage-costs-using-aws-cost-and-usage-reports-amazon-s3-inventory-and-amazon-athena/

upvoted 6 times

LeonSauveterre Most Recent 1 year, 6 months ago

Selected Answer: B

A - EMR is powerful for large-scale data processing, but is overkill for a monthly query since neither Kinesis nor EMR is cost-effective for infrequent processing.
B - Cost and Usage Reports are already optimized for S3 delivery, and Athena directly integrates with S3.
C & D - Delivering reports for each member account creates redundant data. That mean extra time, extra work to do manually, extra money to cost, and extra account to manage.

upvoted 1 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: B

Scalable and cost-effective way = Enable Cost and Usage Reports in the management account. Deliver the reports to Amazon S3 Use Amazon Athen for analysis

upvoted 4 times

potomac 1 year, 7 months ago

Selected Answer: B

B
once a month

upvoted 4 times

Question #642 *Topic 1*

A company wants to run a gaming application on Amazon EC2 instances that are part of an Auto Scaling group in the AWS Cloud. The application will transmit data by using UDP packets. The company wants to ensure that the application can scale out and in as traffic increases and decreases.

What should a solutions architect do to meet these requirements?

A. Attach a Network Load Balancer to the Auto Scaling group. **Most Voted**

B. Attach an Application Load Balancer to the Auto Scaling group.

C. Deploy an Amazon Route 53 record set with a weighted policy to route traffic appropriately.

D. Deploy a NAT instance that is configured with port forwarding to the EC2 instances in the Auto Scaling group.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Sugarbear_01 Highly Voted 2 years, 7 months ago

Selected Answer: A

https://docs.aws.amazon.com/autoscaling/ec2/userguide/autoscaling-load-balancer.html

upvoted 11 times

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: A

UDP can only be monitored by NLB.
ALB is for application layer (HTTP etc)
R53 is DNS
NAT is for port forwarding/address translation etc which is not going to help with scaling

A is correct

upvoted 5 times

KennethNg923 Most Recent 1 year, 11 months ago

Selected Answer: A

UDP packets can scale out and in

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

UDP packets = Network Load Balancer

upvoted 3 times

Question #643 Topic 1

A company runs several websites on AWS for its different brands. Each website generates tens of gigabytes of web traffic logs each day. A solutions architect needs to design a scalable solution to give the company's developers the ability to analyze traffic patterns across all the company's websites. This analysis by the developers will occur on demand once a week over the course of several months. The solution must support queries with standard SQL.

Which solution will meet these requirements MOST cost-effectively?

A. Store the logs in Amazon S3. Use Amazon Athena tor analysis. **Most Voted**

B. Store the logs in Amazon RDS. Use a database client for analysis.

C. Store the logs in Amazon OpenSearch Service. Use OpenSearch Service for analysis.

D. Store the logs in an Amazon EMR cluster Use a supported open-source framework for SQL-based analysis.

Correct Answer: A

Community vote distribution

A (100%)

Comments

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: A

Scalable + "The solution must support queries with standard SQL" = A
B not scalable
C OpenSearch is like ElasticSearch so does not support SQL syntax
D EMR is processing not storage. Map-Reduce can use SQL like syntax but this option does not solve scalable storage issues. You normally run EMR on some stored data

upvoted 6 times

cedser8 2 years, 3 months ago

OpenSearch can support SQL queries, https://docs.aws.amazon.com/opensearch-service/latest/developerguide/sql-support.html

upvoted 2 times

tch Most Recent 1 year, 4 months ago

Selected Answer: A

Amazon Athena is an interactive analytics service that makes it simple to analyze data in Amazon Simple Storage Service (S3) using SQL. Athena is serverless, so there is no infrastructure to set up or manage, and you can start analyzing data immediately. You don't even need to load your data into Athena; it works directly with data stored in Amazon S3. Amazon Athena for SQL uses Trino and Presto with full standard SQL support and works with various standard data formats, including CSV, JSON, Apache ORC, Apache Parquet, and Apache Avro. Athena for Apache Spark supports SQL and allows you to use Apache Spark, an open-source, distributed processing system used for big data workloads. To get started, log in to the Athena Management Console and start interacting with your data using the query editor or notebooks.

upvoted 1 times

KennethNg923 1 year, 11 months ago

Selected Answer: A

standard SQL + analyze traffic

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: A

Difficult question because both A and C meet the requirements. (OpenSearch does "support queries with standard SQL".)

Still, native S3 storage is slightly cheaper than storage for OpenSearch. Also, Athena does not incur additional cost while OpenSearch does. Question asks for cost efficiency, thus A.

D is out, not only because of the cost but also because you do not 'store logs in (!) an Amazon EMR cluster'; you can use (!) an EMR cluster to analyze data that is stored elsewhere.
upvoted 3 times

pentium75 2 years, 5 months ago
And descriptions of both products, Athena as well as OpenSearch, state that you can use them to "analyze" data.
upvoted 3 times

TariqKipkemei 2 years, 6 months ago
Selected Answer: A
solution must support queries with standard SQL = Amazon S3 with Athena
upvoted 4 times

NickGordon 2 years, 7 months ago
Selected Answer: A
A, most cost effective
upvoted 3 times

potomac 2 years, 7 months ago
Selected Answer: A
option D (using Amazon EMR with an open-source framework) may be overkill for the relatively simple SQL-based analysis.
upvoted 2 times

Question #644 Topic 1

An international company has a subdomain for each country that the company operates in. The subdomains are formatted as example.com, country1.example.com, and country2.example.com. The company's workloads are behind an Application Load Balancer. The company wants to encrypt the website data that is in transit.

Which combination of steps will meet these requirements? (Choose two.)

A. Use the AWS Certificate Manager (ACM) console to request a public certificate for the apex top domain example com and a wildcard certificate for *.example.com. **Most Voted**

B. Use the AWS Certificate Manager (ACM) console to request a private certificate for the apex top domain example.com and a wildcard certificate for *.example.com.

C. Use the AWS Certificate Manager (ACM) console to request a public and private certificate for the apex top domain example.com.

D. Validate domain ownership by email address. Switch to DNS validation by adding the required DNS records to the DNS provider.

E. Validate domain ownership for the domain by adding the required DNS records to the DNS provider. **Most Voted**

Correct Answer: AE

Community vote distribution

AE (100%)

Comments

awsgeek75 Highly Voted 1 year, 11 months ago

Selected Answer: AE

B is private certificate so won't help as that is for internal use
C is for apex domain only and won't help with wildcard domain
A is correct

DE are both doable as per these articles

D: https://docs.aws.amazon.com/acm/latest/userguide/dns-validation.html
E: https://docs.aws.amazon.com/acm/latest/userguide/domain-ownership-validation.html

D is less applicable because it does not say if R53 is being used for DNS. You only validate ownership to R53
C makes more sense as it applies to both R53 and other DNS providers

upvoted 9 times

TariqKipkemei Highly Voted 2 years ago

Selected Answer: AE

Validate domain ownership for the domain by adding the required DNS records to the DNS provider then use the AWS Certificate Manager (ACM) console to request a public certificate for the apex top domain example com and a wildcard certificate for *.example.com

upvoted 5 times

cciesam Most Recent 2 years, 1 month ago

Selected Answer: AE

AE correct

upvoted 2 times

potomac 2 years, 1 month ago

Selected Answer: AE

BCD are wrong

upvoted 3 times

t0nx 2 years ago
Why E and not D ?
upvoted 1 times

Cyberkayu 1 year, 11 months ago
need to put A-record and CNAME in public DNS record to proof you are the legal owner of the domain name.
upvoted 4 times

Question #645 Topic 1

A company is required to use cryptographic keys in its on-premises key manager. The key manager is outside of the AWS Cloud because of regulatory and compliance requirements. The company wants to manage encryption and decryption by using cryptographic keys that are retained outside of the AWS Cloud and that support a variety of external key managers from different vendors.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS CloudHSM key store backed by a CloudHSM cluster.

B. Use an AWS Key Management Service (AWS KMS) external key store backed by an external key manager. **Most Voted**

C. Use the default AWS Key Management Service (AWS KMS) managed key store.

D. Use a custom key store backed by an AWS CloudHSM cluster.

Correct Answer: B

Community vote distribution

B (95%) 5%

Comments

pentium75 Highly Voted 1 year, 11 months ago

Selected Answer: B

Keys are supposed to be managed "outside of the AWS cloud", thus A, C and D are out.

upvoted 7 times

potomac Highly Voted 2 years, 1 month ago

Selected Answer: B

https://docs.aws.amazon.com/kms/latest/developerguide/keystore-external.html

upvoted 5 times

evelynsun Most Recent 1 year, 12 months ago

Selected Answer: A

it's A.

This solution is the LEAST operational overhead because it does not require the company to manage any infrastructure or software outside of the AWS Cloud. The AWS CloudHSM key store is managed by AWS, and the company can use it to store and manage its cryptographic keys without having to worry about the underlying infrastructure or software. The CloudHSM cluster is managed by AWS, and the company can use it to create and manage its cryptographic keys without having to worry about the hardware or software.

the AWS CloudHSM key store can also be used for external key managers. The AWS CloudHSM key store is a managed key store that is backed by ar AWS CloudHSM cluster. The AWS CloudHSM cluster is a managed service that is provided by AWS.

upvoted 1 times

pentium75 1 year, 11 months ago

"The AWS CloudHSM key store is managed by AWS" which is exactly what this company does NOT want.

upvoted 7 times

evelynsun 1 year, 12 months ago

it's A.

This solution is the LEAST operational overhead because it does not require the company to manage any infrastructure or software outside of the AWS Cloud. The AWS CloudHSM key store is managed by AWS, and the company can use it to store and manage its cryptographic keys without having to worry about the underlying infrastructure or software. The CloudHSM cluster is managed by AWS, and the company can use it to create and manage its cryptographic keys without having to worry about the hardware or software.

the AWS CloudHSM key store can also be used for external key managers. The AWS CloudHSM key store is a managed key store that is backed by ar AWS CloudHSM cluster. The AWS CloudHSM cluster is a managed service that is provided by AWS.

upvoted 1 times

pentium75 1 year, 11 months ago

"The AWS CloudHSM key store is managed by AWS" which is exactly what this company does NOT want.

upvoted 3 times

SHAAHIBHUSHANAWS 2 years ago

B

https://docs.aws.amazon.com/kms/latest/developerguide/keystore-external.html

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: B

https://docs.aws.amazon.com/kms/latest/developerguide/keystore-external.html#:~:text=Document%20history-,External%20key%20stores,-PDF

upvoted 3 times

1rob 2 years ago

Selected Answer: B

Answer A does not comply because aws cloudHSM is within aws
Answer B is the correct answer because the company is required to use its on-premises key manager. Following
https://docs.aws.amazon.com/kms/latest/developerguide/custom-key-store-overview.html gives :An external key store is an AWS KMS custom key store backed by an external key manager outside of AWS that you own and control.(...)

Answer C and D are both solutions in the aws cloud so that does not fit.

upvoted 3 times

Question #646 Topic 1

A solutions architect needs to host a high performance computing (HPC) workload in the AWS Cloud. The workload will run on hundreds of Amazon EC2 instances and will require parallel access to a shared file system to enable distributed processing of large datasets. Datasets will be accessed across multiple instances simultaneously. The workload requires access latency within 1 ms. After processing has completed, engineers will need access to the dataset for manual postprocessing.

Which solution will meet these requirements?

A. Use Amazon Elastic File System (Amazon EFS) as a shared file system. Access the dataset from Amazon EFS.

B. Mount an Amazon S3 bucket to serve as the shared file system. Perform postprocessing directly from the S3 bucket.

C. Use Amazon FSx for Lustre as a shared file system. Link the file system to an Amazon S3 bucket for postprocessing.
Most Voted

D. Configure AWS Resource Access Manager to share an Amazon S3 bucket so that it can be mounted to all instances for processing and postprocessing.

Correct Answer: C

Community vote distribution

C (100%)

Comments

potomac Highly Voted 2 years, 7 months ago

Selected Answer: C

Amazon FSx for Lustre is a fully managed, high-performance file system optimized for HPC workloads. It is designed to deliver sub-millisecond latencies and high throughput, making it ideal for applications that require parallel access to shared storage, such as simulations and data analytics.

upvoted 9 times

KennethNg923 Most Recent 1 year, 11 months ago

Selected Answer: C

host a high performance computing (HPC) workload -> FSx lustre

upvoted 2 times

zinabu 2 years, 2 months ago

FSx luster for HPC

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: C

EFS could meet the latency requirement for most (!) read (!) operations, but this is not enough here. FSx for Lustre ist specifically designed for HPC.

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

high performance computing (HPC) workloads, shared file system= Amazon FSx for Lustre

upvoted 4 times

Question #647 *Topic 1*

A gaming company is building an application with Voice over IP capabilities. The application will serve traffic to users across the world. The application needs to be highly available with an automated failover across AWS Regions. The company wants to minimize the latency of users without relying on IP address caching on user devices.

What should a solutions architect do to meet these requirements?

A. Use AWS Global Accelerator with health checks. **Most Voted**

B. Use Amazon Route 53 with a geolocation routing policy.

C. Create an Amazon CloudFront distribution that includes multiple origins.

D. Create an Application Load Balancer that uses path-based routing.

Correct Answer: A

Community vote distribution

A (97%) 3%

Comments

potomac Highly Voted 2 years, 7 months ago

Selected Answer: A

Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses or deterministic, fast regional failover.

upvoted 11 times

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: A

A - does exactly what is required
Not B - Would rely on DNS caching (as it should not)
Not C - CloudFront is not for VoIP
Not D - ALB does not address any of the issues and would not support VoIP

upvoted 7 times

KennethNg923 Most Recent 1 year, 11 months ago

Selected Answer: A

automated failover across AWS Region + minimize latency -> Global Accelerator

upvoted 2 times

Murtadhaceit 2 years, 5 months ago

Selected Answer: A

VoIP ==> UDP ==> Global Accelerator.

upvoted 3 times

kaleemanjum 2 years, 6 months ago

Selected Answer: A

AWS Global Accelerator: AWS Global Accelerator is a service that uses static IP addresses (Anycast IPs) to provide a global entry point for your applications. It routes traffic over the AWS global network to the optimal AWS endpoint based on health, geography, and routing policies.

Health Checks: AWS Global Accelerator supports health checks, allowing it to route traffic only to healthy endpoints. This helps in achieving high availability and automated failover across AWS Regions.

upvoted 2 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

A
https://aws.amazon.com/global-accelerator/faqs/#:~:text=Global%20Accelerator%20is%20a%20good,AWS%20Shield%20for%20DDoS%20protection.

upvoted 2 times

ekisako 2 years, 7 months ago

Selected Answer: A

https://docs.aws.amazon.com/global-accelerator/latest/dg/introduction-benefits-of-migrating.html

upvoted 3 times

cciesam 2 years, 7 months ago

Selected Answer: A

Global Accelerator is the answer as it can handle both TCP and UDP

upvoted 2 times

Sugarbear_01 2 years, 7 months ago

Selected Answer: C

This answer should be C

upvoted 1 times

pentium75 2 years, 5 months ago

CloudFront is not for VoIP (which usually uses UDP).

upvoted 1 times

Question #648 Topic 1

A weather forecasting company needs to process hundreds of gigabytes of data with sub-millisecond latency. The company has a high performance computing (HPC) environment in its data center and wants to expand its forecasting capabilities.

A solutions architect must identify a highly available cloud storage solution that can handle large amounts of sustained throughput. Files that are stored in the solution should be accessible to thousands of compute instances that will simultaneously access and process the entire dataset.

What should the solutions architect do to meet these requirements?

A. Use Amazon FSx for Lustre scratch file systems.

B. Use Amazon FSx for Lustre persistent file systems. Most Voted

C. Use Amazon Elastic File System (Amazon EFS) with Bursting Throughput mode.

D. Use Amazon Elastic File System (Amazon EFS) with Provisioned Throughput mode.

Correct Answer: B

Community vote distribution

B (100%)

Comments

potomac Highly Voted 2 years, 7 months ago

Selected Answer: B

Option A (Amazon FSx for Lustre scratch file systems) is designed for temporary data storage and does not provide the data persistence required for this scenario.

upvoted 9 times

FlyingHawk Most Recent 1 year, 4 months ago

Selected Answer: B

Amazon FSx for Lustre is a fully managed file system optimized for high-performance computing (HPC) workloads.

It provides sub-millisecond latency and high throughput, making it ideal for processing large datasets like weather forecasting data.

It supports thousands of compute instances accessing the file system simultaneously, ensuring scalability for the company's HPC environment.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

A persistent file system is recommended because it provides high durability and availability.

Data is replicated within an Availability Zone (AZ) and automatically backed up to Amazon S3, ensuring data is not lost in case of failures.

This is critical for a weather forecasting company that needs to ensure data integrity and availability.

upvoted 1 times

KennethNg923 1 year, 11 months ago

Selected Answer: B

HPC + the entire dataset -> FSx Lustre presistence

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

https://docs.aws.amazon.com/fsx/latest/LustreGuide/using-fsx-lustre.html

Both AB can handle the processing requirements but B is Highly Available which is also a requirement not met by A.

CD won't mee the performance requirements

upvoted 4 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

high performance computing, highly available cloud storage solution = Amazon FSx for Lustre persistent file systems

upvoted 3 times

Question #649 Topic 1

An ecommerce company runs a PostgreSQL database on premises. The database stores data by using high IOPS Amazon Elastic Block Store (Amazon EBS) block storage. The daily peak I/O transactions per second do not exceed 15,000 IOPS. The company wants to migrate the database to Amazon RDS for PostgreSQL and provision disk IOPS performance independent of disk storage capacity.

Which solution will meet these requirements MOST cost-effectively?

A. Configure the General Purpose SSD (gp2) EBS volume storage type and provision 15,000 IOPS.

B. Configure the Provisioned IOPS SSD (io1) EBS volume storage type and provision 15,000 IOPS.

C. Configure the General Purpose SSD (gp3) EBS volume storage type and provision 15,000 IOPS. Most Voted

D. Configure the EBS magnetic volume type to achieve maximum IOPS.

Correct Answer: C

Community vote distribution

C (100%)

Comments

BillaRanga Highly Voted 1 year, 10 months ago

GP2 - • Size of the volume and IOPS are linked, max IOPS is 16,000
GP3 - Can increase IOPS up to 16,000 and throughput up to 1000 MiB/s independently

GP3 is 20% cheaper than GP2

upvoted 6 times

Oblako Highly Voted 2 years ago

Selected Answer: C

Both gp2 and gp3 can provision up to 16.000 IOPS. gp3 is cheaper than gp2.

upvoted 5 times

TariqKipkemei Most Recent 2 years ago

Selected Answer: C

MOST cost-effective =GP3

upvoted 4 times

SHAAHIBHUSHANAWS 2 years ago

C
https://aws.amazon.com/ebs/general-purpose/

upvoted 3 times

lagorb 2 years ago

gp2 and gp3 can provision up to 16.000 IOPS, and gp3 is cheaper than gp2

upvoted 3 times

potomac 2 years, 1 month ago

Selected Answer: C

GP3 is better and cheaper than GP2

upvoted 4 times

Question #650 *Topic 1*

A company wants to migrate its on-premises Microsoft SQL Server Enterprise edition database to AWS. The company's online application uses the database to process transactions. The data analysis team uses the same production database to run reports for analytical processing. The company wants to reduce operational overhead by moving to managed services wherever possible.

Which solution will meet these requirements with the LEAST operational overhead?

A. Migrate to Amazon RDS for Microsoft SOL Server. Use read replicas for reporting purposes **Most Voted**

B. Migrate to Microsoft SQL Server on Amazon EC2. Use Always On read replicas for reporting purposes

C. Migrate to Amazon DynamoDB. Use DynamoDB on-demand replicas for reporting purposes

D. Migrate to Amazon Aurora MySQL. Use Aurora read replicas for reporting purposes

Correct Answer: A

Community vote distribution

A (100%)

Comments

BillaRanga Highly Voted 1 year, 10 months ago

Selected Answer: A

B - Not the LEAST operational Overhead.
C - It is No-Sql - Not compatible with SQL server which is SQL
D - MS Sql Server to MySQL may miss out some SQL Server functionalities.

A - Read replicas for RDS is easy to create and also it is Asynchronous which should not be a problem for the analytics teams as they can bear 2-3 minutes delay

upvoted 7 times

potomac Highly Voted 2 years, 1 month ago

Selected Answer: A

A is the only choice

upvoted 6 times

Firdous586 Most Recent 1 year, 10 months ago

A is the correct answer since RDS supports OLAP
And aurora OLTP

upvoted 5 times

superalaga 1 year, 11 months ago

Selected Answer: A

You can migrate with both A&B but option A is LEAST operational overhead/
A: https://aws.amazon.com/tutorials/move-to-managed/migrate-sql-server-to-amazon-rds/
B: https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/migrate-a-microsoft-sql-server-database-to-aurora-mysql-by-using-aws-dms and-aws-sct.html

upvoted 6 times

TariqKipkemei 2 years ago

Selected Answer: A

Only Amazon RDS allows the creation of readable standby DB instances.

upvoted 4 times

Question #651 Topic 1

A company stores a large volume of image files in an Amazon S3 bucket. The images need to be readily available for the first 180 days. The images are infrequently accessed for the next 180 days. After 360 days, the images need to be archived but must be available instantly upon request. After 5 years, only auditors can access the images. The auditors must be able to retrieve the images within 12 hours. The images cannot be lost during this process.

A developer will use S3 Standard storage for the first 180 days. The developer needs to configure an S3 Lifecycle rule.

Which solution will meet these requirements MOST cost-effectively?

A. Transition the objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 180 days. S3 Glacier Instant Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

B. Transition the objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 180 days. S3 Glacier Flexible Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

C. Transition the objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 180 days, S3 Glacier Instant Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years. Most Voted

D. Transition the objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 180 days, S3 Glacier Flexible Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

Correct Answer: C

Community vote distribution

C (85%) Other

Comments

TariqKipkemei Highly Voted 2 years ago

Selected Answer: C

Images cannot be lost = high availability.
Transition the objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 180 days, S3 Glacier Instant Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

upvoted 11 times

allanm 10 months ago

Yes but flexible retrieval is lower in cost than instant retrieval while meeting the SLA of 12 hours for retrieval. Answer is D

upvoted 3 times

dilaaziz Highly Voted 2 years, 1 month ago

Selected Answer: C

https://aws.amazon.com/s3/storage-classes/glacier/

upvoted 6 times

Salilgen Most Recent 1 year, 5 months ago

Selected Answer: C

IMO answer is C

upvoted 1 times

Linuslin 1 year, 6 months ago

Selected Answer: C

"The developer needs to configure an S3 Lifecycle rule."--->One Zone-IA can't transfer to Glacier Instant Retrieval--->A is out.
Check - Unsupported lifecycle transitions
https://docs.aws.amazon.com/AmazonS3/latest/userguide/lifecycle-transition-general-considerations.html
"Images cannot be lost = high availability"--->Can't be One Zone-IA--->B is out.
"the images need to be archived but must be available instantly upon request"--->Can't be "Flexible" Retrieval--->D is out.

Only C is the correct answer.
upvoted 3 times

Neung983 1 year, 9 months ago

Selected Answer: A

A.
Here's why this option is the most cost-effective:
+S3 One Zone-IA (after 180 days): Offers lower storage costs compared to S3 Standard for infrequently accessed data (180 - 360 days) while maintaining good availability for retrieval.
+S3 Glacier Instant Retrieval (after 360 days): Provides immediate access to archived images (360 - 5 years) at a significantly lower cost than S3 Standard storage. Retrieval costs are incurred but typically lower than keeping the data in S3 Standard.
+S3 Glacier Deep Archive (after 5 years): Offers the lowest storage cost for long-term archival (beyond 5 years) with retrieval times within 12 hours, meeting the auditor access requirement and minimizing ongoing storage costs.
upvoted 1 times

Salilgen 1 year, 5 months ago

A is wrong because "The images cannot be lost during this process" then you can use OneZone
upvoted 1 times

Antitouch 1 year, 11 months ago

Selected Answer: B

https://aws.amazon.com/s3/storage-classes/glacier/#:~:text=S3%20Glacier%20Flexible%20Retrieval%20delivers,year%20and%20is%20retrieved%20asynchronously.
S3 Glacier Flexible Retrieval delivers low-cost storage, up to 10% lower cost than S3 Glacier Instant Retrieval. Flexible retrieval is cheaper than Instant retrieval.
S3 Glacier Flexible retrieval storage class provides minutes to 12 hours retrieval of data. Which is within the required time by auditors.
--> We should select flexible retrieval.

The design is not caring about the high availability. The design is caring about cost. One zone-IA is cheaper than standard IA.
--> We should select One Zone IA.
upvoted 1 times

awsgeek75 1 year, 11 months ago

"The images cannot be lost during this process" is a core requirement.
The design cares about data loss and 5 years is a long time and AZ failure will result in data loss.
upvoted 4 times

pentium75 1 year, 11 months ago

Selected Answer: C

A, B impose risk of the images being lost in case of AZ failure
D does not allow instant access after 180 days
upvoted 4 times

ale_brd_111 1 year, 11 months ago

Selected Answer: C

Images cannot be lost = high availability. A exposes images to risk
upvoted 2 times

Alex1atd 2 years ago

Selected Answer: C

The images cannot be lost during this process.
upvoted 3 times

1rob 2 years ago

Selected Answer: C

"The images cannot be lost during this process" , imho this rules out S3 One zone infrequent access. S3 Glacier Instant Retrieval gives immediate access. S3 Glacier Flexible Retrieval does not give immediate access. so C.
upvoted 5 times

EdenWang 2 years ago

Selected Answer: A

high availability is not mentioned, thus I go for A
upvoted 1 times

pentium75 1 year, 11 months ago

"The images cannot be lost during this process."
upvoted 5 times

TheLaPlanta 1 year, 8 months ago

That's not HA

upvoted 1 times

cciesam 2 years, 1 month ago

Selected Answer: A

I'll go for A as it doesn't talk about High availability. Considering cost. I'll go for A

upvoted 3 times

ekisako 2 years, 1 month ago

"The images cannot be lost during this process."

upvoted 5 times

Question #652 *Topic 1*

A company has a large data workload that runs for 6 hours each day. The company cannot lose any data while the process is running. A solutions architect is designing an Amazon EMR cluster configuration to support this critical data workload.

Which solution will meet these requirements MOST cost-effectively?

A. Configure a long-running cluster that runs the primary node and core nodes on On-Demand Instances and the task nodes on Spot Instances.

B. Configure a transient cluster that runs the primary node and core nodes on On-Demand Instances and the task nodes on Spot Instances. **Most Voted**

C. Configure a transient cluster that runs the primary node on an On-Demand Instance and the core nodes and task nodes on Spot Instances.

D. Configure a long-running cluster that runs the primary node on an On-Demand Instance, the core nodes on Spot Instances, and the task nodes on Spot Instances.

Correct Answer: B

Community vote distribution

B (100%)

Comments

louisaok Highly Voted 2 years, 7 months ago
Relax man. take a break since you have made this far so far.
upvoted 58 times

AWSSURI 1 year, 9 months ago
Rest at the end NOT in the middle
-- Kobe Bryant
upvoted 12 times

potomac Highly Voted 2 years, 7 months ago
Selected Answer: B
A transient cluster provides cost savings because it runs only during the computation time, and it provides scalability and flexibility in a cloud environment.

Option C (transient cluster with On-Demand primary node and Spot core and task nodes) exposes the core nodes to Spot Instance interruptions, which may not be acceptable for a workload that cannot lose any data.
upvoted 15 times

EllenLiu Most Recent 1 year, 5 months ago
Selected Answer: B
take a break and feel better now. thanks for encouraging with each other!
daily batch handling within several hours => transient cluster
data-critical application => spot for task node only
upvoted 2 times

awsgeek75 2 years, 4 months ago
Selected Answer: B
AD are long-running so don't fit in with 6 hours schedule
BC are ideal for scheduled EMR activities
C is wrong as running core node on Spot instance has a risk of data loss https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-master-core-task-nodes.html
B is correct because primary, core will be stable on on-demand as recommended by AWS and task can go on spot instances as task nodes are short lived by nature anyway
upvoted 6 times

pentium75 2 years, 5 months ago

Selected Answer: B

"Long-running cluster" = runs until you shut it down
"Transient cluster" = runs until the workload is completed
This runs only 6 hours each day -> transient -> B or C

"Cannot lose any data while the process is running" -> Primary and core nodes cannot be Spot instances -> A or B
https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-plan-longrunning-transient.html
https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-plan-instances-guidelines.html

upvoted 10 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

Cannot loose data = ondemand primary + core nodes
Save on costs = spot task nodes
Runs for 6 hours = transient cluster

upvoted 6 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

A
https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-plan-instances-guidelines.html
It's long running and no data loss is needed.

upvoted 2 times

pentium75 2 years, 5 months ago

The link says you can lose data if you are running a transient cluster WITH ONLY Spot instances. "Long-running" = runs until you shut it down, "Transient" = Runs until the workload is completed

upvoted 1 times

whiterick 2 years, 4 months ago

Option A suggests a long-running cluster, which continues to run until manually terminated. This means that even if tasks are rerouted due to Spo Instance interruptions, the cluster itself remains active, allowing the rerouted tasks to complete on other nodes.

Option B suggests a transient cluster, which is terminated after all steps are completed. If the Spot Instances are interrupted and tasks are not completed, the cluster might still terminate after the steps are deemed complete, potentially leading to incomplete processing of data.

upvoted 1 times

MFKang 2 years, 6 months ago

Get up Stand up

upvoted 3 times

Question #653 *Topic 1*

A company maintains an Amazon RDS database that maps users to cost centers. The company has accounts in an organization in AWS Organizations. The company needs a solution that will tag all resources that are created in a specific AWS account in the organization. The solution must tag each resource with the cost center ID of the user who created the resource.

Which solution will meet these requirements?

A. Move the specific AWS account to a new organizational unit (OU) in Organizations from the management account. Create a service control policy (SCP) that requires all existing resources to have the correct cost center tag before the resources are created. Apply the SCP to the new OU.

B. Create an AWS Lambda function to tag the resources after the Lambda function looks up the appropriate cost center from the RDS database. Configure an Amazon EventBridge rule that reacts to AWS CloudTrail events to invoke the Lambda function. **Most Voted**

C. Create an AWS CloudFormation stack to deploy an AWS Lambda function. Configure the Lambda function to look up the appropriate cost center from the RDS database and to tag resources. Create an Amazon EventBridge scheduled rule to invoke the CloudFormation stack.

D. Create an AWS Lambda function to tag the resources with a default value. Configure an Amazon EventBridge rule that reacts to AWS CloudTrail events to invoke the Lambda function when a resource is missing the cost center tag.

Correct Answer: B

Community vote distribution

B (62%) A (38%)

Comments

1Alpha1 Highly Voted 2 years, 4 months ago

Selected Answer: A

I'm not sure, but I think this question is from professional solution architect question pool.
Please have a look at this one as well.
https://www.examtopics.com/discussions/amazon/view/112780-exam-aws-certified-solutions-architect-professional-sap-c02/

upvoted 11 times

JohnYu 1 year, 7 months ago

SCPs cannot directly apply tags to resources; they can only restrict actions based on policies. They do not automatically tag resources. SCPs are also used to enforce policies across accounts but cannot look up values from an RDS database or dynamically assign tags.

upvoted 3 times

pentium75 Highly Voted 2 years, 5 months ago

Selected Answer: B

A policy cannot look up "the cost center ID of the user who created the resource", we need Lambda to do that. Thus A is out.

C would work but runs on a schedule which doesn't make sense (and we would temporarily have untagged resources).

D tags resources "with a default value" which is not what we want.

upvoted 8 times

Ernestokoro 2 years, 5 months ago

Please how do you account for this part of the question with option B "The solution must tag each resource with the cost center ID of the user who created the resource." ? For me this typically what SCP would handle.

upvoted 1 times

awsgeek75 2 years, 5 months ago

That SCP won't know the cost centre. "RDS database that maps users to cost centers"
Unless the solution can read the RDS, it won't work and SCP cannot be programmed to read from RDS before applying the cost centre.

upvoted 3 times

ln2718 Most Recent 10 months, 3 weeks ago

Selected Answer: A

I like the responses to answer A. That looks correct

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: B

A is incorrect : SCPs are used to enforce permissions and restrictions across accounts in an organization. They cannot be used to dynamically tag resources based on user information stored in an RDS database. This option does not meet the requirement to tag resources with the cost center ID of the user who created them. C incorrect as it only deploy the CF stack with Lambda, but no mention how lambda function will be triggered. D default value is incorrect.

upvoted 2 times

bujuman 1 year, 7 months ago

Selected Answer: B

SCP can not be used to Tag resources

upvoted 2 times

tonybuivannghia 1 year, 7 months ago

Selected Answer: B

B is my choice

upvoted 2 times

1e22522 1 year, 10 months ago

Selected Answer: B

The best solution that meets all the requirements is option B. Here's why:

It can tag all resources created in a specific AWS account within the organization.
It uses a Lambda function to look up the appropriate cost center from the RDS database, ensuring accurate tagging.
The EventBridge rule reacting to CloudTrail events ensures that resources are tagged as they are created.
This approach can dynamically tag each resource with the cost center ID of the user who created it.

upvoted 5 times

sheilawu 2 years ago

Selected Answer: A

I will vote for A

upvoted 1 times

sandordini 2 years, 1 month ago

Selected Answer: B

We need a solution, to automatically tag, also the existing resources. A,C, are more or less working solutions for new resources, but neither can do the tagging of existing resources. D would add a default tag instead of the specific CC.

upvoted 2 times

gsgdga 2 years, 2 months ago

Selected Answer: A

A is right
https://docs.aws.amazon.com/ko_kr/organizations/latest/userguide/orgs_tagging_abac.html

upvoted 4 times

fea9bdf 2 years, 5 months ago

Answer is A, SCP handles the assignment, no need for a Lambda function, that's unnecessary t seems like
Service Control Policies (SCPs)

SCPs are a policy type that you can utilize to manage permissions across accounts in your AWS Organization.
Using SCPs lets you ensure that your accounts stay within your organization's access control guidelines.
SCPs can be used along-side tag policies to ensure that the tags are applied at the resource creation time and remain attached to the resource.

upvoted 1 times

pentium75 2 years, 5 months ago

How would an SCP look up the correct cost center in the database?

upvoted 4 times

ale_brd_111 2 years, 5 months ago

Selected Answer: B

the company still maintains the RDS, nowhere was asked to drop using it, therefore we shall use a solution that takes advantages of it.

upvoted 4 times

ftaws 2 years, 5 months ago

Selected Answer: A

I also choose A.

upvoted 2 times

awsgeek75 2 years, 5 months ago

SCP cannot connect to RDS where the cost centre information is stored so A won't work.

upvoted 2 times

Oluwatosin09 2 years, 1 month ago

Awsgeek75 and Pentium must be the same person.
They always have the same answers with contributions always on point.
Great Job!

upvoted 2 times

Cyberkayu 2 years, 5 months ago

Selected Answer: A

Company have Organization. A specific AWS account need to ensure all resources were tagged.

Move this specific AWS account under the company OU, use SCP to enforce top down policies that every member account to adhere.

Answer A.

upvoted 1 times

pentium75 2 years, 5 months ago

How would an SCP look up the correct cost center in the database?

upvoted 2 times

evelynsun 2 years, 5 months ago

Selected Answer: B

sorry, i would choose B.
because it allows you to tag resources as they are created, without requiring you to move existing resources.

upvoted 2 times

evelynsun 2 years, 5 months ago

Selected Answer: A

This solution is the best way to meet the requirements of the company. It ensures that all resources in the specific AWS account are tagged with the cost center ID of the user who created the resource. It also allows the company to easily manage and enforce compliance with its tagging policies.

upvoted 1 times

pentium75 2 years, 5 months ago

How would an SCP look up the correct cost center in the database?

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: B

Create an AWS Lambda function to tag the resources after the Lambda function looks up the appropriate cost center from the RDS database.
Configure an Amazon EventBridge rule that reacts to AWS CloudTrail events to invoke the Lambda function.

upvoted 2 times

Question #654 *Topic 1*

A company recently migrated its web application to the AWS Cloud. The company uses an Amazon EC2 instance to run multiple processes to host the application. The processes include an Apache web server that serves static content. The Apache web server makes requests to a PHP application that uses a local Redis server for user sessions.

The company wants to redesign the architecture to be highly available and to use AWS managed solutions.

Which solution will meet these requirements?

A. Use AWS Elastic Beanstalk to host the static content and the PHP application. Configure Elastic Beanstalk to deploy its EC2 instance into a public subnet. Assign a public IP address.

B. Use AWS Lambda to host the static content and the PHP application. Use an Amazon API Gateway REST API to proxy requests to the Lambda function. Set the API Gateway CORS configuration to respond to the domain name. Configure Amazon ElastiCache for Redis to handle session information.

C. Keep the backend code on the EC2 instance. Create an Amazon ElastiCache for Redis cluster that has Multi-AZ enabled. Configure the ElastiCache for Redis cluster in cluster mode. Copy the frontend resources to Amazon S3. Configure the backend code to reference the EC2 instance.

D. Configure an Amazon CloudFront distribution with an Amazon S3 endpoint to an S3 bucket that is configured to host the static content. Configure an Application Load Balancer that targets an Amazon Elastic Container Service (Amazon ECS) service that runs AWS Fargate tasks for the PHP application. Configure the PHP application to use an Amazon ElastiCache for Redis cluster that runs in multiple Availability Zones. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

awsgeek75 **Highly Voted** 1 year, 11 months ago

Selected Answer: D

Key requirements: HA and Managed Services
Key components: PHP, Static content, Redis Elasticache
AB are instantly useless for static content scaling
C could work but is less managed and "configure the backend code to reference EC2 instance" makes no sense
D ECS+Linux+PHP is good managed combination when used with Fargate. S3 for static is well-architected. Multi-AZ ECache for Redis is HA also.
Good managed solution for all purposes.

upvoted 5 times

ferdzcruz **Most Recent** 1 year, 10 months ago

D. ECS + Fargate
Company wants to redesign the architecture = from Server to serverless, and managed by AWS .

upvoted 4 times

evelynsun 1 year, 12 months ago

Selected Answer: D

This solution meets the requirements because it uses AWS managed solutions for hosting the static content and the PHP application. It also uses Amazon ECS to run the PHP application in a highly available and scalable manner. The solution also uses Amazon ElastiCache for Redis to handle session information, which is highly available and scalable. The solution also uses Amazon CloudFront to provide a secure and reliable way to deliver the static content to users.

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: D

Configure an Amazon CloudFront distribution with an Amazon S3 endpoint to an S3 bucket that is configured to host the static content. Configure an Application Load Balancer that targets an Amazon Elastic Container Service (Amazon ECS) service that runs AWS Fargate tasks for the PHP application. Configure the PHP application to use an Amazon ElastiCache for Redis cluster that runs in multiple Availability Zones.

upvoted 3 times

Question #655 Topic 1

A company runs a web application on Amazon EC2 instances in an Auto Scaling group that has a target group. The company designed the application to work with session affinity (sticky sessions) for a better user experience.

The application must be available publicly over the internet as an endpoint. A WAF must be applied to the endpoint for additional security. Session affinity (sticky sessions) must be configured on the endpoint.

Which combination of steps will meet these requirements? (Choose two.)

A. Create a public Network Load Balancer. Specify the application target group.

B. Create a Gateway Load Balancer. Specify the application target group.

C. Create a public Application Load Balancer. Specify the application target group. **Most Voted**

D. Create a second target group. Add Elastic IP addresses to the EC2 instances.

E. Create a web ACL in AWS WAF. Associate the web ACL with the endpoint **Most Voted**

Correct Answer: CE

Community vote distribution

CE (100%)

Comments

ferdzcruz 2 years, 4 months ago

CE.
C. application = ALB
E. WAF to endpoint

upvoted 4 times

awsgeek75 2 years, 5 months ago

Selected Answer: CE

NLB and GLB cannot handle sticky sessions. It's an application level concept (Cookies) so ALB works.
Elastic IP will negate sticky sessions and this combination won't work.
E give proper permissions to WAF

upvoted 4 times

tonybuivannghia 1 year, 7 months ago

I agree with you CE is correct but your explanation is wrong. NLB, ALB and CLB can handle sticky sessions. But the WAF is just only working with ALB, so ALB is correct.

upvoted 1 times

LeonSauveterre 1 year, 6 months ago

After a little digging, turns out NLBs only offer IP-based stickiness, which is not ideal for web applications where many users might share the same public IP address (due to NAT oh well). Cookie-based stickiness offered by ALBs is more appropriate.

PS. Sticky sessions are not supported if you are using TLS termination on NLB. If you need TLS termination and sticky sessions, you should use ALB as well.

upvoted 1 times

Mikado211 2 years, 5 months ago

Selected Answer: CE

- Make it accessible from the web + sticky session == Public ALB
- Additional security == web ACL in WAF (and integrate the web ACL to the ALB)

upvoted 2 times

ZZZ_Sleep 2 years, 5 months ago

Selected Answer: CE

session affinity (sticky sessions) = Application Load Balancer

WAF must be applied to the endpoint for additional security = web ACL in WAF

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: CE

Session Affinity = Application Load Balancer
Create a public Application Load Balancer. Specify the application target group then create a web ACL in AWS WAF. Associate the web ACL with the ALB endpoint.

upvoted 3 times

Question #656 *Topic 1*

A company runs a website that stores images of historical events. Website users need the ability to search and view images based on the year that the event in the image occurred. On average, users request each image only once or twice a year. The company wants a highly available solution to store and deliver the images to users.

Which solution will meet these requirements MOST cost-effectively?

A. Store images in Amazon Elastic Block Store (Amazon EBS). Use a web server that runs on Amazon EC2.

B. Store images in Amazon Elastic File System (Amazon EFS). Use a web server that runs on Amazon EC2.

C. Store images in Amazon S3 Standard. Use S3 Standard to directly deliver images by using a static website.

D. Store images in Amazon S3 Standard-Infrequent Access (S3 Standard-IA). Use S3 Standard-IA to directly deliver images by using a static website. Most Voted

Correct Answer: D

Community vote distribution

D (93%) 7%

Comments

chikuwan Highly Voted 2 years, 6 months ago

Selected Answer: D

users request each image only once or twice a year
So the answer is D

upvoted 10 times

Abbas_Abi_AWS Most Recent 1 year, 10 months ago

Selected Answer: D

D is correct

upvoted 1 times

KennethNg923 1 year, 11 months ago

Selected Answer: D

users request each image only once or twice a year, so infrequent is enough and cheaper

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

"On average, users request each image only once or twice a year."
S3 Infrequent Access is more than enough for this.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: D

Say, you have 1 TB of files that you access twice a year. Yearly cost:
C, S3 Standard: 276 USD for storage, free retrieval = 276 USD
D, S3 Standard-IA: 138 USD for storage, 20 € for retrieval = 158 USD

upvoted 2 times

Kumar05162 2 years, 5 months ago

Option D: Store images in Amazon S3 Standard-Infrequent Access (S3 Standard-IA). Use S3 Standard-IA to directly deliver images by using a static website.

S3 Standard-IA is designed specifically for infrequently accessed data, offering lower storage costs compared to S3 Standard while still providing the necessary durability and availability.

upvoted 2 times

ZZZ_Sleep 2 years, 5 months ago

Selected Answer: D

High Availability = excluded A (EBS)
cost-effective = excluded B (EFS)
only once or twice a year = S3 Standard-IA, excluded C (S3 Standard, frequent access)

Left D, answer

upvoted 4 times

LuADS 2 years, 6 months ago

Selected Answer: C

Suppose there are thousands or millions of users, each image should be recovered once or twice a year X total users... makes it more expensive than the standard class since the recovery price of Standard-IA is $0.01 per GB + price of the requests which is also more expensive too.

upvoted 2 times

pentium75 2 years, 5 months ago

Not sure if you understood what IA class does. "Recovery price is 0.001 per GB", what's the issue with that if images are requested "only once or twice a year"?

Say, you have 1 TB of files that you access twice a year.
S3 Standard: 276 USD for storage, free retrieval = 276 USD
S3 Standard-IA: 138 USD for storage, 20 € for retrieval = 158 USD

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: D

MOST cost-effectively, request each image only once or twice a year= S3 Standard-Infrequent Access

upvoted 2 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

D
https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-class-intro.html
Look at table

upvoted 2 times

achechen 2 years, 6 months ago

Selected Answer: D

if the images are accessed once or twice a year, then it is cheaper to use infrequent access tier

upvoted 4 times

aragornfsm 2 years, 6 months ago

I believe the correct answer is option D, but ChatGPT mentioned option C. I didn't understand. I'm curious about the actual correct answer.

upvoted 2 times

AndreiWebNet 2 years, 6 months ago

Might be the fact the a user is requesting to view a image once or twice a year but how many users are there ? :) that's why it points to C i think. I still think that the correct answer is D due to lack of information in the description

upvoted 1 times

pentium75 2 years, 5 months ago

"Users request each image only once or twice per year", this refers to all users together, it does not say "EACH user". In other words, every image is accessed once or twice a year.

upvoted 2 times

Question #657 Topic 1

A company has multiple AWS accounts in an organization in AWS Organizations that different business units use. The company has multiple offices around the world. The company needs to update security group rules to allow new office CIDR ranges or to remove old CIDR ranges across the organization. The company wants to centralize the management of security group rules to minimize the administrative overhead that updating CIDR ranges requires.

Which solution will meet these requirements MOST cost-effectively?

A. Create VPC security groups in the organization's management account. Update the security groups when a CIDR range update is necessary.

B. Create a VPC customer managed prefix list that contains the list of CIDRs. Use AWS Resource Access Manager (AWS RAM) to share the prefix list across the organization. Use the prefix list in the security groups across the organization. Most Voted

C. Create an AWS managed prefix list. Use an AWS Security Hub policy to enforce the security group update across the organization. Use an AWS Lambda function to update the prefix list automatically when the CIDR ranges change.

D. Create security groups in a central administrative AWS account. Create an AWS Firewall Manager common security group policy for the whole organization. Select the previously created security groups as primary groups in the policy.

Correct Answer: B

Community vote distribution

B (100%)

Comments

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: B

A managed prefix list is a set of one or more CIDR blocks. You can use prefix lists to make it easier to configure and maintain your security groups and route tables. You can create a prefix list from the IP addresses that you frequently use, and reference them as a set in security group rules and routes instead of referencing them individually. If you scale your network and need to allow traffic from another CIDR block, you can update the relevant prefix list and all security groups that use the prefix list are updated. You can also use managed prefix lists with other AWS accounts using Resource Access Manager (RAM).

https://docs.aws.amazon.com/vpc/latest/userguide/managed-prefix-lists.html#:~:text=A-,managed%20prefix,-list%20is%20a

upvoted 8 times

awsgeek75 Highly Voted 2 years, 4 months ago

Such a badly worded question:
"The company has multiple offices around the world. The company needs to update security group rules to allow new office CIDR ranges or to remove old CIDR ranges across the organization."

Are the CIDR groups associated to offices? That will be illogical. I think it should be VPC and not offices.

upvoted 6 times

Gape4 Most Recent 1 year, 11 months ago

Selected Answer: B

I will go for B

upvoted 2 times

KennethNg923 1 year, 11 months ago

Selected Answer: B

prefix list for CIDR blocks

upvoted 2 times

avdxeqtr 2 years, 4 months ago

Selected Answer: B

https://docs.aws.amazon.com/vpc/latest/userguide/managed-prefix-lists.html

upvoted 2 times

ale_brd_111 2 years, 5 months ago

Selected Answer: B

Answer is B

upvoted 2 times

achechen 2 years, 6 months ago

Selected Answer: B

looks like B is the answer. Reference: https://docs.aws.amazon.com/vpc/latest/userguide/managed-prefix-lists.html

upvoted 3 times